
Beyond Augmented-Action Surrogates for Multi-Expert Learning-to-Defer

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Abstract

A learning-to-defer (L2D) system decides, for each input, whether to predict on its own or to hand it to one of several available experts. The very well established recipe trains classifier and router jointly by treating the K classes and J experts as competing actions in one shared $(K+J)$ -action geometry. Subsequent work has proposed a series of incremental fixes within this geometry; we show that each still suffers, to varying severity, from an optimization-level pathology (target distortion, gradient amplification, winner-take-all starvation, set-mass collapse, or class-expert coupling) even under statistical consistency. We step outside the augmented-action family entirely and propose a *decoupled surrogate*: a softmax classifier head and an independent sigmoid head per expert, mirroring the two natural objects of the problem. We show that per-sample updates are then coordinatewise and the class-expert Hessian block is identically zero, and prove an excess-risk bound with calibration constant $\max\{2\sqrt{2}, \sqrt{2J/\lambda}\}$ —to our knowledge the first multi-expert L2D guarantee whose constant does not grow with the expert pool when the per-expert weight is held fixed. On controlled synthetic studies and on CIFAR-10, CIFAR-10H, and Covertypes, it is the only method in our comparison that remains stable as the expert pool grows, preserves rare specialists, and improves over a standalone classifier on every real-data benchmark.

1 Introduction

Learning-to-Defer (L2D) augments a classifier with the option of handing an input to an expert (Madras et al., 2018; Mozannar and Sontag, 2020). In the multi-expert setting, this means deciding, for each example, whether to predict directly or defer to one among several available experts. The Bayes rule has a transparent form: it weighs two conditional quantities against each other — the class posterior $\eta_k(x) = \Pr(Y=k \mid X=x)$ and the expert utilities $\alpha_j(x) = \Pr(M_j=Y \mid X=x)$ — predicting when the best class posterior exceeds the best expert utility and deferring otherwise. The challenge is purely on the surrogate side: learning these two quantities, and the comparison between them, faithfully from finite data.

Most existing multi-expert surrogates follow one common route. They introduce an augmented action space containing the K class actions and the J defer-to-expert actions, learn a single score vector $a(x) \in \mathbb{R}^{K+J}$ over that space, and derive both class and expert supervision from those same shared scores (Mozannar and Sontag, 2020; Verma et al., 2022; Cao et al., 2024; Mao et al., 2025d; Liu et al., 2026). This route has delivered statistically consistent surrogates, but it is also a strong modeling choice. The Bayes rule compares a categorical posterior with expert utilities, whereas the augmented-action reduction places classes and experts inside one shared action geometry. Prior work already shows that this geometry can produce unbounded and misaligned defer-probability estimates (Verma et al., 2022; Cao et al., 2024), and that adding experts can cause systematic underfitting as the surrogate reward structure distorts the gradient budget (Liu et al., 2026). Existing fixes restore the

38 correct probability range or cap the reward to one expert, but they address only one symptom at a
 39 time without examining the *optimization* path that leads there.

40 This is the distinction that motivates the paper. A surrogate can be judged along two axes: what it
 41 learns at the population level, and how it distributes supervision during training. Consistency tells
 42 us about the first axis: the conditional minimizer recovers the Bayes decision. Bounded probability
 43 estimation tells us that the learned scores live on the correct range. Neither says anything about how
 44 overlap among experts reshapes the gradient budget of a sample, whether correct but non-winning
 45 experts are actively suppressed, or whether the class and expert updates remain entangled. In multi-
 46 expert L2D, these local effects are consequences of the surrogate design itself, and they become more
 47 pronounced as the number of experts increases.

48 Our analysis makes this concrete. For additive augmented cross-entropy, the conditional target
 49 is distorted by the total expert utility $\sum_j \alpha_j(x)$: the surrogate learns a flattened version of the
 50 Bayes quantities rather than the quantities themselves (Proposition 14). At the sample level, the
 51 gradient budget is amplified by $1 + |\mathcal{J}(y, m)|$, where $\mathcal{J}(y, m)$ is the realized set of correct experts
 52 (Proposition 4). The distortion is therefore statistical and optimization-level at once. The loss
 53 targets the wrong object, and it allocates more update mass to regions where experts already agree.
 54 PiCCE (Liu et al., 2026) removes the amplification by selecting a single correct expert per sample,
 55 but then pushes every other correct expert *down*, creating a starvation effect that can suppress rare
 56 specialists (Proposition 5). Mao et al. (2025d) avoid amplification but learn only the total acceptable-
 57 set mass, with no ranking inside that set (Proposition 6). A-SM (Cao et al., 2024) corrects the
 58 target and produces bounded estimates, but retains a class–expert gradient coupling that can be
 59 $O(J)$ (Proposition 7). OvA (Verma et al., 2022) fully decouples expert gradients, but treats the class
 60 posterior as K independent binary tasks rather than one categorical distribution (Proposition 19). The
 61 issue is therefore not an isolated defect in any single loss: the augmented-action family repeatedly
 62 trades one mismatch for another.

63 We propose a *decoupled surrogate* that departs from the augmented-action family entirely. The
 64 class head uses a softmax, producing $p(x) \in \Delta^K$. Each expert head uses an independent sigmoid,
 65 producing $u_j(x) \in (0, 1)$. This factorization aligns both axes at once and is, to our knowledge, the
 66 first multi-expert L2D surrogate to simultaneously: (i) identify (η, α) at the population optimum with
 67 a categorical class posterior $p \in \Delta^K$; (ii) have fully decoupled, uniformly bounded optimization
 68 geometry (no amplification, no starvation, no class–expert coupling); (iii) admit a consistency bound
 69 with calibration constant $\max\{2\sqrt{2}, \sqrt{2J/\lambda}\}$, which is invariant in J whenever the per-expert weight
 70 λ/J is held fixed (equivalently, λ scales linearly with J), and equals $2\sqrt{2}$ whenever $\lambda/J \geq 1/4$.

71 Contributions.

- 72 – We identify a common failure mode of augmented-action multi-expert surrogates. For five represen-
 73 tative losses, we prove exact target and gradient/curvature statements showing how shared action
 74 geometry causes target distortion, gradient amplification, winner-take-all starvation, acceptable-set
 75 ranking loss, class–expert coupling, or improper class geometry (Section 4).
- 76 – We propose a decoupled surrogate that estimates the two Bayes objects on their native scales: a
 77 categorical class posterior and independent expert utilities. We prove that its population minimizer
 78 is exactly (η, α) and that its gradient and Hessian are block-decoupled and uniformly bounded
 79 (Section 5).
- 80 – We prove a quantitative consistency bound: decoupled surrogate excess risk controls L2D excess
 81 risk with calibration constant $\max\{2\sqrt{2}, \sqrt{2J/\lambda}\}$. Hence the constant is independent of J when
 82 the per-expert weight λ/J is fixed, unlike the available quantitative augmented-action bounds
 83 (Theorem 10).
- 84 – We validate the decoupled surrogate on synthetic benchmarks targeting each predicted pathology
 85 and on three real-data benchmarks, showing that the predicted pathologies appear in practice and
 86 that the decoupled surrogate remains stable as the expert pool grows (Section 6).

87 2 Related Work

88 Learning-to-defer extends selective prediction and classification with a reject option (Chow, 1970;
 89 Bartlett and Wegkamp, 2008; Geifman and El-Yaniv, 2017; Cortes et al., 2016; Cao et al., 2022;
 90 Cortes et al., 2024b) by allowing a learner to route inputs to external experts rather than abstaining

altogether. The modern L2D formulation was introduced for single-expert systems by Madras et al. (2018) and Mozannar and Sontag (2020), and has since grown into a broader literature on surrogate design and statistical guarantees (Mao et al., 2024a; Charusaie et al., 2022; Mozannar et al., 2023; Mao et al., 2024e, 2025d; Montreuil et al., 2025b, 2026a). Appendix A gives an extended related-work discussion.

Surrogate design and underfitting. Early predictor-rejector frameworks separate the classification and rejection functions, but Ni et al. (2019) show that such surrogates are inconsistent in the multiclass case. Mozannar and Sontag (2020) introduce the first consistent multiclass surrogate via an augmented action space that jointly scores the K class actions and J defer-to-expert actions. Subsequent work refines this shared-vector design. Verma et al. (2022) compare coupled softmax and one-vs-all (OvA) parameterizations and show that the softmax defer-probability estimate is unbounded, whereas OvA remains bounded. Cao et al. (2024) show that symmetric reductions can produce invalid defer probabilities and propose an asymmetric softmax parameterization with a multi-expert extension. Mao et al. (2025d) design a single-stage surrogate family with formal consistency guarantees for multi-action deferral. A complementary line studies optimization pathologies: Narasimhan et al. (2022) first identify classifier underfitting in the single-expert setting and propose a post-hoc correction. Still in the single-expert setting, Liu et al. (2024) propose a training-time mitigation. More recently, Liu et al. (2026) show that in the multi-expert setting, even redundant experts can induce underfitting through the additive expert reward, and propose a winner-take-all correction (PiCCE).

Positioning. Our work differs from the above in scope: rather than proposing an incremental fix within the augmented-action family, we show that the family as a whole exhibits a pattern in which each surrogate trades a fix on one axis (statistical target or optimization geometry) for a failure on the other. The decoupled surrogate departs from the shared-action-space design entirely and avoids this trade-off by construction.

3 Preliminaries

Multi-expert learning to defer. Following the modern learning-to-defer literature (Madras et al., 2018; Mozannar and Sontag, 2020; Mao et al., 2024e; Montreuil et al., 2026a), let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space supporting a triple $(X, Y, \mathbf{M})$ with distribution $\mathcal{D}$ on $\mathcal{X} \times [K] \times [K]^J$, where $\mathcal{X} \subseteq \mathbb{R}^d$ is the input space, $[K] := \{1, \dots, K\}$ is the label space, and $\mathbf{M} = (M_1, \dots, M_J)$ records the predictions of J experts. We write (x, y, m) for a realization of $(X, Y, \mathbf{M})$.

The learner consists of a classifier $h : \mathcal{X} \rightarrow [K]$ and a router $r : \mathcal{X} \rightarrow \{0\} \cup [J]$. The decision $r(x) = 0$ means that the learner predicts with $h(x)$, whereas $r(x) = j$ means that the instance is routed to expert j . In the classification-specialized setting considered throughout the paper (Mozannar and Sontag, 2020; Verma et al., 2022; Cao et al., 2024; Mao et al., 2024e; Liu et al., 2026), the only question is therefore which of the K class predictions and the J expert actions yields the smallest conditional error.

Definition 1 (Multi-Expert Defer Loss). The multi-expert defer loss incurred on a realization (x, y, m) is

$$\ell_{\perp}(h, r; x, y, m) = \mathbf{1}\{h(x) \neq y\} \mathbf{1}\{r(x) = 0\} + \sum_{j \in [J]} \mathbf{1}\{m_j \neq y\} \mathbf{1}\{r(x) = j\}. \quad (1)$$

Its population risk is $\mathcal{E}_{\perp}(h, r) := \mathbb{E}[\ell_{\perp}(h, r; X, Y, \mathbf{M})]$.

At each input, the learner either predicts on its own and incurs the usual misclassification loss, or it transfers control to one of the experts and pays that expert’s error on the same example. The population objective is to choose the better option pointwise in expectation.

Bayes quantities. The Bayes rule is governed by two families of conditional probabilities. The first is the class posterior $\eta_k(x) := \Pr(Y = k \mid X = x)$ for $k \in [K]$. The second is the vector of expert utilities $\alpha_j(x) := \Pr(M_j = Y \mid X = x)$ for $j \in [J]$. These quantities are the natural objects of the defer problem. Predicting class k incurs conditional risk $1 - \eta_k(x)$, while deferring to expert j incurs conditional risk $1 - \alpha_j(x)$. Thus the optimal action depends only on the largest class posterior and the largest expert utility, not on an augmented distribution over $K + J$ actions.

139 **Lemma 2** (Bayes Rule for Multi-Expert L2D). *For every $x \in \mathcal{X}$, a Bayes-optimal decision is*
 140 *obtained by*

$$r^*(x) \in \begin{cases} \{0\}, & \text{if } \max_{k \in [K]} \eta_k(x) \geq \max_{j \in [J]} \alpha_j(x), \\ \arg \max_{j \in [J]} \alpha_j(x), & \text{otherwise,} \end{cases} \quad (2)$$

141 with $h^*(x) \in \arg \max_{k \in [K]} \eta_k(x)$.

142 Lemma 2 reveals the basic statistical structure of the problem. The class posterior $\eta(x)$ is a categorical
 143 distribution and therefore lies in the simplex Δ^K . By contrast, the expert utilities $\alpha_1(x), \dots, \alpha_J(x)$
 144 are J conditional probabilities in $[0, 1]$ with no analogous normalization constraint. The Bayes-
 145 sufficient object is therefore the pair $(\eta(x), \alpha(x)) \in \Delta^K \times [0, 1]^J$.

146 **Augmented-action viewpoint.** Because the loss in Definition 1 is discontinuous, gradient-based
 147 learning must proceed through a surrogate (Zhang, 2002; Bartlett et al., 2006; Steinwart, 2007). The
 148 dominant one-stage reduction, introduced by Mozannar and Sontag (2020), casts the problem as
 149 prediction over one enlarged action space $\mathcal{A} := [K] \cup \{\perp_1, \dots, \perp_J\}$ and learns one score for each
 150 action.

151 **Definition 3** (Augmented-Action Surrogate Family). An *augmented-action surrogate* is specified by
 152 a score map $a : \mathcal{X} \rightarrow \mathbb{R}^{K+J}$ together with a samplewise loss of the form

$$\Phi^{\text{aug}}(a; x, y, m) = \Phi_{\text{cls}}^{\text{aug}}(a; x, y) + \Phi_{\text{exp}}^{\text{aug}}(a; x, y, m),$$

153 and a prediction rule that selects one augmented action from the same shared score vector $a(x)$, either
 154 by $\arg \max_i a_i(x)$ or by the argmax of a maxima-preserving transform of $a(x)$.

155 This definition covers the baselines studied in the paper: additive cross-entropy (Mozannar and
 156 Sontag, 2020), Verma’s OvA surrogate (Verma et al., 2022), A-SM (Cao et al., 2024), PiCCE (Liu
 157 et al., 2026), and Mao25 (Mao et al., 2025d). They differ in how they build the class and expert terms,
 158 but they all learn defer decisions through one shared action geometry over classes and experts.

159 4 The Augmented-Action Mismatch

160 We now analyze the augmented-action surrogates of Definition 3 along two axes: (i) the *conditional*
 161 *target* at the population optimum, and (ii) the *local geometry* — how the surrogate distributes gradient
 162 mass during training. Consistency and bounded estimation address axis (i); we show that axis (ii)
 163 matters equally, and that the augmented-action family repeatedly trades a failure on one axis for
 164 a failure on the other. Throughout, $\mathcal{J}(y, m) := \{j \in [J] : m_j = y\}$ denotes the set of correct
 165 experts and $q_i(x) := \exp(a_i) / \sum_{r=1}^{K+J} \exp(a_r)$. Detailed proofs, curvature calculations, and worked
 166 examples for the baseline analyses are collected in Appendix D–I.

167 4.1 Additive Cross-Entropy: Target Distortion and Gradient Amplification

168 We begin with the simplest augmented-action surrogate, additive cross-entropy (Mozannar and
 169 Sontag, 2020), which rewards the correct class and every correct expert through one shared softmax:

$$\Phi^{\text{CE}}(a; x, y, m) = -\log q_y(x) - \sum_{j \in \mathcal{J}(y, m)} \log q_{K+j}(x). \quad (3)$$

170 Prior work has already shown that additive CE can produce unbounded defer-probability estimates
 171 (Verma et al., 2022) and can underfit when experts are redundant (Liu et al., 2026). This makes it a
 172 useful lens for the two mismatches studied in this paper: on axis (i), it distorts the *target*; on axis (ii), it
 173 induces a multiplicity pathology that worsens as the number of experts grows. The resulting failure is
 174 therefore not merely one of probability validity. Equally importantly, the shared augmented geometry
 175 alters the per-sample gradients and curvature in a manner that systematically favors regions of high
 176 expert overlap.

177 At the population level, additive CE does not target (η, α) directly. Its conditional optimum normalizes
 178 both the class and expert terms by the same factor $1 + U(x)$, where $U(x) = \sum_j \alpha_j(x)$, so the target
 179 itself changes with the total expert overlap. If experts $2, \dots, J$ are redundant copies of expert 1, the
 180 Bayes decision is unchanged, yet the normalization grows linearly in J and pushes every softmax
 181 probability toward zero. This same normalization underlies the unbounded reconstruction issue
 182 identified by Cao et al. (2024). We defer the exact conditional minimizer statement to Appendix D.2.

183 **Gradient and curvature amplification (axis (ii)).** The target distortion is only part of the issue.
 184 The samplewise geometry is at least as important, because it determines how the surrogate allocates
 185 update mass during training.

186 **Proposition 4** (Samplewise amplification in additive CE). *For one sample (x, y, m) with $\mathcal{J} =$
 187 $\mathcal{J}(y, m)$, the gradient and Hessian of Φ^{CE} satisfy*

$$\frac{\partial \Phi^{\text{CE}}}{\partial a_i} = (1 + |\mathcal{J}|) q_i(x) - \mathbf{1}\{i = y\} - \mathbf{1}\{i = K + j, j \in \mathcal{J}\}. \quad (4)$$

$$\nabla_a^2 \Phi^{\text{CE}} = (1 + |\mathcal{J}|) (\text{Diag}(q) - qq^\top), \quad (5)$$

188 with largest eigenvalue $\lambda_{\max}(\nabla_a^2 \Phi^{\text{CE}}) \leq (1 + |\mathcal{J}|)/2$.

190 Both parts carry the same multiplicity factor $1 + |\mathcal{J}|$. If six of eight experts are correct on a sample,
 191 additive CE treats it as $7\times$ more important than standard CE would, and the curvature jumps from
 192 $1/2$ to $7/2$. Since samples with many correct experts tend to be easy, high-agreement regions, the
 193 optimizer is both biased *toward* those regions and poorly conditioned there — pulled away from the
 194 decision boundaries where the classify-versus-defer choice is most consequential.

195 4.2 Why Winner-Take-All Fails

196 The amplification of §4.1 arises because the additive reward fires once for the true class *and* once
 197 for every correct expert. A natural repair is to reward exactly one correct expert per sample. This is
 198 the strategy of PiCCE (Liu et al., 2026), instantiated with cross-entropy as the base multiclass loss.
 199 Let j^* be the deterministically tie-broken maximizer of a_{K+j} over $j \in \mathcal{J}(y, m)$ when this set is
 200 nonempty. Then

$$\Phi^{\text{PiCCE}}(a; x, y, m) = -\log q_y(x) - \mathbf{1}\{\mathcal{J} \neq \emptyset\} \log q_{K+j^*}(x). \quad (6)$$

201 With at most two log-loss terms active, the total positive gradient mass is capped at 2 regardless of
 202 how many experts are correct, and the Hessian is $\nabla_a^2 \Phi^{\text{PiCCE}} = 2(\text{Diag}(q) - qq^\top)$, so $\lambda_{\max} \leq 1$
 203 uniformly (Appendix E.4). On axis (ii), the multiplicity explosion of (5) is gone.

204 **Starvation of non-winning experts (axis (ii)).** The $\arg \max$ selection solves the amplification
 205 problem but creates a new gradient pathology.

206 **Proposition 5** (Winner-take-all starvation in PiCCE). *Under PiCCE with $\mathcal{J}(y, m) \neq \emptyset$, the gradient
 207 on any logit is $\partial \Phi^{\text{PiCCE}} / \partial a_i = 2q_i - \mathbf{1}\{i = y\} - \mathbf{1}\{i = K + j^*\}$. For any correct expert $k \in \mathcal{J}$
 208 with $k \neq j^*$,*

$$\frac{\partial \Phi^{\text{PiCCE}}}{\partial a_{K+k}} = 2q_{K+k}(x) > 0. \quad (7)$$

209 *Gradient descent pushes expert k 's logit downward, even though it is correct.*

210 The mechanism is simple: in the shared softmax, the two log-loss terms reward only the true class
 211 y and the winning expert j^* . Every other action receives only the repulsive softmax redistribution
 212 $2q_i$, which pushes its logit down. For classes $k \neq y$ this repulsion is desirable. For correct experts
 213 $k \in \mathcal{J} \setminus \{j^*\}$ it is not: these experts are being punished despite being right, purely because a rival
 214 expert had a higher logit.

215 **Self-reinforcing lock-in and a moving target.** Starvation compounds across iterations. If j_1 starts
 216 with a marginally higher logit than j_2 on a shared region, j_1 wins the $\arg \max$ and receives gradient
 217 $2q_{K+j_1} - 1$, while j_2 is pushed down ($2q_{K+j_2} > 0$). In the non-saturated regime $q_{K+j_1} < 1/2$,
 218 the winner is pulled up and the gap widens; even after saturation, the non-winning correct expert
 219 continues to receive the repulsive positive gradient. This positive feedback converts initialization
 220 noise into deterministic suppression. This also corrupts the statistical target on axis (i). PiCCE does
 221 not estimate (η, α) but $(\eta, \gamma(a, \cdot))$, where $\gamma_j(a, x) = \Pr(j = j^* \text{ and } j \in \mathcal{J} \mid X = x)$ depends on the
 222 current scores through the winner map (Appendix E.3). As starvation suppresses experts, γ_j shifts
 223 mass toward surviving winners, further entrenching their dominance. The target is therefore *moving*
 224 rather than fixed by the data distribution.

225 4.3 Three Refined Surrogates, Three Residual Gaps

226 Three further baselines target different defects. Each closes one gap but opens another.

227 **Mao25: correct set, no ranking.** The surrogate of Mao et al. (2025d) penalizes the total mass
 228 assigned outside the acceptable set: $\Phi^{\text{Mao}} = 1 - (q_y + \sum_{j \in \mathcal{J}} q_{K+j})$. It comes with formal
 229 consistency guarantees (Mao et al., 2025d). Beyond those guarantees, we observe that the loss is
 230 linear in the total acceptable mass S_S , making it a set-valued analogue of multiclass MAE (Ghosh
 231 et al., 2017).

232 **Proposition 6** (Acceptable-set mass in Mao25). *The gradient is $\partial\Phi^{\text{Mao}}/\partial a_i = q_i(S_S - \mathbf{1}\{i \in$
 233 $\mathcal{S}(y, m)\})$, where $\mathcal{S}(y, m) = \{y\} \cup \{K + j : j \in \mathcal{J}\}$ and $S_S = \sum_{c \in \mathcal{S}} q_c$. The loss depends on
 234 the acceptable set only through S_S , and for acceptable coordinates the gradient has the common
 235 factor $S_S - 1$. Thus the samplewise update distinguishes acceptable from unacceptable actions, but
 236 it contains no term that prefers one acceptable expert over another on the basis of expert quality.*

237 Because the samplewise supervision cannot express preferences inside the acceptable set, Mao25
 238 does not recover expert utilities directly. The surrogate can identify a Bayes-optimal action, but
 239 it does not estimate the full vector (η, α) needed for utility-level comparison among experts — a
 240 limitation on axis (i). The same MAE-like structure also weakens optimization: the total attractive
 241 gradient mass on acceptable actions is $\sum_{i \in \mathcal{S}} -\partial\Phi^{\text{Mao}}/\partial a_i = S_S(1 - S_S)$, which is at most 1/4
 242 and vanishes as $S_S \rightarrow 0$. So when the model currently assigns little mass to the acceptable set,
 243 the corrective signal is weak, consistent with the optimization behavior commonly associated with
 244 MAE-like losses (Zhang and Sabuncu, 2018).

245 **A-SM: correct target, coupled geometry.** The asymmetric softmax of Cao et al. (2024) defines
 246 class probabilities $\xi_k = \exp(a_k) / \sum_{k'} \exp(a_{k'})$ over classes only and expert estimates via a sigmoid
 247 $\psi_j = \sigma(a_{K+j} - \log \sum_{k' \neq k^*} \exp(a_{k'}))$ with $k^* \in \arg \max_k a_k$ (Definition 17). At the population
 248 optimum $\xi_k^* = \eta_k$ and $\psi_j^* = \alpha_j$: axis (i) is fully corrected. But ψ_j still depends on the class logits,
 249 coupling the two estimation problems on axis (ii).

250 **Proposition 7** (Class-expert gradient coupling in A-SM). *Fix $k^* \in \arg \max_{k \in [K]} a_k$. For any $r \neq k^*$,*
 251 *let $\pi_r = \exp(a_r) / \sum_{k \neq k^*} \exp(a_k)$. Then*

$$\frac{\partial\Phi^{\text{ASM}}}{\partial a_r} = \underbrace{(\xi_r - \mathbf{1}\{r=y\})}_{\text{class term}} - \pi_r \underbrace{\sum_{j=1}^J (\psi_j - \mathbf{1}\{m_j=y\})}_{\text{expert leakage}}. \quad (8)$$

252 The leakage term vanishes in expectation at the optimum but can be $O(J)$ on individual samples. The
 253 mixed Hessian block has operator norm $\leq \sqrt{J}/4$ (Appendix G.4), so the coupling grows with the
 254 expert pool. A-SM is the strongest augmented baseline on axis (i); its residual weakness is entirely
 255 on axis (ii).

256 **OvA: decoupled gradients, improper class posterior.** OvA (Verma et al., 2022) abandons the
 257 shared simplex entirely, replacing it by $K + J$ independent binary logistic losses (Definition 18).
 258 Because every coordinate is optimized independently, the amplification, starvation, and coupling
 259 pathologies are all absent: axis (ii) is fully resolved. Its remaining gap is on axis (i): the class head
 260 consists of K independent sigmoids, so the estimates need not lie in Δ^K away from the optimum
 261 (Cao et al., 2024).

262 **Synthesis.** Table 1 summarizes the picture. Every baseline repairs some properties but introduces
 263 or retains others; all five pathologies trace to one architectural root — entanglement of classes and
 264 experts inside the shared $(K+J)$ -simplex. Two design principles emerge, one for each axis: *separate*
 265 the class and expert estimation problems (no shared normalization, no gradient coupling), addressing
 266 axis (ii); and *match* each head to the statistical type of its target (softmax for $\eta \in \Delta^K$, independent
 267 sigmoids for $\alpha_j \in [0, 1]$), addressing axis (i). The next section shows that the decoupled surrogate
 268 satisfies both.

269 5 The Decoupled Surrogate

270 The analysis of Section 4 traces every pathology to one architectural choice: entangling class and
 271 expert estimates inside a shared normalization. Two design principles emerged: *separate* the two
 272 estimation problems so that gradients do not interfere, and *match* each head to the statistical type of its
 273 target so that the conditional minimizer recovers (η, α) directly. The decoupled surrogate implements
 274 both.

Table 1: Structural properties of multi-expert L2D surrogates. $\checkmark$ = satisfied; $\times$ = violated. Columns (I)/(II)/(TH): statistical target / local geometry / theoretical guarantee. Each entry cites the proposition, definition, equation, or appendix subsection that establishes it.

		Add. CE	PiCCE	Mao25	A-SM	OvA	Decoupled
Conditional minimizer recovers the Bayes pair (η, α)	(I)	$\times$ (14)	$\times$ (E.3)	$\times$ (6)	$\checkmark$ (17)	$\checkmark$ (19)	$\checkmark$ (10)
Per-sample gradient and Hessian free of $O(\mathcal{J})$ prefactor	(II)	$\times$ (4)	$\checkmark$ (E.4)	$\checkmark$ (6)	$\checkmark$ (7)	$\checkmark$ (H.3)	$\checkmark$ (9)
No winner-take-all suppression of correct experts	(II)	$\checkmark$ (4)	$\times$ (5)	$\checkmark$ (6)	$\checkmark$ (7)	$\checkmark$ (19)	$\checkmark$ (9)
No class-expert coupling in the per-sample gradient	(II)	$\times$ (4)	$\times$ (5)	$\times$ (6)	$\times$ (7)	$\checkmark$ (H.3)	$\checkmark$ (12)
Class head is a categorical posterior throughout training	(I)	$\times$ (3)	$\times$ (3)	$\times$ (3)	$\checkmark$ (17)	$\times$ (19)	$\checkmark$ (8)
Quantitative surrogate-to-target excess-risk bound	(TH)	$\checkmark^\dagger$	$\times$	$\checkmark^\ddagger$	$\times$	$\times$	$\checkmark$ (10)

† Comp-sum reduction (Mao et al., 2024e), constant $\sqrt{2(J+1)}$. ‡ Single-stage family (Mao et al., 2025d), constant $K+J$.

275 5.1 Formulation and Prediction

276 Let $w : \mathcal{X} \rightarrow \mathbb{R}^K$ and $s : \mathcal{X} \rightarrow \mathbb{R}^J$ denote the classifier and expert score maps. The decoupled
277 surrogate estimates the class posterior and the expert utilities on their native probability scales:
278 $p_k(x) = \exp(w_k(x)) / \sum_{\ell=1}^K \exp(w_\ell(x))$ and $u_j(x) = \sigma(s_j(x))$, where σ is the logistic sigmoid.
279 The class estimate $p(x) \in \Delta^K$ is a proper categorical distribution; each expert estimate $u_j(x) \in (0, 1)$
280 is an independent probability. These types mirror the Bayes-sufficient pair $(\eta, \alpha) \in \Delta^K \times [0, 1]^J$
281 exactly — no shared simplex forces classes and experts to compete for probability mass. We record
282 the corresponding bounded cost-sensitive formulation in Appendix C; throughout the main text we
283 restrict attention to the classification-specialized zero-one defer case.
284 **Definition 8** (Decoupled surrogate). For $\lambda > 0$, the decoupled surrogate is

$$\Phi_\lambda^{\text{dec}}(w, s; x, y, m) := -\log p_y(x) - \frac{\lambda}{J} \sum_{j=1}^J \left(t_j \log u_j(x) + (1 - t_j) \log(1 - u_j(x)) \right), \quad (9)$$

285 where $t_j := \mathbf{1}\{m_j = y\}$.

286 The first term is the standard multiclass log-loss for the classifier. The second is an average of J
287 independent Bernoulli log-losses, one per expert, with binary targets t_j . The λ/J weight has a single
288 interpretation: λ is the total expert-side budget, split evenly across the J experts so that the per-expert
289 weight λ/J is the quantity that actually controls gradients, curvature, and the surrogate-regret transfer
290 constant (Section 5.3). Any $\lambda > 0$ yields the same conditional minimizer, so λ governs only the
291 relative learning speed between the two heads, not the target being estimated. Appendix C.4 discusses
292 this choice in detail.

293 At test time, the decoupled surrogate applies the plug-in Bayes comparison directly in probability
294 space:

$$r_{\text{dec}}(x) \in \begin{cases} \{0\}, & \text{if } \max_{k \in [K]} p_k(x) \geq \max_{j \in [J]} u_j(x), \\ \arg \max_{j \in [J]} u_j(x), & \text{otherwise,} \end{cases} \quad (10)$$

295 5.2 Gradient Structure

296 The per-sample gradient of the decoupled surrogate resolves every pathology identified in Section 4.
297 We prove every result in Appendix C.2.

298 **Proposition 9** (Per-sample gradient of the decoupled surrogate). *For one sample (x, y, m) ,*

$$\frac{\partial \Phi_\lambda^{\text{dec}}}{\partial w_r} = p_r(x) - \mathbf{1}\{r=y\}, \quad \frac{\partial \Phi_\lambda^{\text{dec}}}{\partial s_j} = \frac{\lambda}{J} (u_j(x) - t_j). \quad (11)$$

299 Three properties follow immediately, each directly addressing a failure mode of the augmented-action
300 family:

- 301 – *No amplification.* The gradient for expert j depends only on j 's own prediction u_j and target t_j .
302 Adding more correct experts changes nothing about any individual gradient. The $O(|\mathcal{J}|)$ factor of
303 Proposition 4 is absent.
- 304 – *No starvation.* If expert j is correct, then $\partial \Phi / \partial s_j = (\lambda/J)(u_j - 1) < 0$ whenever $u_j < 1$:
305 expert j 's logit is pushed *up*, regardless of what other experts predict or whether a rival has a higher

score. Every correct expert receives positive reinforcement. The winner-take-all suppression of Proposition 5 is absent.

– *No coupling.* The classifier gradient involves only the class probabilities $p(x)$ and the label y ; it is completely independent of the expert predictions and targets. Conversely, each expert gradient depends only on its own head. The class–expert leakage of Proposition 7 is absent.

The second-order geometry confirms this decoupling (Appendix C.3). Because the class and expert terms depend on disjoint parameters, the Hessian is block-diagonal with a *zero* mixed block:

$$\nabla_{(w,s)}^2 \Phi_\lambda^{\text{dec}} = \begin{pmatrix} \text{Diag}(p) - pp^\top & 0 \\ 0 & \frac{\lambda}{J} \text{Diag}(u_j(1-u_j)) \end{pmatrix}. \quad (12)$$

The class block is the standard softmax covariance with $\lambda_{\max} \leq 1/2$; the expert block is diagonal with entries bounded by $\lambda/(4J)$. Contrast this with additive CE, whose curvature scales as $(1 + |\mathcal{J}|)/2$ (Proposition 4), and with A-SM, whose mixed block grows as $\sqrt{J}$ (Proposition 7). OvA shares the decoupled geometry, but its class head consists of K independent sigmoids whose outputs need not lie in Δ^K : away from the population optimum, several class estimates can simultaneously exceed 0.5, so $\max_k \hat{\eta}_k$ is uncalibrated against $\max_j u_j$ in the classify-versus-defer comparison. The decoupled surrogate is the only one in our comparison that combines full decoupling with a categorical class posterior $p \in \Delta^K$.

5.3 Consistency and Surrogate-Regret Transfer

The structural separation translates into a clean statistical guarantee. Because the conditional surrogate risk decomposes into a multiclass cross-entropy for p plus J independent Bernoulli cross-entropies for $(u_1, \dots, u_J)$, each subproblem has a unique minimizer: $p^*(x) = \eta(x)$ and $u_j^*(x) = \alpha_j(x)$ for all $j \in [J]$ (Appendix C.5).

The following theorem makes this guarantee quantitative: any reduction in the decoupled surrogate excess risk transfers, at a controlled rate, to the true L2D excess risk.

Theorem 10 (Consistency bound for the decoupled surrogate). *Let $p: \mathcal{X} \rightarrow \Delta^K$ and $u: \mathcal{X} \rightarrow [0, 1]^J$ be measurable, and let $f_{p,u} = (h_{\text{dec}}, r_{\text{dec}})$ be the plug-in policy of (10). For any distribution $\mathcal{D}$,*

$$\mathcal{E}_\perp(f_{p,u}) - \mathcal{E}_\perp^* \leq \max \left\{ 2\sqrt{2}, \sqrt{\frac{2J}{\lambda}} \right\} \sqrt{\mathcal{E}_{\text{dec}}(p, u) - \mathcal{E}_{\text{dec}}^*}. \quad (13)$$

In particular, the theorem implies Bayes consistency: along any sequence of measurable estimates (p_n, u_n) whose decoupled surrogate excess risk $\mathcal{E}_{\text{dec}}(p_n, u_n) - \mathcal{E}_{\text{dec}}^*$ tends to zero, the right-hand side of (13) tends to zero, and therefore the L2D excess risk $\mathcal{E}_\perp(f_{p_n, u_n}) - \mathcal{E}_\perp^*$ also tends to zero.

The bound has the $\sqrt{\cdot}$ calibration form (Mao et al., 2023b). The constant $\max\{2\sqrt{2}, \sqrt{2J/\lambda}\}$ equals $2\sqrt{2}$ whenever the per-expert weight satisfies $\lambda/J \geq 1/4$, and grows as $\sqrt{2J/\lambda}$ otherwise. The $1/J$ normalization in the surrogate (9) is precisely what makes this constant J -independent: scaling λ with J (equivalently, holding the per-expert weight λ/J fixed) leaves the transfer guarantee unchanged as the expert pool grows, mirroring the gradient stability of Proposition 9.

Comparison with existing bounds. Two prior quantitative references are relevant. First, Mao et al. (2024e) analyze the earlier comp-sum reduction that includes the Add. CE surrogate (Mozannar and Sontag, 2020); in our classification-specialized setting, that yields the transfer constant $\sqrt{2(J+1)}$. Second, the Mao25 (Mao et al., 2025d) single-stage surrogate has an excess-risk constant that scales with the augmented action dimension $K+J$. The decoupled surrogate is the only multi-expert surrogate in our comparison whose transfer constant is $O(1)$ in J (for fixed per-expert weight λ/J). For the remaining baselines (PicCE, A-SM, OvA), the cited analyses establish Bayes consistency but do not give an explicit transfer constant (Appendix I).

6 Experiments

We evaluate the decoupled surrogate on four synthetic suites (each targeting one proposition from Section 4) and three real-data benchmarks spanning vision, human annotators, and tabular model routing. All methods share the same backbone and training budget within each benchmark; full details are in the appendix.

Synthetic suites (Appendix K). Each suite isolates one surrogate pathology by varying only the structural quantity relevant to the corresponding proposition while keeping the Bayes problem analytically known. (i) Nested redundant experts, targeting Add. CE amplification (Prop. 4): at $J=24$, defer regret 0.0002 vs. 0.348 for Add. CE (best baseline: PiCCE at 0.238). (ii) Rare specialist, targeting PiCCE starvation (Prop. 5): on shared-correct samples, the decoupled surrogate selects the specialist with probability 0.994 vs. 0 for PiCCE. (iii) Shared acceptability, targeting Mao25’s set-mass collapse (Prop. 6): the decoupled surrogate reaches defer regret 0.0008 vs. 0.154 for Mao25. (iv) Class–expert coupling, targeting A-SM (Prop. 7): A-SM’s mixed class–expert Hessian block grows from 0.12 at $J=1$ to 0.48 at $J=17$ ($\approx 0.12\sqrt{J}$), while the decoupled surrogate’s is identically zero by construction (Appendix K.4).

CIFAR-10 with synthetic experts (Appendix L). We train a ResNet-12 end-to-end on CIFAR-10 with J nested redundant experts whose correctness is drawn from a shared latent (Section K.1). Table 2 reports system and classifier accuracy. The decoupled surrogate is the only method in our comparison that *improves* over the standalone classifier at every J ; Add. CE, A-SM, and OvA degrade as the redundant pool grows. A-SM’s classifier accuracy collapses from .831 ($J=8$) to .680 ($J=32$), consistent with the softmax-coupling pathology of Proposition 7.

Table 2: CIFAR-10 with redundant synthetic experts (mean $\pm$ std, 4 seeds). Best in **bold**; second-best underlined.

Method	Sys. Acc. $\uparrow$			Cls. Acc. $\uparrow$		
	$J=8$	$J=16$	$J=32$	$J=8$	$J=16$	$J=32$
Decoupled	.918 $\pm$.001	.919 $\pm$.003	.919 $\pm$.002	.909 $\pm$.001	.909 $\pm$.002	.908 $\pm$.003
Classif. only	<u>.910</u> $\pm$.002	<u>.911</u> $\pm$.001	<u>.908</u> $\pm$.002	.910 $\pm$.002	.911 $\pm$.001	.908 $\pm$.002
OvA	.901 $\pm$.001	.890 $\pm$.002	.880 $\pm$.003	.875 $\pm$.001	.859 $\pm$.004	.844 $\pm$.003
PiCCE	.858 $\pm$.007	.860 $\pm$.011	.859 $\pm$.008	.888 $\pm$.008	.895 $\pm$.006	.880 $\pm$.014
A-SM	.860 $\pm$.002	.827 $\pm$.003	.783 $\pm$.006	.831 $\pm$.002	.772 $\pm$.003	.680 $\pm$.009
Add. CE	.815 $\pm$.002	.790 $\pm$.013	.712 $\pm$.015	.902 $\pm$.004	.895 $\pm$.003	.872 $\pm$.009
Mao25	.703 $\pm$.000	.703 $\pm$.000	.703 $\pm$.000	.105 $\pm$.024	.111 $\pm$.010	.105 $\pm$.025

CIFAR-10H with human annotators (Appendix M). We freeze a pretrained ResNet-12 and train only the L2D head on CIFAR-10H (Peterson et al., 2019) with J real annotators subsampled from the crowd (5 draws, Table 3). The decoupled surrogate leads system accuracy at every J with stable classifier quality ($\geq .889$); A-SM collapses from .890 ($J=5$) to .470 ($J=20$) and OvA becomes unstable (std .203 at $J=10$).

Table 3: CIFAR-10H with real human annotators (mean $\pm$ std, 5 draws). Best in **bold**; second-best underlined.

Method	Sys. Acc. $\uparrow$			Cls. Acc. $\uparrow$		
	$J=5$	$J=10$	$J=20$	$J=5$	$J=10$	$J=20$
Decoupled	.960 $\pm$.005	.958 $\pm$.003	.961 $\pm$.004	.892 $\pm$.001	.892 $\pm$.003	.889 $\pm$.001
OvA	<u>.959</u> $\pm$.005	.954 $\pm$.008	<u>.956</u> $\pm$.006	.889 $\pm$.001	.799 $\pm$.203	.890 $\pm$.002
A-SM	.959 $\pm$.004	.952 $\pm$.007	.949 $\pm$.005	<u>.890</u> $\pm$.002	.747 $\pm$.313	.470 $\pm$.435
PiCCE	.954 $\pm$.003	<u>.957</u> $\pm$.004	.953 $\pm$.002	.889 $\pm$.001	<u>.890</u> $\pm$.002	<u>.890</u> $\pm$.001
Add. CE	.953 $\pm$.004	.952 $\pm$.004	.951 $\pm$.003	.889 $\pm$.002	.888 $\pm$.001	.742 $\pm$.331
Mao25	.950 $\pm$.005	.948 $\pm$.003	.948 $\pm$.004	.116 $\pm$.133	.077 $\pm$.024	.045 $\pm$.014

Covertypes with model experts (Appendix N). On Forest CoverType (Blackard and Dean, 1999) with $J=5$ heterogeneous pretrained model experts (oracle +9.8 pp over the best single model), the decoupled surrogate reaches system accuracy 0.934 and classifier accuracy 0.941 — *the only L2D method in our comparison whose classifier improves over the standalone classifier* (0.929); every augmented-action baseline degrades the classifier, and Mao25 collapses to 100% deferral.

7 Conclusion

Every augmented-action surrogate for multi-expert L2D trades a fix on one axis for a failure on the other. The decoupled surrogate resolves both by estimating η with a softmax and each α_j with an independent sigmoid, yielding coordinatewise gradients, a zero class–expert Hessian block, and a consistency bound whose constant is J -independent for fixed λ/J . Across the synthetic suites and real-data benchmarks, it is the only method in our comparison that avoids amplification and preserves rare specialists, while improving over the standalone classifier on the real-data benchmarks.

8 Limitations

We see no limitations specific to the decoupled surrogate beyond those inherent to the multi-expert L2D setting itself: reliance on observed expert predictions over labeled training data, and the assumption that each expert remains available at inference. Relaxing these assumptions—missing or non-stationary experts, partial annotation, expert availability constraints, calibration of human trust—is not the aim of this work, but is an important direction that subsequent work can build on top of the decoupled formulation.

9 Impact Statement

This work contributes foundational theory and algorithms for learning-to-defer. Better-calibrated deferral policies can improve the safety and accuracy of human–AI systems in high-stakes settings such as healthcare triage, content moderation, and financial decisioning by routing difficult cases to qualified human experts. As with any deferral system, risks include over-reliance on imperfect experts, unequal error distributions across subgroups, and accountability ambiguity when decisions are shared between model and human.

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575 A Extended Related Work

576 Learning-to-defer extends selective prediction and classification with a reject option (Chow, 1970;
577 Bartlett and Wegkamp, 2008; Geifman and El-Yaniv, 2017; Cortes et al., 2016; Cao et al., 2022;
578 Cortes et al., 2024a) and is also closely related to the abstention literature (Mao et al., 2024a,i; Cortes
579 et al., 2024b). The distinguishing feature of L2D is that abstention is replaced by routing to an
580 external expert. The modern L2D literature begins with adaptive deferral in the single-expert setting
581 (Madras et al., 2018; Mozannar and Sontag, 2020).

582 **One-stage methods.** One-stage methods learn prediction and deferral jointly. Madras et al. (2018)
583 introduced adaptive deferral in the single-expert setting, while Mozannar and Sontag (2020) gave
584 the modern multiclass formulation by casting learning to defer as prediction over an augmented
585 action space. Most subsequent one-stage work refines this augmented-action view rather than
586 departing from it. Verma et al. (2022) compare coupled softmax and one-vs-all surrogates for
587 multiple experts; Cao et al. (2024) analyze the calibration pathology of symmetric softmax reductions
588 and propose an asymmetric softmax alternative; Charusaie et al. (2022) study complementarity
589 and sample efficiency; Mozannar et al. (2023) analyze realizability and exact optimization; Wei
590 et al. (2024) incorporate human–AI dependence into surrogate design; and Liu et al. (2024) show
591 that even statistically consistent single-expert surrogates can underfit when non-zero expert-query
592 costs introduce redundant label smoothing. In the multi-expert setting, Mao et al. (2024e) derive
593 quantitative guarantees for comp-sum-style reductions, Mao et al. (2024f, 2025d) develop consistent
594 loss families for learning to defer, and Liu et al. (2026) identify redundancy-driven underfitting and
595 propose PiCCE as a correction.

596 More recent extensions that continue to optimize prediction and deferral jointly also fit naturally in
597 the one-stage category. These include top- k deferral (Montreuil et al., 2026a), adversarially robust
598 one-stage L2D (Montreuil et al., 2026d), non-stationary time-series deferral (Montreuil et al., 2026e),
599 missing-annotation variants (Nguyen et al., 2025), deferral to a population or to unseen experts (Tailor
600 et al., 2024; Strong et al., 2026), and application-driven one-stage systems in sequential medicine and
601 healthcare (Joshi et al., 2021; Strong et al., 2025). Overall, the one-stage literature is best viewed as a
602 sequence of refinements and extensions of the augmented-action formulation introduced by Mozannar
603 and Sontag (2020), first in the single-expert $K + 1$ setting and then in the multi-expert $K + J$ setting.

604 **Two-stage methods.** By contrast, *two-stage* methods first train a predictor and then learn a defer
605 rule or router on top of the frozen predictor. This includes post-hoc estimators for deferring to a
606 single expert (Narasimhan et al., 2022), principled two-stage formulations for multiple experts (Mao
607 et al., 2023a), and extensions to multitask and regression settings (Montreuil et al., 2025b; Mao
608 et al., 2024g). Recent extensions of the two-stage line also include adversarial robustness guarantees
609 (Montreuil et al., 2025a), online learning with varying experts (Montreuil et al., 2026b), budgeted
610 L2D (DeSalvo et al., 2025), and extractive question answering (Montreuil et al., 2026c).

611 The distinction is important: one-stage methods redesign the joint prediction–deferral surrogate,
612 whereas two-stage methods optimize only the routing rule after the predictor has been fixed. Our
613 decoupled parameterization is introduced here as a one-stage surrogate, but the same separation
614 between class prediction and expert-utility estimation can also be used within a two-stage pipeline.

615 **Quantitative consistency bounds.** Classical Bayes consistency (Bartlett et al., 2006; Tewari and
616 Bartlett, 2007; Zhang, 2002; Steinwart, 2007) guarantees that the conditional minimizer of the
617 surrogate recovers the Bayes decision. Quantitative versions turn surrogate excess risk into target
618 excess risk through an explicit calibration function. This line includes multiclass and cross-entropy
619 analyses (Awasthi et al., 2022; Mao et al., 2024b, 2023b), smooth surrogates (Mao et al., 2024h),
620 regression (Mao et al., 2024c), multi-label learning (Mao et al., 2024d), abstention (Mao et al.,
621 2024i,a; Cortes et al., 2024b), imbalanced and balanced binary classification (Cortes et al., 2025a,b;
622 Mao et al., 2025c), sharper enhanced bounds (Mao et al., 2025b), and noise-adaptive refinements
623 (Mao et al., 2025a); see Zhong (2025); Mao (2025) for unified treatments. Theorem 10 gives such
624 a $\sqrt{\cdot}$ -form bound for the decoupled surrogate, with calibration constant J -independent for fixed
625 per-expert weight λ/J .

626 **Positioning.** Our work is closest to the one-stage literature, but differs in aim. Rather than proposing
627 another repair within an augmented-action parameterization, we compare representative one-stage
628 surrogates along two axes—population target and local optimization geometry—and then propose a
629 decoupled surrogate outside the shared-action-space design. In that sense, our contribution is both

comparative and constructive: we isolate a recurrent tradeoff within existing multi-expert surrogates and show that the decoupled surrogate avoids it while retaining a quantitative consistency guarantee.

B Proof of Lemma 2

Lemma 2 (Bayes Rule for Multi-Expert L2D). *For every $x \in \mathcal{X}$, a Bayes-optimal decision is obtained by*

$$r^*(x) \in \begin{cases} \{0\}, & \text{if } \max_{k \in [K]} \eta_k(x) \geq \max_{j \in [J]} \alpha_j(x), \\ \arg \max_{j \in [J]} \alpha_j(x), & \text{otherwise,} \end{cases} \quad (2)$$

with $h^*(x) \in \arg \max_{k \in [K]} \eta_k(x)$.

Proof We prove the statement directly from the true defer loss (1). Fix an input $x \in \mathcal{X}$ and condition on the event $X = x$. For a decision pair (h, r) , the corresponding conditional true deferral risk is

$$\mathcal{C}_\perp(h, r \mid x) := \mathbb{E}[\ell_\perp(h, r; X, Y, \mathbf{M}) \mid X = x].$$

Substituting the definition of $\ell_\perp$ gives

$$\begin{aligned} \mathcal{C}_\perp(h, r \mid x) &= \mathbb{E} \left[\mathbf{1}\{h(x) \neq Y\} \mathbf{1}\{r(x) = 0\} + \sum_{j=1}^J \mathbf{1}\{M_j \neq Y\} \mathbf{1}\{r(x) = j\} \mid X = x \right] \\ &= \mathbf{1}\{r(x) = 0\} \Pr(h(x) \neq Y \mid X = x) + \sum_{j=1}^J \mathbf{1}\{r(x) = j\} \Pr(M_j \neq Y \mid X = x). \end{aligned} \quad (14)$$

Thus, once x is fixed, the learner is simply choosing one action among the K class actions and the J defer actions, and each action has an explicit conditional error probability.

Take a class label $k \in [K]$ and consider the action “predict class k ”, that is, choose $h(x) = k$ and $r(x) = 0$. By (14), its conditional risk is

$$\mathcal{C}_\perp(k, 0 \mid x) = \Pr(Y \neq k \mid X = x).$$

Since $\eta_k(x) = \Pr(Y = k \mid X = x)$ and the label space is multiclass,

$$\Pr(Y \neq k \mid X = x) = 1 - \Pr(Y = k \mid X = x) = 1 - \eta_k(x).$$

Therefore the best classifier action at x is any

$$h^*(x) \in \arg \max_{k \in [K]} \eta_k(x),$$

and its conditional risk is

$$\min_{k \in [K]} \mathcal{C}_\perp(k, 0 \mid x) = 1 - \max_{k \in [K]} \eta_k(x).$$

Now take an expert index $j \in [J]$ and consider the action “defer to expert j ”, that is, choose $r(x) = j$. Again by (14),

$$\mathcal{C}_\perp(\perp_j \mid x) = \Pr(M_j \neq Y \mid X = x).$$

By definition of the expert utility $\alpha_j(x) = \Pr(M_j = Y \mid X = x)$, we obtain

$$\Pr(M_j \neq Y \mid X = x) = 1 - \alpha_j(x).$$

Hence the best defer action at x is any

$$j^*(x) \in \arg \max_{j \in [J]} \alpha_j(x),$$

and its conditional risk is

$$\min_{j \in [J]} \mathcal{C}_\perp(\perp_j \mid x) = 1 - \max_{j \in [J]} \alpha_j(x).$$

651 **Step 3: compare the best class action and the best defer action.** The Bayes decision at x is the
 652 action with the smallest conditional true deferral risk among all $K + J$ available actions. Combining
 653 the two displays above,

$$\begin{aligned} \inf_{h,r} \mathcal{C}_\perp(h, r \mid x) &= \min \left\{ 1 - \max_{k \in [K]} \eta_k(x), 1 - \max_{j \in [J]} \alpha_j(x) \right\} \\ &= 1 - \max \left\{ \max_{k \in [K]} \eta_k(x), \max_{j \in [J]} \alpha_j(x) \right\}. \end{aligned}$$

654 Therefore:

- 655 • if $\max_{k \in [K]} \eta_k(x) \geq \max_{j \in [J]} \alpha_j(x)$, then the best class action is at least as good as every defer
 656 action, so one may choose $r^*(x) = 0$ together with any $h^*(x) \in \arg \max_{k \in [K]} \eta_k(x)$;
- 657 • if $\max_{k \in [K]} \eta_k(x) < \max_{j \in [J]} \alpha_j(x)$, then deferring is strictly better, and one may choose $r^*(x) \in$
 658 $\arg \max_{j \in [J]} \alpha_j(x)$.

659 This is exactly the rule stated in (2). ■

660 C The Decoupled Surrogate

661 We present the decoupled surrogate first because it is the reference point for all of the baseline
 662 diagnoses that follow. The later sections can be read as controlled departures from this aligned
 663 formulation. The decoupled surrogate abandons the augmented-action reduction and instead learns
 664 the class posterior and the expert utilities in their native forms.

665 C.1 Formulation

666 **Cost-sensitive formulation.** Before specializing to the zero-one defer loss, it is useful to record the
 667 more general bounded-cost version of the decoupled construction. Let $c_j(x, y) \in [0, 1]$ denote the
 668 cost of routing (x, y) to expert j , and let

$$\tau_j(x, y) := 1 - c_j(x, y) \in [0, 1]$$

669 denote the corresponding utility target. The cost-sensitive decoupled surrogate is

$$\Phi_\lambda^{\text{dec,cs}}(w, s; x, y) := -\log p_y(x) - \frac{\lambda}{J} \sum_{j=1}^J \left(\tau_j(x, y) \log u_j(x) + (1 - \tau_j(x, y)) \log(1 - u_j(x)) \right).$$

670 Thus the class side is unchanged, while each expert head is trained with a Bernoulli cross-entropy
 671 against a possibly soft utility target $\tau_j(x, y)$.

672 The decoupled surrogate uses one softmax head for the classifier and one sigmoid head for each
 673 expert. The softmax head models the categorical class posterior, while the expert heads model the
 674 Bernoulli events “expert j is correct on this input”.

675 **Definition 11** (Decoupled surrogate). Let $w(x) \in \mathbb{R}^K$ denote class logits and $s(x) \in \mathbb{R}^J$ denote
 676 expert logits. Define

$$p_k(x) := \frac{\exp(w_k(x))}{\sum_{\ell=1}^K \exp(w_\ell(x))}, \quad u_j(x) := \sigma(s_j(x)).$$

677 For $\lambda > 0$, define

$$\Phi_\lambda^{\text{dec}}(w, s; x, y, m) := -\log p_y(x) - \frac{\lambda}{J} \sum_{j=1}^J \left(t_j \log u_j(x) + (1 - t_j) \log(1 - u_j(x)) \right),$$

678 where $t_j := \mathbf{1}\{m_j = y\}$.

679 Prediction compares the classifier confidence $\max_{k \in [K]} p_k(x)$ with the largest estimated expert utility
 680 $\max_{j \in [J]} u_j(x)$ and defers whenever the latter is larger. This is the classification-specialized version
 681 of the general bounded-cost formulation above. In the present paper,

$$c_j(x, y) = \mathbf{1}\{m_j \neq y\}, \quad \tau_j(x, y) = \mathbf{1}\{m_j = y\},$$

so the expert-side target reduces exactly to binary correctness.

Two structural features are already visible from the definition. First, the class side remains a categorical distribution throughout training because $p(x) \in \Delta^K$ for every parameter value. Second, the expert side is coordinatewise: each $u_j(x)$ is a bounded Bernoulli probability attached to one expert only. The rest of the section formalizes the consequences of this design.

C.2 Per-Sample Gradient Structure

Proposition 9 (Per-sample gradient of the decoupled surrogate). *For one sample (x, y, m) ,*

$$\frac{\partial \Phi_\lambda^{\text{dec}}}{\partial w_r} = p_r(x) - \mathbf{1}\{r=y\}, \quad \frac{\partial \Phi_\lambda^{\text{dec}}}{\partial s_j} = \frac{\lambda}{J}(u_j(x) - t_j). \quad (11)$$

Proof The samplewise decoupled loss is the sum of one multiclass cross-entropy term and J independent Bernoulli cross-entropy terms. We derive the two pieces separately.

Classifier head. The classifier contribution is

$$\Phi_{\text{cls}}(w; y) = -\log p_y = -w_y + \log \sum_{\ell=1}^K e^{w_\ell}.$$

Differentiate with respect to w_r :

$$\frac{\partial \Phi_{\text{cls}}}{\partial w_r} = -\mathbf{1}\{r=y\} + \frac{e^{w_r}}{\sum_{\ell=1}^K e^{w_\ell}} = p_r - \mathbf{1}\{r=y\}.$$

This is the usual multiclass cross-entropy gradient.

Expert heads. For expert j , write $t_j = \mathbf{1}\{m_j = y\}$. Its contribution is

$$\Phi_j(s_j; t_j) = -t_j \log u_j - (1 - t_j) \log(1 - u_j), \quad u_j = \sigma(s_j).$$

Differentiate with respect to s_j . By the chain rule,

$$\frac{\partial \Phi_j}{\partial s_j} = \left(-\frac{t_j}{u_j} + \frac{1 - t_j}{1 - u_j} \right) \frac{du_j}{ds_j}.$$

Since $\frac{du_j}{ds_j} = u_j(1 - u_j)$, we obtain

$$\frac{\partial \Phi_j}{\partial s_j} = \left(-\frac{t_j}{u_j} + \frac{1 - t_j}{1 - u_j} \right) u_j(1 - u_j) = u_j - t_j.$$

Multiplying by the prefactor λ/J gives

$$\frac{\partial \Phi_\lambda^{\text{dec}}}{\partial s_j} = \frac{\lambda}{J}(u_j - t_j).$$

This completes the gradient calculation. The important structural point is that the classifier residuals and the expert residuals live on disjoint coordinates: the class gradient depends only on (w, y) , and the j -th expert gradient depends only on (s_j, t_j) . ■

This proposition isolates the first-order reason the decoupled surrogate behaves differently from the augmented-action surrogates. A sample contributes one classifier residual and J expert residuals, but these residuals do not compete through a shared normalization. Correct experts are reinforced independently of one another, and expert-side errors cannot directly distort the class-side update.

C.3 Per-Sample Curvature Structure

We now derive the Hessian explicitly. The goal is to show not only that the curvature is bounded, but also that the class and expert directions remain decoupled at second order.

708 Start with the classifier block. From the softmax Jacobian,

$$\frac{\partial p_r}{\partial w_\ell} = p_r(\mathbf{1}\{r = \ell\} - p_\ell),$$

709 so the classifier Hessian is

$$\nabla_w^2(-\log p_y) = \text{Diag}(p) - pp^\top.$$

710 This is the usual softmax covariance matrix. It captures competition only among the class coordinates
711 themselves. For the expert side, differentiate the gradient $\frac{\lambda}{J}(u_j - t_j)$ once more:

$$\frac{\partial^2 \Phi_\lambda^{\text{dec}}}{\partial s_j^2} = \frac{\lambda}{J} \frac{du_j}{ds_j} = \frac{\lambda}{J} u_j(1 - u_j),$$

712 and for $j \neq \ell$,

$$\frac{\partial^2 \Phi_\lambda^{\text{dec}}}{\partial s_j \partial s_\ell} = 0,$$

713 because Φ_j depends only on s_j .

714 Finally, the class term depends only on w , and the expert terms depend only on s . Therefore every
715 mixed class–expert derivative is zero:

$$\frac{\partial^2 \Phi_\lambda^{\text{dec}}}{\partial w_r \partial s_j} = 0 \quad \text{for all } r \in [K], j \in [J].$$

716 Hence the full Hessian is block diagonal:

$$\nabla_{(w,s)}^2 \Phi_\lambda^{\text{dec}} = \begin{pmatrix} \text{Diag}(p) - pp^\top & 0 \\ 0 & \frac{\lambda}{J} \text{Diag}(u_j(1 - u_j)) \end{pmatrix}.$$

717 The block-diagonal form makes the spectral analysis immediate. The eigenvalues of a block-diagonal
718 matrix are the union of the eigenvalues of its diagonal blocks, so the largest eigenvalue of the full
719 Hessian is the larger of the largest class-block eigenvalue and the largest expert-block eigenvalue.

720 For the class block, $\text{Diag}(p) - pp^\top$ is a softmax covariance matrix. Its largest eigenvalue is at
721 most $1/2$ (Mohri et al., 2012). Intuitively, this is the intrinsic curvature scale of ordinary multiclass
722 cross-entropy. The decoupled surrogate does not increase that scale.

723 For the expert block, the matrix is diagonal, so its eigenvalues are exactly the diagonal entries

$$\frac{\lambda}{J} u_j(1 - u_j), \quad j \in [J].$$

724 Since $0 \leq u_j(1 - u_j) \leq 1/4$ for every j , each expert-side eigenvalue is bounded by $\lambda/(4J)$.
725 Therefore

$$\lambda_{\max}(\nabla_{(w,s)}^2 \Phi_\lambda^{\text{dec}}) \leq \max\left\{\frac{1}{2}, \frac{\lambda}{4J}\right\}.$$

726 This inequality is informative in two ways. First, the class-side curvature is uniformly bounded by
727 the same constant as ordinary multiclass cross-entropy. Second, the expert-side curvature becomes
728 smaller as the number of experts grows if the total expert weight λ is held fixed. In particular, adding
729 experts does not create the kind of growing stiffness that appears in additive CE, nor the mixed
730 class–expert second-order interaction that appears in A-SM.

731 This is the local-geometric signature of the decoupled surrogate. There is no multiplicity factor,
732 no winner-take-all routing effect, and no mixed class–expert block. Each expert curvature term is
733 also bounded by $(\lambda/J)/4$, so adding more experts does not by itself create an increasingly stiff
734 optimization landscape. The classifier head and expert heads interact only through the final routing
735 rule, not through the samplewise training geometry.

736 C.4 The Role of λ/J and How to Choose λ

737 The previous two subsections already show that the quantity that matters structurally is not λ by itself,
 738 but the per-expert weight λ/J . This quantity controls the scale of every expert-side update. In the
 739 gradient formula,

$$\frac{\partial \Phi_{\lambda}^{\text{dec}}}{\partial s_j} = \frac{\lambda}{J} (u_j - t_j),$$

740 the magnitude of the j -th expert residual is proportional to λ/J . In the Hessian,

$$\frac{\partial^2 \Phi_{\lambda}^{\text{dec}}}{\partial s_j^2} = \frac{\lambda}{J} u_j (1 - u_j),$$

741 the expert-side curvature is also proportional to λ/J . Thus the same quantity governs both first-order
 742 and second-order expert-side scale.

743 This observation is practically important when the number of experts varies. If one were to keep λ
 744 fixed while increasing J , then λ/J would decrease. Every expert gradient would shrink, every expert
 745 curvature term would shrink, and the optimization would place progressively less emphasis on fitting
 746 the expert utilities. In other words, a fixed λ implicitly weakens the expert task as more experts are
 747 added. That is usually not the intended comparison across different expert pools.

748 By contrast, keeping the per-expert weight λ/J fixed means scaling λ with J . Under this scaling, the
 749 per-expert gradient magnitude remains on the same order, the per-expert curvature bound remains
 750 $(\lambda/J)/4$, and the transfer constant in Theorem 10 remains

$$\max \left\{ 2\sqrt{2}, \sqrt{\frac{2J}{\lambda}} \right\},$$

751 which is independent of J as long as λ/J is held fixed.

752 **One hyperparameter, two lenses.** It is worth emphasizing that λ and λ/J are not two independent
 753 knobs: they are the same quantity, parametrized differently. We keep both names because they serve
 754 different roles:

- 755 • λ is the user-facing coefficient that appears in the loss definition (9) and is tuned on a validation
 756 split, matching the convention of single-expert L2D ($J=1$, $\lambda = \lambda/J$).
- 757 • λ/J is the analysis-facing per-expert weight that appears in gradients, curvature bounds, and the
 758 transfer constant, and is the quantity one should keep fixed when comparing settings with different
 759 numbers of experts.

760 In practice, the user tunes λ and reads the theory in terms of λ/J ; the two views coincide whenever
 761 J is fixed.

762 The existing theory does *not* identify a universally optimal choice of λ . The conditional minimizer is
 763 the same for every $\lambda > 0$, so there is no population-level “best” coefficient in the sense of changing
 764 the target being estimated. What λ/J changes is the relative optimization emphasis between the
 765 classifier head and the expert heads in finite samples and under function-class constraints. That
 766 tradeoff depends on the optimization method, the capacity of the two heads, the amount of data, and
 767 the statistical difficulty of estimating η versus the α_j ’s. Without additional assumptions on those
 768 ingredients, one should not expect a theorem to single out one universally optimal value.

769 The present theory nevertheless gives useful guidance.

770 First, Theorem 10 shows that the transfer constant equals $2\sqrt{2}$ whenever $\lambda/J \geq 1/4$, and deteriorates
 771 only when $\lambda/J < 1/4$. So values below $1/4$ are theoretically less attractive: they downweight the
 772 expert task and simultaneously weaken the calibration constant.

773 Second, the gradient and curvature formulas show what increasing λ/J does. Larger λ/J strengthens
 774 expert-side updates and increases expert-side curvature linearly, but it does *not* introduce class–expert
 775 coupling or amplification. So increasing λ/J is not dangerous in the way increasing the number
 776 of experts is for the augmented-action baselines; it simply gives the expert task more optimization
 777 weight.

778 Third, because the class gradient is unaffected by λ/J , changing λ/J only changes the balance
 779 between the classifier loss and the average expert loss. This suggests a simple practical rule:

- tune the per-expert weight λ/J , not λ alone;
- keep λ/J fixed when comparing settings with different numbers of experts;
- use $\lambda/J \geq 1/4$ as a natural default range if no stronger prior is available.

Among such values, the final choice should be selected by validation on the target routing objective. If the classifier is already strong and the main challenge is accurate expert selection, a larger λ/J is often sensible. If the classifier head is harder to fit or the expert labels are especially noisy, a more moderate λ/J may work better. The key point is that these are finite-sample optimization considerations, not changes in the population target.

C.5 Consistency Bound

We now prove the main statistical guarantee of the decoupled surrogate.

Theorem 10 (Consistency bound for the decoupled surrogate). *Let $p: \mathcal{X} \rightarrow \Delta^K$ and $u: \mathcal{X} \rightarrow [0, 1]^J$ be measurable, and let $f_{p,u} = (h_{\text{dec}}, r_{\text{dec}})$ be the plug-in policy of (10). For any distribution $\mathcal{D}$,*

$$\mathcal{E}_{\perp}(f_{p,u}) - \mathcal{E}_{\perp}^* \leq \max \left\{ 2\sqrt{2}, \sqrt{\frac{2J}{\lambda}} \right\} \sqrt{\mathcal{E}_{\text{dec}}(p, u) - \mathcal{E}_{\text{dec}}^*}. \quad (13)$$

Proof Throughout this proof we abbreviate the per-expert weight as $\beta := \lambda/J$; this is a local shorthand for the fraction appearing in (9), not a separate hyperparameter. Fix x . We define

$$\delta_{\perp}(p, u \mid x) := \mathcal{C}_{\perp}(f_{p,u} \mid x) - \mathcal{C}_{\perp}^*(x)$$

for the conditional L2D regret and

$$\delta_{\text{dec}}(p, u \mid x) := \mathcal{C}_{\text{dec}}(p, u \mid x) - \mathcal{C}_{\text{dec}}^*(x)$$

for the conditional surrogate excess.

Conditioning on $X = x$ in the samplewise surrogate gives

$$\begin{aligned} \mathcal{C}_{\text{dec}}(p, u \mid x) &= - \sum_{k=1}^K \eta_k(x) \log p_k(x) \\ &\quad - \beta \sum_{j=1}^J \left(\alpha_j(x) \log u_j(x) + (1 - \alpha_j(x)) \log(1 - u_j(x)) \right), \end{aligned} \quad (15)$$

The class term depends only on $p(x)$, and the j -th expert term depends only on $u_j(x)$.

For any $q \in \Delta^K$,

$$- \sum_{k=1}^K \eta_k(x) \log q_k = H(\eta(x)) + \text{KL}(\eta(x) \parallel q),$$

where $H(\eta(x)) = - \sum_k \eta_k(x) \log \eta_k(x)$. For any $v \in (0, 1)$,

$$- \alpha_j(x) \log v - (1 - \alpha_j(x)) \log(1 - v) = H_{\text{B}}(\alpha_j(x)) + \text{KL}(\text{Bern}(\alpha_j(x)) \parallel \text{Bern}(v)),$$

where $H_{\text{B}}(t) = -t \log t - (1 - t) \log(1 - t)$ is the Bernoulli entropy. Substituting these identities into (15) yields

$$\begin{aligned} \mathcal{C}_{\text{dec}}(p, u \mid x) &= H(\eta(x)) + \beta \sum_{j=1}^J H_{\text{B}}(\alpha_j(x)) \\ &\quad + \text{KL}(\eta(x) \parallel p(x)) + \beta \sum_{j=1}^J \text{KL}(\text{Bern}(\alpha_j(x)) \parallel \text{Bern}(u_j(x))). \end{aligned} \quad (16)$$

The first line is independent of (p, u) , so it is exactly the conditional minimum. Therefore

$$\delta_{\text{dec}}(p, u \mid x) = \text{KL}(\eta(x) \parallel p(x)) + \beta \sum_{j=1}^J \text{KL}(\text{Bern}(\alpha_j(x)) \parallel \text{Bern}(u_j(x))). \quad (17)$$

803 This identity shows that the conditional surrogate excess is exactly the sum of one class estimation
 804 error and J expert estimation errors, each measured in the natural KL divergence of its own subprob-
 805 lem. Since every KL term is nonnegative and vanishes only when its two arguments coincide, the
 806 conditional surrogate risk is minimized exactly when $p(x) = \eta(x)$ and $u_j(x) = \alpha_j(x)$ for every j .

807 We now translate this conditional estimation error into conditional L2D regret. Define

$$\varepsilon_{\text{cls}}(x) := \|p(x) - \eta(x)\|_\infty, \quad \varepsilon_{\text{exp}}(x) := \|u(x) - \alpha(x)\|_\infty.$$

808 Also let $y^* \in \arg \max_y \eta_y(x)$, $\hat{y} \in \arg \max_y p_y(x)$, $j^* \in \arg \max_j \alpha_j(x)$, $\hat{j} \in \arg \max_j u_j(x)$,
 809 $a^* = \eta_{y^*}(x)$, and $b^* = \alpha_{j^*}(x)$.

810 If both the plug-in rule and the Bayes rule predict, then

$$\delta_\perp(p, u | x) = \eta_{y^*}(x) - \eta_{\hat{y}}(x).$$

811 Since $\hat{y}$ maximizes p , we have $p_{\hat{y}} \geq p_{y^*}$, and hence

$$\eta_{y^*} - \eta_{\hat{y}} = (\eta_{y^*} - p_{y^*}) + (p_{y^*} - p_{\hat{y}}) + (p_{\hat{y}} - \eta_{\hat{y}}) \leq 2\varepsilon_{\text{cls}}(x).$$

812 If the plug-in rule predicts while the Bayes rule defers, then $b^* > a^*$ and $p_{\hat{y}} \geq u_{\hat{j}} \geq u_{j^*}$. Therefore

$$\delta_\perp(p, u | x) = \alpha_{j^*} - \eta_{\hat{y}} \leq (\alpha_{j^*} - u_{j^*}) + (u_{j^*} - p_{\hat{y}}) + (p_{\hat{y}} - \eta_{\hat{y}}) \leq \varepsilon_{\text{exp}}(x) + \varepsilon_{\text{cls}}(x).$$

813 If both rules defer, then

$$\delta_\perp(p, u | x) = \alpha_{j^*}(x) - \alpha_{\hat{j}}(x).$$

814 Since $\hat{j}$ maximizes u , the same argument gives

$$\alpha_{j^*} - \alpha_{\hat{j}} \leq 2\varepsilon_{\text{exp}}(x).$$

815 If the plug-in rule defers while the Bayes rule predicts, then $a^* > b^*$ and $u_{\hat{j}} > p_{\hat{y}} \geq p_{y^*}$. Hence

$$\delta_\perp(p, u | x) = \eta_{y^*} - \alpha_{\hat{j}} \leq (\eta_{y^*} - p_{y^*}) + (p_{y^*} - u_{\hat{j}}) + (u_{\hat{j}} - \alpha_{\hat{j}}) \leq \varepsilon_{\text{cls}}(x) + \varepsilon_{\text{exp}}(x).$$

816 All four cases are therefore covered by

$$\delta_\perp(p, u | x) \leq 2 \max\{\varepsilon_{\text{cls}}(x), \varepsilon_{\text{exp}}(x)\}. \quad (18)$$

817 We next bound these two sup-norm errors by the conditional surrogate excess. Since $\|\cdot\|_\infty \leq \|\cdot\|_1$,
 818 Pinsker's inequality for categorical distributions gives

$$\varepsilon_{\text{cls}}(x) \leq \|p(x) - \eta(x)\|_1 \leq \sqrt{2 \text{KL}(\eta(x) \| p(x))}.$$

819 For each expert coordinate, Pinsker's inequality for Bernoulli distributions gives

$$|u_j(x) - \alpha_j(x)| \leq \sqrt{\frac{1}{2} \text{KL}(\text{Bern}(\alpha_j(x)) \| \text{Bern}(u_j(x)))}.$$

820 Taking the maximum over j and upper bounding the maximum by the Euclidean norm yields

$$\varepsilon_{\text{exp}}(x) = \max_j |u_j(x) - \alpha_j(x)| \leq \sqrt{\frac{1}{2} \sum_{j=1}^J \text{KL}(\text{Bern}(\alpha_j(x)) \| \text{Bern}(u_j(x)))}.$$

821 Combining these inequalities with (17) gives

$$\varepsilon_{\text{cls}}(x) \leq \sqrt{2 \delta_{\text{dec}}(p, u | x)}, \quad \varepsilon_{\text{exp}}(x) \leq \sqrt{\frac{\delta_{\text{dec}}(p, u | x)}{2\beta}}.$$

822 Substituting into (18),

$$\begin{aligned} \delta_\perp(p, u | x) &\leq 2 \max\{\varepsilon_{\text{cls}}(x), \varepsilon_{\text{exp}}(x)\} = \max\{2\varepsilon_{\text{cls}}(x), 2\varepsilon_{\text{exp}}(x)\} \\ &\leq \max\left\{2\sqrt{2 \delta_{\text{dec}}(p, u | x)}, 2\sqrt{\frac{\delta_{\text{dec}}(p, u | x)}{2\beta}}\right\} = \max\left\{2\sqrt{2}, \sqrt{\frac{2}{\beta}}\right\} \sqrt{\delta_{\text{dec}}(p, u | x)}. \end{aligned}$$

823 Defining $\Gamma(t) := \max\{2\sqrt{2}, \sqrt{2/\beta}\}\sqrt{t}$, this is the pointwise calibration inequality

$$\delta_{\perp}(p, u \mid x) \leq \Gamma(\delta_{\text{dec}}(p, u \mid x)). \quad (19)$$

824 Taking expectations over X and using Jensen’s inequality, with the concavity of Γ , gives

$$\mathcal{E}_{\perp}(f_{p,u}) - \mathcal{E}_{\perp}^* = \mathbb{E}_X[\delta_{\perp}(p, u \mid X)] \leq \mathbb{E}_X[\Gamma(\delta_{\text{dec}}(p, u \mid X))] \leq \Gamma(\mathbb{E}_X[\delta_{\text{dec}}(p, u \mid X)]).$$

825 Since

$$\mathbb{E}_X[\delta_{\text{dec}}(p, u \mid X)] = \mathcal{E}_{\text{dec}}(p, u) - \mathcal{E}_{\text{dec}}^*,$$

826 the claimed bound follows from the definition of Γ and $\beta = \lambda/J$. ■

827 This theorem contains the whole statistical story of the decoupled surrogate. The pointwise KL
828 decomposition gives exact conditional alignment, the calibration inequality turns this alignment into
829 regret control, and Jensen’s inequality lifts the pointwise statement to population excess risk.

830 C.6 Bayes-Consistency Corollary

831 **Corollary 12** (Bayes consistency of the decoupled surrogate). *Every exact population minimizer of*
832 *the decoupled surrogate satisfies $p = \eta$ and $u_j = \alpha_j$ almost surely for all $j \in [J]$, and the induced*
833 *plug-in rule is Bayes-optimal. More generally, if $\mathcal{E}_{\text{dec}}(p_n, u_n) \rightarrow \mathcal{E}_{\text{dec}}^*$, then $\mathcal{E}_{\perp}(f_{p_n, u_n}) \rightarrow \mathcal{E}_{\perp}^*$.*

834 **Proof** The unrestricted class contains (η, α) . Any exact minimizer (p, u) achieves the population
835 infimum, hence its conditional excess $\delta_{\text{dec}}(p, u \mid x) = 0$ almost surely. By the KL decomposition
836 (17), this forces $p = \eta$ and $u_j = \alpha_j$ a.s. for all j , so the plug-in rule is Bayes-optimal by Lemma 2.
837 The approximate statement follows from Theorem 10. ■

838 D Additive Cross-Entropy (Mozannar and Sontag, 2020)

839 This section collects everything needed to understand the additive cross-entropy baseline in one place.
840 The key point is that additive CE is misaligned on both axes simultaneously: it changes the population
841 target and it rescales the local geometry by the realized number of correct experts.

842 The section proceeds in the same order as the main paper’s two-axis analysis. We first restate
843 the surrogate and show exactly what it estimates at the conditional optimum. We then derive its
844 samplewise gradient and Hessian and make the multiplicity factor $1 + |\mathcal{J}|$ explicit. The closing
845 comparison with the decoupled surrogate and the worked example then summarize why both effects
846 matter.

847 D.1 Formulation

848 The standard multi-expert cross-entropy baseline treats the K classifier actions and the J defer actions
849 as one augmented action space and rewards the true class together with every correct expert.

850 **Definition 13** (Additive augmented cross-entropy, Mozannar and Sontag (2020)). Let $a(x) \in \mathbb{R}^{K+J}$
851 be an augmented score vector and define

$$q_i(x) := \frac{\exp(a_i(x))}{\sum_{r=1}^{K+J} \exp(a_r(x))}, \quad i \in [K+J].$$

$$\Phi^{\text{CE}}(a; x, y, m) := -\log q_y(x) - \sum_{j: m_j=y} \log q_{K+j}(x).$$

853 Prediction is made by the augmented argmax of $a(x)$. This is the baseline whose additive reward
854 structure leads to the redundancy-driven underfitting effect highlighted in the main text.

855 D.2 Conditional Target Distortion

856 **Proposition 14** (Conditional minimizer of additive CE). *The conditional risk of CE at x is uniquely*
857 *minimized at*

$$q_y^*(x) = \frac{\eta_y(x)}{1 + U(x)}, \quad q_{K+j}^*(x) = \frac{\alpha_j(x)}{1 + U(x)}, \quad y \in [K], j \in [J]. \quad (20)$$

Proof We begin by writing the conditional risk as a cross-entropy against a normalized target. Once this normalization is explicit, the optimizer follows from the standard uniqueness of the cross-entropy minimizer over a simplex.

The conditional surrogate risk at a fixed x is

$$\mathcal{C}_{\text{add}}(q \mid x) = - \sum_{y=1}^K \eta_y(x) \log q_y - \sum_{j=1}^J \alpha_j(x) \log q_{K+j},$$

subject to $q \in \Delta^{K+J}$. Since $\sum_y \eta_y(x) = 1$, the total target mass is $1 + U(x)$ where $U(x) = \sum_j \alpha_j(x)$. Define the normalized target

$$\tilde{t}_y(x) = \frac{\eta_y(x)}{1 + U(x)}, \quad \tilde{t}_{K+j}(x) = \frac{\alpha_j(x)}{1 + U(x)}.$$

Then $\mathcal{C}_{\text{add}}(q \mid x) = (1 + U(x))H(\tilde{t}(x), q)$, where H denotes cross-entropy. Cross-entropy over a simplex is uniquely minimized at $q = \tilde{t}$, which gives the stated formula. ■

D.3 Gradient Amplification and Curvature

Proposition 4 (Samplewise amplification in additive CE). *For one sample (x, y, m) with $\mathcal{J} = \mathcal{J}(y, m)$, the gradient and Hessian of Φ^{CE} satisfy*

$$\frac{\partial \Phi^{\text{CE}}}{\partial a_i} = (1 + |\mathcal{J}|) q_i(x) - \mathbf{1}\{i = y\} - \mathbf{1}\{i = K + j, j \in \mathcal{J}\}. \quad (4)$$

$$\nabla_a^2 \Phi^{\text{CE}} = (1 + |\mathcal{J}|) (\text{Diag}(q) - qq^\top), \quad (5)$$

with largest eigenvalue $\lambda_{\max}(\nabla_a^2 \Phi^{\text{CE}}) \leq (1 + |\mathcal{J}|)/2$.

Proof The loss is a scaled log-sum-exp plus linear terms. We therefore derive the gradient first and then the Hessian, keeping track of where the factor $1 + |\mathcal{J}|$ enters.

Write $\mathcal{J} = \mathcal{J}(y, m)$. The loss can be rewritten as

$$\Phi^{\text{CE}}(a; x, y, m) = (1 + |\mathcal{J}|) \log \sum_{r=1}^{K+J} e^{a_r} - a_y - \sum_{j \in \mathcal{J}} a_{K+j}.$$

Part (a): gradient. Since $\partial \log \sum_r e^{a_r} / \partial a_i = q_i$, differentiating gives

$$\frac{\partial \Phi^{\text{CE}}}{\partial a_i} = (1 + |\mathcal{J}|) q_i - \mathbf{1}\{i = y\} - \mathbf{1}\{i = K + j, j \in \mathcal{J}\}.$$

The total positive mass is the sum of the indicator coefficients: $1 + |\mathcal{J}|$.

Part (b): curvature. We now differentiate once more. Recall that

$$q_i = \frac{e^{a_i}}{\sum_{r=1}^{K+J} e^{a_r}}.$$

Differentiating q_i with respect to a_r gives the standard softmax Jacobian

$$\frac{\partial q_i}{\partial a_r} = q_i(\mathbf{1}\{i = r\} - q_r).$$

Equivalently,

$$\frac{\partial q_i}{\partial a_i} = q_i(1 - q_i), \quad \frac{\partial q_i}{\partial a_r} = -q_i q_r \quad (r \neq i).$$

Therefore the Hessian of the log-sum-exp term $\log \sum_r e^{a_r}$ is the matrix whose (i, r) entry is $q_i(\mathbf{1}\{i = r\} - q_r)$, namely

$$\nabla_a^2 \log \sum_{r=1}^{K+J} e^{a_r} = \text{Diag}(q) - qq^\top.$$

881 This matrix is the covariance matrix of a categorical random variable with probabilities q , which is
 882 why it is positive semidefinite. Since the terms $-a_y$ and $-\sum_{j \in \mathcal{J}} a_{K+j}$ are linear in a , they disappear
 883 under second differentiation. Hence

$$\nabla_a^2 \text{CE} = (1 + |\mathcal{J}|)(\text{Diag}(q) - qq^\top).$$

884 It remains to bound the largest eigenvalue. Let $M := \text{Diag}(q) - qq^\top$. We apply Gershgorin's circle
 885 theorem row by row.

886 For row i , the diagonal entry is

$$M_{ii} = q_i - q_i^2 = q_i(1 - q_i),$$

887 and every off-diagonal entry is

$$M_{ij} = -q_i q_j, \quad j \neq i.$$

888 The Gershgorin radius of row i is the sum of the absolute values of the off-diagonal entries:

$$R_i = \sum_{j \neq i} |M_{ij}| = \sum_{j \neq i} q_i q_j = q_i \sum_{j \neq i} q_j = q_i(1 - q_i),$$

889 because $\sum_j q_j = 1$. Gershgorin's theorem therefore says that every eigenvalue λ of M lies in at least
 890 one interval

$$[M_{ii} - R_i, M_{ii} + R_i] = [q_i(1 - q_i) - q_i(1 - q_i), q_i(1 - q_i) + q_i(1 - q_i)] = [0, 2q_i(1 - q_i)].$$

891 Since the scalar function $u \mapsto u(1 - u)$ is maximized at $u = 1/2$, we have $q_i(1 - q_i) \leq 1/4$ for
 892 every i . Hence every eigenvalue of M lies in $[0, 1/2]$, and in particular

$$\lambda_{\max}(M) \leq \frac{1}{2}.$$

893 Finally, multiplying by the prefactor $1 + |\mathcal{J}|$ yields

$$\lambda_{\max}(\nabla_a^2 \text{CE}) \leq \frac{1 + |\mathcal{J}|}{2}.$$

894

■

895 D.4 Head-to-Head with the Decoupled Surrogate

896 Additive CE and the decoupled surrogate differ on both analytical axes, and the contrast is useful
 897 because the two losses are trying to solve the same decision problem in very different ways. Additive
 898 CE folds classes and experts into one augmented softmax. The decoupled surrogate instead keeps the
 899 class posterior and the expert utilities as separate objects throughout training and prediction. The
 900 next two paragraphs make precise why this distinction matters.

901 **Axis (i): statistical target.** Additive CE does not recover (η, α) directly. By Proposition 14, it learns
 902 the normalized object

$$\left(\frac{\eta_1}{1 + U}, \dots, \frac{\eta_K}{1 + U}, \frac{\alpha_1}{1 + U}, \dots, \frac{\alpha_J}{1 + U} \right),$$

903 so the target itself moves with the total expert overlap $U(x)$. This is the first misalignment. If one
 904 duplicates an already-correct expert several times, the underlying Bayes decision problem does not
 905 change, but the additive-CE target does: every class and expert coordinate is shrunk by the larger
 906 denominator $1 + U(x)$. In other words, additive CE is not estimating the native decision quantities
 907 on their original scale.

908 The decoupled surrogate has no such dependence. Its conditional minimizer is exactly $p^*(x) = \eta(x)$
 909 and $u_j^*(x) = \alpha_j(x)$ for every expert (Appendix C.5). Duplicating one expert leaves the target for
 910 every other coordinate unchanged. This is the sense in which the decoupled surrogate is target-aligned:
 911 the statistical object being estimated is the Bayes object itself, not a renormalized proxy.

912 **Axis (ii): local geometry.** Additive CE allocates one positive unit of gradient mass to the true class
 913 and one additional unit to every correct expert, which yields the multiplicative factor $1 + |\mathcal{J}|$ in
 914 both the gradient and the Hessian. The practical meaning is simple: samples on which many experts
 915 happen to be correct are up-weighted in both first-order and second-order optimization, regardless of
 916 whether those samples are actually more informative.

917 The decoupled surrogate eliminates this samplewise multiplicity entirely. Its class gradient is
 918 the standard multiclass cross-entropy gradient, and each expert receives an independent Bernoulli
 919 gradient of size $(\lambda/J)(u_j - t_j)$, unaffected by how many other experts are correct. Thus the decoupled
 920 surrogate keeps the optimization problem on the same scale across low-overlap and high-overlap
 921 regions of the input space.

922 Additive CE is problematic for two separate reasons: it aims at the wrong population target, and it
 923 also distorts the optimization landscape around each sample. The decoupled surrogate fixes both
 924 issues at once by preserving the native target (η, α) and by using coordinatewise gradients with no
 925 overlap-dependent amplification.

926 D.5 Worked Example: Redundant Experts

927 Consider $K = 2$ classes and $J = 4$ redundant experts. Suppose the Bayes-optimal expert behavior is
 928 represented by expert 1 and experts 2,3,4 are exact copies of it. On a sample with true label $y = 1$,
 929 assume all four experts are correct, so $|\mathcal{J}| = 4$.

930 This is the cleanest possible redundancy scenario. From the perspective of the decision problem,
 931 there is really only one useful expert behavior here, copied four times. A reasonable surrogate should
 932 therefore treat this sample almost the same way as it would treat a sample with only one correct
 933 expert. The calculation below shows that additive CE does not.

934 It is useful to make the axis-(i) issue explicit first. Suppose the class posterior is

$$\eta(x) = (0.6, 0.4),$$

935 and the underlying useful expert behavior has success probability 0.9. If there is only one such expert,
 936 additive CE targets

$$\left(\frac{0.6}{1 + 0.9}, \frac{0.4}{1 + 0.9}, \frac{0.9}{1 + 0.9} \right) = (0.316, 0.211, 0.474).$$

937 If we now replace that one expert by four exact copies, the Bayes decision does not change: the best
 938 class utility is still 0.6 and the best expert utility is still 0.9. However, additive CE now targets

$$\left(\frac{0.6}{1 + 4 \cdot 0.9}, \frac{0.4}{1 + 4 \cdot 0.9}, \frac{0.9}{1 + 4 \cdot 0.9}, \frac{0.9}{1 + 4 \cdot 0.9}, \frac{0.9}{1 + 4 \cdot 0.9}, \frac{0.9}{1 + 4 \cdot 0.9} \right)$$

$$= (0.130, 0.087, 0.196, 0.196, 0.196, 0.196).$$

940 So even before discussing gradients, the surrogate target has already moved. Nothing about the
 941 underlying decision problem changed; only the number of redundant copies changed.

942 Take an uninformative additive-CE initialization with equal logits on the $K + J = 6$ actions, hence
 943 $q_i = 1/6$ for all i . Proposition 4 gives

$$\frac{\partial \Phi^{\text{CE}}}{\partial a_1} = 5 \cdot \frac{1}{6} - 1 = -\frac{1}{6}, \quad \frac{\partial \Phi^{\text{CE}}}{\partial a_2} = 5 \cdot \frac{1}{6} = \frac{5}{6},$$

944 and for every correct expert $j \in \{1, 2, 3, 4\}$,

$$\frac{\partial \Phi^{\text{CE}}}{\partial a_{K+j}} = 5 \cdot \frac{1}{6} - 1 = -\frac{1}{6}.$$

945 Thus the sample contributes one rewarded class term and four rewarded expert terms, for a total
 946 positive mass of 5. By the Hessian formula proved above, the same sample also carries curvature
 947 prefactor 5. Both quantities are inflated by the redundancy count.

948 To see why this is a pathology rather than a harmless rescaling, compare it with the single-copy
 949 version of the same problem. If only expert 1 were present, then the same derivation would give total

positive mass 2 rather than 5, because the sample would reward one class and one expert. So merely replacing one expert by four exact copies multiplies the local scale of the update by a factor of 5/2, even though no new information has been added.

Now compare the decoupled surrogate on the same sample. Let the class head start at $p = (1/2, 1/2)$ and each expert head at $u_j = 1/2$. Then

$$\frac{\partial \Phi^{\text{dec}}}{\partial w_1} = -\frac{1}{2}, \quad \frac{\partial \Phi^{\text{dec}}}{\partial w_2} = \frac{1}{2}, \quad \frac{\partial \Phi^{\text{dec}}}{\partial s_j} = \frac{\lambda}{4} \left(\frac{1}{2} - 1 \right) = -\frac{\lambda}{8},$$

for all $j = 1, \dots, 4$. If the same sample had only one correct expert instead of four, the decoupled surrogate class gradient would be unchanged and the nonzero expert gradient would still be of the same per-coordinate form. In other words, the decoupled surrogate updates each relevant coordinate because it is correct, not because many other coordinates happen to agree with it.

Under additive CE, duplicating one correct expert into four copies changes both what the model is trying to estimate and how strongly this sample pulls on the optimizer. Under the decoupled surrogate, duplication does neither: the class task remains the same, and each expert is trained on its own correctness event with the same local scale. That is why the decoupled surrogate remains stable in redundancy-heavy regimes while additive CE can underfit or become dominated by high-overlap regions.

The problem is not merely that its probability estimates can require post-processing. The deeper issue is structural. Redundant experts change the surrogate target and simultaneously change the optimization geometry, even when the Bayes decision problem itself is unchanged. The decoupled surrogate is designed precisely to remove both dependencies.

E PiCCE (Liu et al., 2026)

PiCCE (Liu et al., 2026) was designed to remove the additive amplification of CE (Mozannar and Sontag, 2020) by rewarding only one correct expert per sample. This section shows the resulting tradeoff. The samplewise scale becomes controlled, but the winner-take-all mechanism creates a different pathology: correct experts that fail to win the internal $\arg \max$ receive negative updates.

The logic of the section is parallel to that of additive CE. We first define the surrogate precisely, then isolate the starvation effect at the sample level, then show how the same winner-take-all mechanism induces a moving conditional target, and finally compare the resulting behavior with the decoupled surrogate. Two facts should be kept distinct throughout. First, PiCCE genuinely improves on additive CE in one respect: it removes the overlap-dependent amplification factor. Second, that repair is achieved by forcing correct experts to compete with one another. The point of the section is to show that this second effect is not a minor side effect but the central limitation of the surrogate.

E.1 Formulation

PiCCE modifies the additive surrogate by rewarding only one correct expert rather than all of them.

Definition 15 (PiCCE, Liu et al. (2026)). Let $a(x) \in \mathbb{R}^{K+J}$ be an augmented score vector and define

$$q_i(x) := \frac{\exp(a_i(x))}{\sum_{r=1}^{K+J} \exp(a_r(x))}, \quad i \in [K+J].$$

Let

$$j^*(a; x, y, m) = \arg \max_{j: m_j = y} a_{K+j}(x)$$

with deterministic tie-breaking, whenever at least one expert is correct. Then

$$\Phi^{\text{PiCCE}}(a; x, y, m) := -\log q_y(x) - \mathbf{1}\{\exists j \in [J] : m_j = y\} \log q_{K+j^*(a; x, y, m)}(x).$$

Prediction again uses the augmented $\arg \max$ over $a(x)$. In our analysis, this winner-take-all correction removes additive amplification but can starve useful experts that are correct yet repeatedly unselected.

988 E.2 Winner-Take-All Starvation

989 **Proposition 5** (Winner-take-all starvation in PiCCE). *Under PiCCE with $\mathcal{J}(y, m) \neq \emptyset$, the gradient*
 990 *on any logit is $\partial\Phi^{\text{PiCCE}}/\partial a_i = 2q_i - \mathbf{1}\{i = y\} - \mathbf{1}\{i = K + j^*\}$. For any correct expert $k \in \mathcal{J}$*
 991 *with $k \neq j^*$,*

$$\frac{\partial\Phi^{\text{PiCCE}}}{\partial a_{K+k}} = 2q_{K+k}(x) > 0. \quad (7)$$

992 *Gradient descent pushes expert k 's logit downward, even though it is correct.*

993 **Proof** Once the winning expert is fixed, PiCCE becomes a softmax loss with exactly two rewarded
 994 coordinates: the true class and the selected expert. The sign pattern for every other correct expert can
 995 therefore be read off directly from the gradient.

996 Assume $\mathcal{J}(y, m) \neq \emptyset$ and let j^* be the PiCCE winner. Then

$$\Phi^{\text{PiCCE}}(a; x, y, m) = 2 \log \sum_{r=1}^{K+J} e^{a_r} - a_y - a_{K+j^*}.$$

997 Differentiating with respect to a_i gives $\partial\Phi^{\text{PiCCE}}/\partial a_i = 2q_i - \mathbf{1}\{i = y\} - \mathbf{1}\{i = K + j^*\}$. This
 998 formula already shows the winner-take-all structure: only two coordinates receive a subtractive
 999 reward term, namely the true class and the selected expert. Every other coordinate sees only the
 1000 positive softmax term $2q_i$.

1001 For a correct expert $k \neq j^*$, set $i = K + k$. Since $k \neq j^*$, neither indicator is active, so

$$\frac{\partial\Phi^{\text{PiCCE}}}{\partial a_{K+k}} = 2q_{K+k}(x) > 0,$$

1002 where the strict positivity uses $q_{K+k} > 0$ (softmax outputs are strictly positive). Hence gradient
 1003 descent pushes a_{K+k} downward.

1004 The expert k is correct on the current sample, but because it failed to win the internal competition,
 1005 the update acts as if it were an ordinary non-target coordinate and suppresses its logit. ■

1006 E.3 Score-Dependent Conditional Target

1007 For completeness we derive the conditional risk formula used in §4.2. Condition on $X = x$ and take
 1008 expectations over (Y, M) . The PiCCE loss rewards expert $j^* = \arg \max_{j \in \mathcal{J}(Y, M)} a_{K+j}$ whenever
 1009 $\mathcal{J} \neq \emptyset$. Define $\gamma_j(a, x) = \Pr(j \in \mathcal{J}(Y, M) \text{ and } j = j^* \mid X = x)$. The class term is standard:

$$\mathbb{E}[\log q_Y \mid X = x] = \sum_{k=1}^K \eta_k(x) \log q_k.$$

1010 For the expert term, partition the event $\{\mathcal{J} \neq \emptyset\}$ according to which expert wins the internal
 1011 competition. This yields

$$\mathbb{E}[\mathbf{1}\{\mathcal{J} \neq \emptyset\} \log q_{K+j^*} \mid X = x] = \sum_{j=1}^J \gamma_j(a, x) \log q_{K+j}.$$

1012 Substituting both identities gives

$$\mathcal{C}_{\text{PiCCE}}(q, a \mid x) = - \sum_{k=1}^K \eta_k(x) \log q_k - \sum_{j=1}^J \gamma_j(a, x) \log q_{K+j}.$$

1013 Thus, for a fixed score vector a , PiCCE behaves like an augmented cross-entropy whose class-side
 1014 coefficients are $\eta_k(x)$ and whose expert-side coefficients are $\gamma_j(a, x)$, not $\alpha_j(x)$. Equivalently,
 1015 viewed as a weighted multiclass log-loss over the augmented simplex, its instantaneous target is
 1016 proportional to

$$(\eta_1(x), \dots, \eta_K(x), \gamma_1(a, x), \dots, \gamma_J(a, x)).$$

1017 The coefficients $\gamma_j(a, x)$ depend on the score vector a through the winner map j^* , so the expert-side
 1018 target moves as a is updated. This is the precise sense in which PiCCE chases a moving target. The
 1019 class-side weights are fixed by the data distribution, but the expert-side weights are redistributed by
 1020 the current score ordering. This is in contrast to additive CE, where the coefficients are fixed by the
 1021 distribution alone.

1022 Fix a point x and suppose there is positive conditional probability that experts 1 and 2 are both correct.
 1023 If the current scores satisfy $a_{K+1} > a_{K+2}$ on those outcomes, then that shared probability mass
 1024 contributes to $\gamma_1(a, x)$. If the score ordering is reversed, the same data mass contributes to $\gamma_2(a, x)$
 1025 instead. The data distribution has not changed; only the current score ordering has. That is why the
 1026 PiCCE expert target is score-dependent rather than intrinsic to the underlying decision problem.

1027 E.4 Curvature

1028 The winner-take-all repair eliminates the multiplicity factor of additive CE. Whenever $\mathcal{J}(y, m) \neq \emptyset$,

$$\Phi^{\text{PiCCE}}(a; x, y, m) = 2 \log \sum_{r=1}^{K+J} e^{a_r} - a_y - a_{K+j^*},$$

1029 where j^* is treated as fixed on the current sample. We now differentiate this expression exactly as in
 1030 the additive-CE case, but with prefactor 2 instead of $1 + |\mathcal{J}|$.

1031 The only nonlinear term is the log-sum-exp. Its Hessian is

$$\nabla_a^2 \log \sum_{r=1}^{K+J} e^{a_r} = \text{Diag}(q) - qq^\top,$$

1032 while the linear terms $-a_y$ and $-a_{K+j^*}$ vanish under second differentiation. Therefore

$$\nabla_a^2 \Phi^{\text{PiCCE}} = 2(\text{Diag}(q) - qq^\top),$$

1033 The matrix $\text{Diag}(q) - qq^\top$ is again the softmax covariance matrix. From the additive-CE derivation
 1034 above, every eigenvalue of this matrix lies in $[0, 1/2]$. Multiplying by the prefactor 2 gives

$$\lambda_{\max}(\nabla_a^2 \Phi^{\text{PiCCE}}) \leq 1.$$

1035 So PiCCE is indeed well behaved in the second-order sense that additive CE is not. This point is worth
 1036 stating clearly because it isolates the real failure mode. The problem with PiCCE is not exploding
 1037 curvature or overlap-dependent scaling. The problem is the direction of the updates: the surrogate
 1038 suppresses correct experts that fail to win the internal comparison, and this same winner-take-all
 1039 mechanism makes the expert-side target score-dependent.

1040 E.5 Head-to-Head with the Decoupled Surrogate

1041 **Axis (i): statistical target.** PiCCE preserves the correct class coefficients $\eta_k(x)$, but on the expert
 1042 side it replaces $\alpha_j(x)$ by the score-dependent quantity $\gamma_j(a, x)$. The target therefore depends on
 1043 the optimization trajectory through the current winner map. This is the first key contrast with the
 1044 decoupled surrogate. Under the decoupled surrogate, the expert target for coordinate j is always the
 1045 same object, $\alpha_j(x)$, regardless of which expert currently has the highest score. The data determine
 1046 the target. Under PiCCE, by contrast, shared correct mass is assigned to whichever expert is currently
 1047 winning the internal competition.

1048 The practical implication is that PiCCE does not merely estimate the wrong expert quantity at the
 1049 optimum; it can also reshape the quantity it is trying to estimate while training proceeds. The
 1050 decoupled surrogate does not have this moving-target effect: the class and expert subproblems are
 1051 proper losses with fixed conditional targets (η, α) .

1052 **Axis (ii): local geometry.** PiCCE solves the multiplicity problem by capping the number of rewarded
 1053 terms at two, but it does so by selecting a unique correct expert and repelling every other correct
 1054 expert. This is not just a bookkeeping choice. It means that on a sample where several experts are
 1055 simultaneously correct, PiCCE turns agreement into competition.

The decoupled surrogate makes the opposite design choice. It keeps the per-coordinate updates bounded while reinforcing every correct expert independently. Agreement among correct experts therefore does not create either additive amplification or winner-take-all suppression. PiCCE and the decoupled surrogate agree on the need to avoid additive amplification, but they differ fundamentally on whether multiple correct experts should compete with one another or all receive supportive updates.

PiCCE improves on additive CE along one dimension: it removes the $O(|\mathcal{J}|)$ scaling. But it does so by replacing an overlap-dependent magnitude problem with a winner-dependent direction problem. The decoupled surrogate avoids both failures simultaneously: it keeps the local scale bounded and it preserves supportive updates for every correct expert.

E.6 Worked Example: Generalist and Specialist Both Correct

Consider one generalist expert and one specialist expert. On a sample from the specialist region, suppose both experts are correct, so $\mathcal{J} = \{1, 2\}$, but the current logits satisfy $a_{K+1} > a_{K+2}$, meaning the generalist wins the internal PiCCE competition. This is precisely the situation in which one would like the training signal to preserve both experts: the generalist is correct, but the specialist is also correct on the very region for which it was meant to be useful. PiCCE does not preserve both. It forces them to compete.

To make the signs concrete, suppose the current softmax assigns

$$q_{K+1} = 0.30, \quad q_{K+2} = 0.20.$$

Then $j^* = 1$ and

$$\frac{\partial \Phi^{\text{PiCCE}}}{\partial a_{K+1}} = 2q_{K+1} - 1, \quad \frac{\partial \Phi^{\text{PiCCE}}}{\partial a_{K+2}} = 2q_{K+2}.$$

Substituting the numbers gives

$$\frac{\partial \Phi^{\text{PiCCE}}}{\partial a_{K+1}} = 2(0.30) - 1 = -0.40, \quad \frac{\partial \Phi^{\text{PiCCE}}}{\partial a_{K+2}} = 2(0.20) = 0.40.$$

Under gradient descent, the generalist logit is pushed upward because its gradient is negative, while the specialist logit is pushed downward because its gradient is positive. So the same sample provides evidence for both experts, yet PiCCE uses that evidence to strengthen the winner and weaken the loser.

This same example also reflects the axis-(i) issue. On any population of such samples, the shared “both correct” mass is assigned to whichever expert currently wins the internal comparison. If the generalist is ahead, that mass contributes to $\gamma_1(a, x)$; if the specialist overtakes it, the same mass contributes to $\gamma_2(a, x)$ instead. The target therefore co-moves with the current score ordering.

For the decoupled surrogate, let the same sample have expert probabilities $u_1 = 0.80$ and $u_2 = 0.75$, with targets $t_1 = t_2 = 1$. Then

$$\frac{\partial \Phi^{\text{dec}}}{\partial s_1} = \frac{\lambda}{2}(0.80 - 1) = -0.10\lambda, \quad \frac{\partial \Phi^{\text{dec}}}{\partial s_2} = \frac{\lambda}{2}(0.75 - 1) = -0.125\lambda.$$

Both experts are reinforced. The specialist may still trail the generalist at the current iterate, but it is not punished for being correct. The sample acts as supportive evidence for both experts rather than as an internal tournament.

This example is therefore the PiCCE analogue of the redundant-expert example for additive CE. The key point is not merely that PiCCE uses a different formula. The deeper point is structural. Whenever several experts are correct on the same sample, PiCCE converts agreement into competition. The decoupled surrogate preserves the bounded geometry without introducing that competition, which is exactly why it is more robust in the rare-specialist regime studied in the main text.

F Mao25 (Mao et al., 2025d)

Mao25 moves away from additive rewards and winner-take-all selection by optimizing only the total mass assigned to the acceptable set. This resolves the two most immediate geometric issues, but it also removes the signal needed to rank experts within that set.

1098 The central question in this section is what object Mao25 is actually trying to fit. The answer is
 1099 not the full vector (η, α) , but only the total probability assigned to the acceptable set. That choice
 1100 makes the local geometry comparatively benign, but it also explains why the surrogate cannot express
 1101 preferences among multiple acceptable experts.

1102 The positive theory in Mao et al. (2025d) is important to keep in view. In the single-stage multiple-
 1103 expert setting, their family L_Ψ comes with formal consistency guarantees under mild endpoint
 1104 conditions on Ψ . They further single out the member $\Psi(u) = 1 - u$, denoted L_{mae} in their paper,
 1105 because it also admits quantitative excess-risk control. Our criticism is not about those consistency
 1106 guarantees. It is about the practical optimization behavior of that specific $\Psi(u) = 1 - u$ instantiation:
 1107 its samplewise loss is linear in acceptable-set mass and therefore has an MAE-like optimization
 1108 profile.

1109 F.1 Formulation

1110 The single-stage surrogate of Mao et al. (2025d) defines a family indexed by a decreasing function Ψ .
 1111 In their multiple-expert experiments, they instantiate this family with $\Psi(u) = 1 - u$, which is the
 1112 version we include in our empirical comparison.

1113 **Definition 16** (Mao25 single-stage surrogate with $\Psi(u) = 1 - u$, Mao et al. (2025d)). Let $a(x) \in$
 1114 $\mathbb{R}^{K+J}$ be an augmented score vector and define

$$q_i(x) := \frac{\exp(a_i(x))}{\sum_{r=1}^{K+J} \exp(a_r(x))}, \quad i \in [K + J].$$

1115 Then

$$\Phi^{\text{Mao}}(a; x, y, m) := 1 - \left(q_y(x) + \sum_{j: m_j=y} q_{K+j}(x) \right).$$

1116 Prediction uses the augmented argmax over $a(x)$. This surrogate depends only on the total mass
 1117 assigned to the acceptable set consisting of the true class and all correct experts.

1118 This has a simple consequence at the conditional level. Fix an input x and take expectation over the
 1119 random pair (Y, M) .

1120 The class part contributes

$$\mathbb{E}[q_Y \mid X = x] = \sum_{k=1}^K \eta_k(x) q_k,$$

1121 because q_k is fixed once x is fixed, and only the random label Y remains inside the expectation.
 1122 Likewise, for the expert part,

$$\mathbb{E} \left[\sum_{j: M_j=Y} q_{K+j} \mid X = x \right] = \sum_{j=1}^J \alpha_j(x) q_{K+j},$$

1123 because expert j contributes exactly on the event $\{M_j = Y\}$, whose conditional probability is $\alpha_j(x)$
 1124 by definition.

1125 Substituting these two identities into the loss gives

$$\mathcal{C}_{\text{Mao}}(q \mid x) = 1 - \sum_{k=1}^K \eta_k(x) q_k - \sum_{j=1}^J \alpha_j(x) q_{K+j}.$$

1126 Now the key point is visible. The conditional risk is a linear function of the augmented probabilities
 1127 q . Minimizing it is therefore the same as maximizing

$$\sum_{k=1}^K \eta_k(x) q_k + \sum_{j=1}^J \alpha_j(x) q_{K+j}$$

1128 over the simplex Δ^{K+J} . A linear objective over a simplex is minimized at an extreme point, so there
 1129 always exists a conditional minimizer that puts all mass on one action whose coefficient is maximal
 1130 among

$$(\eta_1(x), \dots, \eta_K(x), \alpha_1(x), \dots, \alpha_J(x)).$$

1131 If the maximizer is unique, that minimizer is unique and is exactly the Bayes-optimal action. If
 1132 several actions tie for the maximum, any convex combination over those maximizers is also optimal.
 1133 So Mao25 can recover a Bayes-optimal action, but it does not recover the full vector (η, α) . It
 1134 identifies which action can be optimal, not the underlying utilities themselves.

1135 F.2 Acceptable-Set Mass Without Ranking

1136 **Proposition 6** (Acceptable-set mass in Mao25). *The gradient is $\partial\Phi^{\text{Mao}}/\partial a_i = q_i(S_S - \mathbf{1}\{i \in$
 1137 $\mathcal{S}(y, m)\})$, where $\mathcal{S}(y, m) = \{y\} \cup \{K + j : j \in \mathcal{J}\}$ and $S_S = \sum_{c \in \mathcal{S}} q_c$. The loss depends on
 1138 the acceptable set only through S_S , and for acceptable coordinates the gradient has the common
 1139 factor $S_S - 1$. Thus the samplewise update distinguishes acceptable from unacceptable actions, but
 1140 it contains no term that prefers one acceptable expert over another on the basis of expert quality.*

1141 **Proof** The argument has two steps. First we show that the loss depends on the score vector only
 1142 through the acceptable-set mass. Then we differentiate that mass to obtain the gradient formula.

1143 Let

$$\mathcal{S}(y, m) := \{y\} \cup \{K + j : j \in \mathcal{J}(y, m)\}$$

1144 denote the acceptable set. By definition,

$$\Phi^{\text{Mao}}(a; x, y, m) = 1 - \sum_{c \in \mathcal{S}(y, m)} q_c(x) = 1 - S_S(x).$$

1145 Thus the loss depends on a only through the total acceptable mass $S_S(x)$.

1146 To differentiate, use the softmax Jacobian $\partial q_c / \partial a_i = q_c(\mathbf{1}\{c = i\} - q_i)$. Then

$$\begin{aligned} \frac{\partial S_S}{\partial a_i} &= \sum_{c \in \mathcal{S}(y, m)} q_c(\mathbf{1}\{c = i\} - q_i) \\ &= \mathbf{1}\{i \in \mathcal{S}(y, m)\} q_i - q_i \sum_{c \in \mathcal{S}(y, m)} q_c \\ &= q_i(\mathbf{1}\{i \in \mathcal{S}(y, m)\} - S_S). \end{aligned}$$

1147 and therefore

$$\frac{\partial \Phi^{\text{Mao}}}{\partial a_i} = -\frac{\partial S_S}{\partial a_i} = q_i(S_S - \mathbf{1}\{i \in \mathcal{S}(y, m)\}).$$

1148 If $i \in \mathcal{S}(y, m)$, then $S_S \leq 1$, so the gradient is non-positive. If $i \notin \mathcal{S}(y, m)$, then the gradient is
 1149 $q_i S_S \geq 0$. Finally, since $\Phi^{\text{Mao}} = 1 - S_S$, any two score vectors with the same S_S incur the same
 1150 loss regardless of their internal allocation inside $\mathcal{S}(y, m)$.

1151 This formula makes the absence of direct within-set ranking explicit. If $i, r \in \mathcal{S}(y, m)$ are both
 1152 acceptable actions on the current sample, then

$$\frac{\partial \Phi^{\text{Mao}}}{\partial a_i} = q_i(S_S - 1), \quad \frac{\partial \Phi^{\text{Mao}}}{\partial a_r} = q_r(S_S - 1).$$

1153 Both coordinates share the same factor $S_S - 1$. Their update magnitudes differ only through the
 1154 current probabilities q_i and q_r , not through any term encoding which acceptable expert is statistically
 1155 better.

1156 The same formula also reveals why optimization can be weak. Summing the attractive gradient mass
 1157 over the acceptable set gives

$$\sum_{i \in \mathcal{S}} -\frac{\partial \Phi^{\text{Mao}}}{\partial a_i} = \sum_{i \in \mathcal{S}} q_i(1 - S_S) = S_S(1 - S_S).$$

1158 This quantity is at most $1/4$ and tends to zero as $S_S \rightarrow 0$. Thus, when the model currently places
 1159 very little mass on the acceptable set, the corrective signal toward that set is weak. In the single-label
 1160 special case $|\mathcal{S}| = 1$, the loss reduces to $1 - q_y$, which is exactly the multiclass MAE form. The same
 1161 weak-gradient phenomenon is well known for MAE-like losses in noisy-label learning (Zhang and
 1162 Sabuncu, 2018). ■

1163 F.3 Curvature

1164 Although Mao25 does not suffer from additive amplification, its second-order structure still depends
 1165 only on acceptable-set membership and the total acceptable mass. We differentiate the gradient
 1166 formula step by step.

1167 Write

$$g_i(a) := \frac{\partial \Phi^{\text{Mao}}}{\partial a_i} = q_i(S_S - \mathbf{1}\{i \in \mathcal{S}\}).$$

1168 Then

$$\frac{\partial g_i}{\partial a_r} = \frac{\partial q_i}{\partial a_r}(S_S - \mathbf{1}\{i \in \mathcal{S}\}) + q_i \frac{\partial S_S}{\partial a_r}.$$

1169 The first term comes from differentiating the prefactor q_i and the second term comes from differenti-
 1170 ating the acceptable-set mass S_S . From the softmax Jacobian,

$$\frac{\partial q_i}{\partial a_r} = q_i(\mathbf{1}\{i = r\} - q_r),$$

1171 and from the previous subsection,

$$\frac{\partial S_S}{\partial a_r} = q_r(\mathbf{1}\{r \in \mathcal{S}\} - S_S).$$

1172 Substituting both identities into the product rule gives the full Hessian entry:

$$\frac{\partial^2 \Phi^{\text{Mao}}}{\partial a_i \partial a_r} = q_i(\mathbf{1}\{i = r\} - q_r)(S_S - \mathbf{1}\{i \in \mathcal{S}\}) + q_i q_r(\mathbf{1}\{r \in \mathcal{S}\} - S_S).$$

1173 No additional term is hidden. This is the complete second derivative obtained by differentiating the
 1174 explicit gradient formula.

1175 It is useful to unpack the main cases.

1176 **Case 1:** $i = r \in \mathcal{S}$. Then

$$\frac{\partial^2 \Phi^{\text{Mao}}}{\partial a_i^2} = q_i(1 - q_i)(S_S - 1) + q_i^2(1 - S_S).$$

1177 This depends on q_i and on the common scalar S_S , but not on which acceptable expert i represents.

1178 **Case 2:** $i \neq r$ and $i, r \in \mathcal{S}$. Then

$$\frac{\partial^2 \Phi^{\text{Mao}}}{\partial a_i \partial a_r} = -q_i q_r(S_S - 1) + q_i q_r(1 - S_S) = 2q_i q_r(1 - S_S).$$

1179 Again the expression depends only on the current masses and on the shared acceptable-set mass.

1180 **Case 3:** $i \in \mathcal{S}$ and $r \notin \mathcal{S}$. Then

$$\frac{\partial^2 \Phi^{\text{Mao}}}{\partial a_i \partial a_r} = -q_i q_r(S_S - 1) - q_i q_r S_S = q_i q_r(1 - 2S_S).$$

1181 **Case 4:** $i, r \notin \mathcal{S}$. Then

$$\frac{\partial^2 \Phi^{\text{Mao}}}{\partial a_i \partial a_r} = q_i(\mathbf{1}\{i = r\} - q_r)S_S - q_i q_r S_S.$$

1182 In particular, for $i \neq r$ both outside the acceptable set,

$$\frac{\partial^2 \Phi^{\text{Mao}}}{\partial a_i \partial a_r} = -2q_i q_r S_S.$$

1183 This expression contains no term that depends on which acceptable expert is statistically better.
 1184 Even at second order, the surrogate sees acceptable-set mass and membership, not expert quality.
 1185 This is the geometric analogue of the first-order statement above: Mao25 distinguishes acceptable
 1186 from unacceptable actions, but its local geometry contains no utility-sensitive comparison inside the
 1187 acceptable set.

F.4 Head-to-Head with the Decoupled Surrogate

Axis (i): statistical target. Mao25 is aligned with the Bayes action only at the coarse level of choosing the best augmented action. Its conditional risk is linear in q , so a minimizer places all mass on one optimal action, but the surrogate does not recover the full utility vector (η, α) . The decoupled surrogate does. At the conditional minimum, the decoupled surrogate returns $p^*(x) = \eta(x)$ and $u_j^*(x) = \alpha_j(x)$ for every expert. The difference is therefore not whether the two surrogates can identify a good action in principle. The difference is that the decoupled surrogate estimates the underlying utilities themselves, whereas Mao25 only optimizes a coarse action-level objective.

Axis (ii): local geometry. Mao25 avoids both additive amplification and winner-take-all starvation. In that sense it is geometrically cleaner than CE and PiCCE. But the geometry is still too coarse for utility estimation. On a sample where several actions are acceptable, the loss depends only on the total acceptable mass and the acceptable/unacceptable partition. It contains no term that says which acceptable expert should be preferred. For the concrete choice $\Psi(u) = 1 - u$, the total attractive gradient mass toward the acceptable set is $S_S(1 - S_S)$. This quantity is maximal at $1/4$ and shrinks to zero as $S_S \rightarrow 0$, so the corrective signal can be weak precisely when the model currently assigns little mass to the right set. The decoupled surrogate keeps the benign bounded geometry while training each expert head against its own correctness process.

The contrast with the decoupled surrogate is therefore different from the PiCCE contrast. Mao25 does not harm correct experts by suppressing them, and it does not amplify high-overlap samples. Its limitation is coarseness: it treats expert supervision as set membership rather than utility estimation, and under the linear choice $\Psi(u) = 1 - u$ this coarse signal can also be weak away from the optimum. The decoupled surrogate retains the stable geometry while preserving expert-specific targets.

F.5 Worked Example: Two Correct Experts of Unequal Quality

Consider a defer region in which expert 1 is substantially better overall than expert 2:

$$\alpha_1(x) = 0.9, \quad \alpha_2(x) = 0.5.$$

Assume the best class utility is smaller, say $\max_k \eta_k(x) = 0.4$, so the Bayes-optimal action is to defer to expert 1.

The conditional Mao25 risk at this x is then

$$\mathcal{C}_{\text{Mao}}(q \mid x) = 1 - \sum_{k=1}^K \eta_k(x) q_k - 0.9 q_{K+1} - 0.5 q_{K+2}.$$

This conditional objective is minimized by placing all mass on expert 1, which indeed recovers the correct action. But the surrogate does not recover the utility values 0.9 and 0.5 themselves; it only prefers the larger coefficient through a linear argmax objective.

Now inspect the samplewise supervision when both experts happen to be correct on a particular draw. Compare two augmented allocations with the same acceptable mass:

$$(q_y, q_{K+1}, q_{K+2}) = (0.10, 0.50, 0.00), \quad (q_y, q_{K+1}, q_{K+2}) = (0.10, 0.25, 0.25).$$

Both have $S_S = 0.60$, so both incur the same Mao25 loss $\Phi^{\text{Mao}} = 0.40$. The surrogate therefore cannot prefer the first allocation over the second on the basis that expert 1 is the reliable one.

The gradients make the same point. Since both experts are acceptable on this sample, the gradient formula gives

$$\frac{\partial \Phi^{\text{Mao}}}{\partial a_i} = q_i(0.60 - 1) = -0.4 q_i, \quad i \in \{y, K+1, K+2\}.$$

For the first allocation,

$$\left(\frac{\partial \Phi^{\text{Mao}}}{\partial a_y}, \frac{\partial \Phi^{\text{Mao}}}{\partial a_{K+1}}, \frac{\partial \Phi^{\text{Mao}}}{\partial a_{K+2}} \right) = (-0.04, -0.20, 0.00),$$

whereas for the second allocation,

$$\left(\frac{\partial \Phi^{\text{Mao}}}{\partial a_y}, \frac{\partial \Phi^{\text{Mao}}}{\partial a_{K+1}}, \frac{\partial \Phi^{\text{Mao}}}{\partial a_{K+2}} \right) = (-0.04, -0.10, -0.10).$$

1226 The difference between the two acceptable experts is driven only by their current masses q_{K+1} and
 1227 q_{K+2} , not by the fact that expert 1 is the better expert overall.

1228 The decoupled surrogate behaves differently. Its expert-side conditional risk is

$$-0.9 \log u_1 - 0.1 \log(1 - u_1) - 0.5 \log u_2 - 0.5 \log(1 - u_2),$$

1229 which is uniquely minimized at $u_1 = 0.9$ and $u_2 = 0.5$. The ranking signal is there by construction.
 1230 Mao25 can recover the correct defer action, but it does not estimate the underlying expert utilities.
 1231 On the sample itself, it only rewards membership in the acceptable set, and the total reward scale
 1232 is capped by $S_S(1 - S_S)$. The decoupled surrogate does both: it preserves the correct action and
 1233 recovers the utility gap that distinguishes expert 1 from expert 2.

1234 G A-SM (Cao et al., 2024)

1235 A-SM (Cao et al., 2024) is the strongest augmented-action baseline on the population axis. It restores
 1236 bounded estimates and recovers the correct target. Its remaining weakness is optimization-level: the
 1237 expert losses still leak into the class gradient through the shared score vector.

1238 This section therefore plays a different role from the earlier baseline sections. For additive CE, PiCCE,
 1239 and Mao25, the main mismatch is already visible in the conditional target. For A-SM, the target is
 1240 largely the right one. The point here is to isolate the remaining axis-(ii) weakness cleanly and show
 1241 that it survives even after boundedness and population alignment are repaired.

1242 G.1 Formulation

1243 A-SM replaces the symmetric augmented normalization by an asymmetric one that keeps the class
 1244 probabilities and expert probabilities bounded and semantically valid. We use the multi-expert
 1245 extension given in Appendix H of Cao et al. (2024). To stay consistent with the main text, we write
 1246 ξ_k for the class estimates and ψ_j for the expert estimates.

1247 The single-expert A-SM paper is written with one defer coordinate, so its expert denominator is
 1248 naturally expressed over $K + 1$ coordinates: the K class logits and the single defer logit. In the
 1249 multi-expert extension, this construction is applied separately to each expert. Thus the class head still
 1250 uses the first K logits only, while expert j is normalized against those same K class logits plus its
 1251 own expert logit a_{K+j} , not against the other expert logits. This point is easy to miss when moving
 1252 from the single-expert notation to the multi-expert one, so we state it explicitly here.

1253 **Definition 17** (Multi-expert A-SM, Cao et al. (2024)). Let $a(x) \in \mathbb{R}^{K+J}$ be an augmented score
 1254 vector. Define the class estimates as the standard softmax over the first K coordinates,

$$\xi_k(a(x)) := \frac{\exp(a_k(x))}{\sum_{k'=1}^K \exp(a_{k'}(x))}, \quad k \in [K],$$

1255 and the expert estimates, for each $j \in [J]$, as

$$\psi_j(a(x)) := \frac{\exp(a_{K+j}(x))}{\sum_{k'=1}^K \exp(a_{k'}(x)) + \exp(a_{K+j}(x)) - \max_{k' \in [K]} \exp(a_{k'}(x))}.$$

1256 The corresponding surrogate is

$$\begin{aligned} \Phi^{\text{ASM}}(a; x, y, m) := & -\log \xi_y(a(x)) - \sum_{j=1}^J \left(\mathbf{1}[m_j = y] \log \psi_j(a(x)) \right. \\ & \left. + \mathbf{1}[m_j \neq y] \log(1 - \psi_j(a(x))) \right). \end{aligned} \quad (21)$$

1257 Prediction uses the maxima-preserving property of (ξ, ψ) , equivalently the argmax of the underlying
 1258 score vector $a(x)$. The expert estimate can be written through the equivalent sigmoid identity

$$\psi_j(a(x)) = \sigma \left(a_{K+j}(x) - \log \sum_{k' \neq k^*} \exp(a_{k'}(x)) \right),$$

1259 where $k^* \in \arg \max_{k \in [K]} a_k(x)$. To see the equivalence directly, write

$$B(a) := \sum_{k=1}^K \exp(a_k) - \max_{k \in [K]} \exp(a_k) = \sum_{k \neq k^*} \exp(a_k).$$

1260 Then the denominator of ψ_j is exactly $B(a) + \exp(a_{K+j})$, so

$$\psi_j(a(x)) = \frac{\exp(a_{K+j}(x))}{\exp(a_{K+j}(x)) + B(a(x))} = \sigma(a_{K+j}(x) - \log B(a(x))).$$

1261 This form is algebraically equivalent to the original max-based denominator and is the convenient
1262 form for the derivative calculations below. It also makes clear that the map is only piecewise smooth:
1263 when the maximizing class changes, the active comparison set changes as well.

1264 This representation is also conceptually useful. It shows that each expert head is, by itself, an ordinary
1265 sigmoid probability whose reference level is set by the non-max class logits. The expert-side issue in
1266 A-SM is therefore not that the expert probability is ill-defined or unbounded. The issue is that the
1267 class logits appear inside that reference level, so class and expert optimization are still entangled.

1268 G.2 Conditional Target

1269 At the population level, A-SM is intended to repair exactly the defect that motivated Cao et al.: the
1270 single augmented softmax of additive CE does not directly produce bounded estimates of the class
1271 posterior and expert correctness. A-SM changes the parameterization so that the class side remains a
1272 categorical distribution over the first K coordinates, while each expert side is a bounded probability
1273 in $[0, 1]$.

1274 The multi-expert extension in Appendix H of [Cao et al. \(2024\)](#) states the corresponding optimality
1275 property directly: at a conditional optimum,

$$\xi_k^*(x) = \eta_k(x), \quad \psi_j^*(x) = \alpha_j(x),$$

1276 for every class $k \in [K]$ and expert $j \in [J]$. This is the precise reason A-SM is the strongest
1277 augmented-action baseline on axis (i). Unlike additive CE, PicCE, and Mao25, its main limitation is
1278 not that it identifies the wrong statistical object. It identifies the right object but reaches it through a
1279 shared geometry that still couples class and expert optimization.

1280 That distinction matters for the rest of the paper. Once the target axis is repaired, any remaining
1281 weakness must come from the optimization geometry rather than from statistical misspecification.
1282 A-SM is therefore the cleanest test case for asking whether “correct target” is sufficient by itself. Our
1283 answer is no: even with the right conditional optimum, the local update structure can remain coupled
1284 in a way that the decoupled surrogate avoids.

1285 G.3 Class–Expert Gradient Coupling

1286 **Proposition 7** (Class–expert gradient coupling in A-SM). *Fix $k^* \in \arg \max_{k \in [K]} a_k$. For any $r \neq k^*$,
1287 let $\pi_r = \exp(a_r) / \sum_{k \neq k^*} \exp(a_k)$. Then*

$$\frac{\partial \Phi^{\text{ASM}}}{\partial a_r} = \underbrace{(\xi_r - \mathbf{1}\{r=y\})}_{\text{class term}} - \pi_r \underbrace{\sum_{j=1}^J (\psi_j - \mathbf{1}\{m_j=y\})}_{\text{expert leakage}}. \quad (8)$$

1288 **Proof** We work on a region where the maximizing class index k^* is fixed. On such a region, the
1289 A-SM map is smooth and the active comparison set in the expert denominators does not change. The
1290 goal of this proof is only the *first-order* coupling statement: we derive the ordinary class term, then
1291 show how each expert residual contributes an additional term to every non-max class logit.

1292 **Class term.** The term $-\log \xi_y$ is the standard multiclass cross-entropy over $a_1, \dots, a_K$. Its gradient
1293 with respect to a_r is $\xi_r - \mathbf{1}\{r=y\}$.

1294 **Expert term.** For a fixed expert j , define

$$t_j := \mathbf{1}\{m_j = y\}, \quad B(a) := \sum_{k \neq k^*} \exp(a_k).$$

1295 Then

$$\psi_j = \sigma(a_{K+j} - \log B(a)) = \frac{\exp(a_{K+j})}{\exp(a_{K+j}) + B(a)}.$$

1296 The corresponding expert contribution to the loss is the Bernoulli cross-entropy

$$L_j = -t_j \log \psi_j - (1 - t_j) \log(1 - \psi_j).$$

1297 Differentiating with respect to the sigmoid argument gives the standard identity

$$\frac{\partial L_j}{\partial v_j} = \psi_j - t_j, \quad v_j := a_{K+j} - \log B(a).$$

1298 This already shows that the expert head itself is well behaved: with respect to its own effective score
1299 v_j , it is just an ordinary Bernoulli logistic loss. So the only remaining task is to differentiate v_j with
1300 respect to the class logits and see how the coupling arises.

1301 If $r \neq k^*$, then

$$\frac{\partial v_j}{\partial a_r} = -\frac{1}{B(a)} \frac{\partial B(a)}{\partial a_r} = -\frac{\exp(a_r)}{\sum_{k \neq k^*} \exp(a_k)} = -\pi_r,$$

1302 where

$$\pi_r := \frac{\exp(a_r)}{\sum_{k \neq k^*} \exp(a_k)}.$$

1303 If instead $r = k^*$, then a_{k^*} does not appear in $B(a)$ on this region, so $\partial v_j / \partial a_{k^*} = 0$.

1304 Combining the two derivatives gives

$$\frac{\partial L_j}{\partial a_r} = \frac{\partial L_j}{\partial v_j} \frac{\partial v_j}{\partial a_r} = -(\psi_j - t_j) \pi_r, \quad r \neq k^*,$$

1305 and

$$\frac{\partial L_j}{\partial a_{k^*}} = 0.$$

1306 This is the leakage mechanism in its simplest form: every expert residual $\psi_j - t_j$ is injected into
1307 every non-max class logit through the same weight π_r .

1308 It is also useful to record the expert-side own derivative. Since $\partial v_j / \partial a_{K+j} = 1$,

$$\frac{\partial L_j}{\partial a_{K+j}} = \frac{\partial L_j}{\partial v_j} = \psi_j - t_j.$$

1309 So the expert block itself behaves exactly like a standard Bernoulli cross-entropy. The pathology is
1310 not on the expert-side update viewed in isolation. The pathology is that the same expert residuals
1311 also enter the class gradient.

1312 Summing over j for $r \neq k^*$ gives

$$\sum_{j=1}^J \frac{\partial L_j}{\partial a_r} = -\pi_r \sum_{j=1}^J (\psi_j - \mathbf{1}\{m_j = y\}).$$

1313 Adding the class term yields the stated formula.

1314 The sign pattern is now transparent. If many experts are correct but currently underestimated, then
1315 many terms $\psi_j - \mathbf{1}\{m_j = y\}$ are negative, and their sum adds an extra negative contribution to each
1316 non-max class gradient. If instead many experts are currently overestimated, the same mechanism
1317 pushes in the opposite direction. Either way, the class update no longer reflects only class-side error.
1318 It is contaminated by the aggregate expert residual. ■

1319 G.4 Curvature

1320 For A-SM the most informative second-order object is the mixed class–expert block. It captures
 1321 exactly the feature that distinguishes A-SM from the decoupled surrogate: even though the target is
 1322 correct, the local quadratic geometry does not separate the class and expert subproblems.

1323 We again work on a region where the maximizing class index k^* is fixed. From Proposition 7, the
 1324 gradient on a non-max class logit a_r is

$$\frac{\partial \Phi^{\text{ASM}}}{\partial a_r} = (\xi_r - \mathbf{1}\{r = y\}) - \pi_r \sum_{j=1}^J (\psi_j - \mathbf{1}\{m_j = y\}), \quad r \neq k^*.$$

1325 To obtain the mixed Hessian entry, differentiate this expression with respect to an expert logit a_{K+j} .

1326 The class term $\xi_r - \mathbf{1}\{r = y\}$ depends only on the class logits, so its derivative with respect to a_{K+j}
 1327 is zero. The coefficient π_r also depends only on the non-max class logits on the fixed- k^* region, so it
 1328 too is constant with respect to a_{K+j} . Thus

$$\frac{\partial^2 \Phi^{\text{ASM}}}{\partial a_r \partial a_{K+j}} = -\pi_r \frac{\partial \psi_j}{\partial a_{K+j}}, \quad r \neq k^*.$$

1329 Now recall that

$$\psi_j = \sigma(v_j), \quad v_j = a_{K+j} - \log \sum_{k \neq k^*} \exp(a_k).$$

1330 Since the second term of v_j involves only class logits, we have

$$\frac{\partial v_j}{\partial a_{K+j}} = 1.$$

1331 Applying the sigmoid derivative then gives

$$\frac{\partial \psi_j}{\partial a_{K+j}} = \psi_j(1 - \psi_j).$$

1332 Substituting this into the previous display yields the full mixed entry:

$$\frac{\partial^2 \Phi^{\text{ASM}}}{\partial a_r \partial a_{K+j}} = -\pi_r \psi_j(1 - \psi_j), \quad r \neq k^*.$$

1333 For $r = k^*$, the expert-leakage term is absent already at first order, so the mixed derivative is zero.

1334 Therefore the non-max-class/expert mixed block is exactly

$$H_{\text{mix}} = -\pi d^\top, \quad \pi_r = \frac{\exp(a_r)}{\sum_{k \neq k^*} \exp(a_k)}, \quad d_j = \psi_j(1 - \psi_j).$$

1335 This formula contains several useful facts at once.

1336 **Rank-one structure.** The mixed block is an outer product, so it has rank at most one. The coupling
 1337 is therefore highly structured, not arbitrary: all non-max class coordinates interact with the expert
 1338 block through the same class-side direction π .

1339 **Operator-norm bound.** Because $H_{\text{mix}} = -\pi d^\top$, its operator norm is

$$\|H_{\text{mix}}\|_{\text{op}} = \|\pi\|_2 \|d\|_2.$$

1340 Now π is a probability vector over the non-max classes, hence $\|\pi\|_2 \leq 1$. Also each entry of d
 1341 satisfies

$$0 \leq d_j = \psi_j(1 - \psi_j) \leq \frac{1}{4},$$

1342 because the scalar function $u \mapsto u(1 - u)$ is maximized at $u = 1/2$. Hence

$$\|d\|_2 \leq \sqrt{\sum_{j=1}^J \left(\frac{1}{4}\right)^2} = \frac{\sqrt{J}}{4},$$

1343 and therefore

$$\|H_{\text{mix}}\|_{\text{op}} \leq \frac{\sqrt{J}}{4}.$$

1344 **Interpretation.** The factor π_r measures how active non-max class r is inside the class-side comparison, while $\psi_j(1 - \psi_j)$ measures how uncertain expert j currently is. The mixed coupling is strongest
 1345 when both are non-negligible: there is nontrivial class competition among the non-max classes, and
 1346 many experts are in the uncertain middle regime rather than near 0 or 1.
 1347

1348 This is exactly the regime in which the decoupled surrogate differs most sharply. For the decoupled
 1349 surrogate, the mixed class–expert block is identically zero. For A-SM, the target is already correct,
 1350 but the optimization geometry still contains a structured interaction whose norm can grow with the
 1351 expert pool.

1352 For completeness, note also what is *not* happening. The expert block does not introduce expert–expert
 1353 coupling through the other expert logits: each expert head still depends only on its own expert
 1354 coordinate and the class-side reference level. The mixed block is therefore the central second-order
 1355 object, because it is the unique place where the class and expert tasks remain tied together.

1356 G.5 Head-to-Head with the Decoupled Surrogate

1357 **Axis (i): statistical target.** A-SM and the decoupled surrogate are aligned on the conditional target.
 1358 Both recover the Bayes-sufficient quantities at the population optimum: a categorical class posterior
 1359 together with bounded expert probabilities. The difference between them is therefore not about
 1360 whether they estimate the right object, but about how they organize the optimization problem that
 1361 reaches that object.

1362 This point is worth stating carefully because it distinguishes A-SM from the earlier baselines. For
 1363 additive CE, PiCCE, and Mao25, the criticism already appears at the level of what the surrogate is
 1364 trying to fit. For A-SM, that is no longer the issue. A-SM repairs the population target. In that sense
 1365 it is much closer to the decoupled surrogate than the other augmented baselines. The remaining gap
 1366 is entirely geometric.

1367 **Axis (ii): local geometry.** A-SM still routes the expert objectives through the class scores, so the
 1368 class gradient contains the expert-leakage term of Proposition 7. The decoupled surrogate removes
 1369 this term completely. Its class gradient depends only on (p, y) and its mixed Hessian block is
 1370 zero. This is the key residual weakness of A-SM. It has already repaired the target distortion and
 1371 boundedness problems, but it has not fully separated the class and expert subproblems. The decoupled
 1372 surrogate is the only method in our comparison that keeps the same target alignment while making
 1373 the mixed block identically zero.

1374 The practical consequence is that the two methods react differently during training. Under the
 1375 decoupled surrogate, an error in expert j changes only expert j ’s update. Under A-SM, that same
 1376 expert error also perturbs the class update whenever the non-max class normalizer is active. So even
 1377 though both methods are aiming for the same conditional optimum, they do not follow the same
 1378 local route toward that optimum. This is the sharpest way to understand the residual gap between
 1379 the two methods: A-SM is target-correct but route-coupled, whereas the decoupled surrogate is both
 1380 target-correct and route-decoupled.

1381 G.6 Worked Example: Complementary Experts

1382 Consider $K = 3$ classes and $J = 4$ complementary experts. Fix a region where the maximizing class
 1383 index is $k^* = 1$, and inspect the gradient on the class-2 logit a_2 for a sample with $y = 2$. Suppose

$$\xi_2 = 0.25, \quad \pi_2 = 0.40, \quad \sum_{j=1}^4 (\psi_j - \mathbf{1}\{m_j = y\}) = 0.80.$$

1384 Then Proposition 7 gives

$$\frac{\partial \Phi^{\text{ASM}}}{\partial a_2} = (\xi_2 - 1) - \pi_2 \sum_{j=1}^4 (\psi_j - \mathbf{1}\{m_j = y\}) = -0.75 - 0.32 = -1.07.$$

1385 The first term, -0.75 , is the ordinary class signal saying that class 2 is the true label and its current
 1386 probability estimate is too small. The second term, -0.32 , is not class information at all. It is
 1387 leakage from the expert heads. It appears only because A-SM routes every expert residual through
 1388 the non-max class normalizer, and it becomes larger as the aggregate expert residual grows.

1389 This example is deliberately chosen so that the issue is easy to see. If A-SM were fully decoupled,
 1390 the class-2 logit would receive exactly the ordinary cross-entropy gradient -0.75 . Instead, the expert
 1391 block contributes an extra -0.32 , which changes both the magnitude and the direction of the local
 1392 class update. The class learner is therefore not reacting purely to class error on this sample; it is also
 1393 reacting to unresolved expert uncertainty.

1394 Under the decoupled surrogate, the same sample would have class gradient

$$\frac{\partial \Phi^{\text{dec}}}{\partial w_2} = p_2 - 1,$$

1395 with no expert contribution at all. So the class update would reflect only the classifier’s own error on
 1396 the sample. This is the clearest numerical illustration of the axis-(ii) difference between A-SM and
 1397 the decoupled surrogate: both can target the right optimum, but only the decoupled surrogate keeps
 1398 the class update free of expert noise.

1399 The example also makes clear why the issue can worsen with many experts. The leakage enters
 1400 through the aggregate residual $\sum_{j=1}^J (\psi_j - \mathbf{1}\{m_j = y\})$. When several experts are still in their
 1401 uncertain regime, those residuals add before being injected into the class gradient. The decoupled
 1402 surrogate has no analogous accumulation because each expert head is optimized in isolation.

1403 The conclusion is therefore narrower than for additive CE or PiCCE, but still important. A-SM shows
 1404 that fixing the statistical target is not enough. One can correct boundedness and recover the right
 1405 population quantities, yet still retain a shared local geometry that mixes class and expert learning.
 1406 The decoupled surrogate closes that final gap by keeping the same target alignment while removing
 1407 the mixed interaction altogether.

1408 H OvA (Verma et al., 2022)

1409 The multi-expert one-vs-all construction of Verma et al. (2022) is the cleanest augmented baseline on
 1410 the optimization axis. It removes the shared softmax entirely and replaces it by independent binary
 1411 logistic tasks. As a result, none of the earlier geometric pathologies remain: there is no multiplicity
 1412 amplification, no winner-take-all starvation, and no mixed class–expert curvature.

1413 The remaining question is different. What is lost when the class side is represented not by one
 1414 categorical distribution in Δ^K , but by K independent Bernoulli scores? At the conditional optimum,
 1415 OvA still identifies the correct class marginals $\eta_k(x)$ and expert utilities $\alpha_j(x)$. The gap appears
 1416 away from that optimum. For arbitrary parameters, the class vector produced by OvA need not lie on
 1417 the simplex, so the largest class score is not automatically the confidence of a coherent categorical
 1418 model. This section makes that distinction explicit.

1419 H.1 Formulation

1420 OvA separates the routing problem into $K + J$ binary tasks: one logistic problem for each class and
 1421 one logistic problem for each expert. The class coordinates ask whether the true label equals a given
 1422 class, while the expert coordinates ask whether a given expert is correct on the current example.

1423 **Definition 18** (Multi-expert OvA, Verma et al. (2022)). Let $g(x) \in \mathbb{R}^K$ denote class scores and
 1424 $s(x) \in \mathbb{R}^J$ denote expert scores. Define

$$\begin{aligned} \gamma_k(y) &:= 2\mathbf{1}[y = k] - 1, & \zeta_j(y, m) &:= 2\mathbf{1}[m_j = y] - 1. \\ \Phi^{\text{OvA}}(g, s; x, y, m) &:= \sum_{k=1}^K \log(1 + \exp(-\gamma_k(y)g_k(x))) + \sum_{j=1}^J \log(1 + \exp(-\zeta_j(y, m)s_j(x))). \end{aligned}$$

1426 This is the signed-label form used in Verma et al. (2022). For the analysis below it is convenient to
 1427 rewrite the same loss using the binary targets

$$b_k(y) := \mathbf{1}\{y = k\}, \quad d_j(y, m) := \mathbf{1}\{m_j = y\}.$$

1428 Using the standard identity

$$\log(1 + \exp(-(2b - 1)v)) = -b \log \sigma(v) - (1 - b) \log(1 - \sigma(v)), \quad b \in \{0, 1\},$$

1429 the OvA loss can be written equivalently as

$$\begin{aligned} \Phi^{\text{OvA}}(g, s; x, y, m) &= \sum_{k=1}^K \left(-b_k(y) \log \sigma(g_k(x)) - (1 - b_k(y)) \log(1 - \sigma(g_k(x))) \right) \\ &+ \sum_{j=1}^J \left(-d_j(y, m) \log \sigma(s_j(x)) - (1 - d_j(y, m)) \log(1 - \sigma(s_j(x))) \right). \end{aligned}$$

1431 This form makes two features explicit. First, every coordinate is optimized independently. Second,
1432 the class head is not a softmax over the K classes; it is a vector of K separate sigmoids.

1433 The prediction rule chooses the classifier if $\max_{k \in [K]} g_k(x) > \max_{j \in [J]} s_j(x)$ and otherwise defers
1434 to the largest expert score. Since the sigmoid is strictly increasing, this is equivalent to comparing
1435 $\max_k \sigma(g_k(x))$ and $\max_j \sigma(s_j(x))$ on the probability scale.

1436 H.2 Conditional Minimizer

1437 **Proposition 19** (OvA conditional minimizer). *The conditional OvA risk is a sum of $K + J$ in-*
1438 *dependent Bernoulli cross-entropies. Its optimal probabilities satisfy $\sigma(g_k^*(x)) = \eta_k(x)$ and*
1439 *$\sigma(s_j^*(x)) = \alpha_j(x)$ whenever the corresponding targets lie in $(0, 1)$. If a target equals 0 or 1,*
1440 *the conditional infimum is approached as the corresponding logit tends to $\mp\infty$ or $\pm\infty$, respectively.*

1441 **Proof** Fix an input x . Taking conditional expectation over $(Y, M) \mid X = x$ gives

$$\begin{aligned} \mathcal{C}_{\text{OvA}}(g, s \mid x) &= \sum_{k=1}^K \left(-\eta_k(x) \log \sigma(g_k) - (1 - \eta_k(x)) \log(1 - \sigma(g_k)) \right) \\ &+ \sum_{j=1}^J \left(-\alpha_j(x) \log \sigma(s_j) - (1 - \alpha_j(x)) \log(1 - \sigma(s_j)) \right). \end{aligned}$$

1443 The key point is that this expression separates completely across coordinates. There is one Bernoulli
1444 cross-entropy for each class coordinate and one Bernoulli cross-entropy for each expert coordinate.
1445 The class-side target for coordinate k is $\eta_k(x)$, while the expert-side target for coordinate j is $\alpha_j(x)$.

1446 To make the argument explicit, consider the generic Bernoulli cross-entropy with target parameter
1447 $\theta \in [0, 1]$:

$$\varphi(v) = -\theta \log \sigma(v) - (1 - \theta) \log(1 - \sigma(v)),$$

1448 where v is the Bernoulli logit. Differentiating once gives

$$\varphi'(v) = \sigma(v) - \theta.$$

1449 Thus any finite stationary point must satisfy $\sigma(v) = \theta$. Differentiating again gives

$$\varphi''(v) = \sigma(v)(1 - \sigma(v)) \geq 0.$$

1450 Hence φ is convex, and it is strictly convex whenever $\sigma(v) \in (0, 1)$. If $\theta \in (0, 1)$, the condition
1451 $\sigma(v) = \theta$ identifies the unique finite minimizer. If $\theta = 1$, then $\varphi'(v) = \sigma(v) - 1 < 0$ for every
1452 finite v , so the risk decreases monotonically and its infimum is approached as $v \rightarrow +\infty$. Similarly, if
1453 $\theta = 0$, then $\varphi'(v) = \sigma(v) > 0$ for every finite v , so the infimum is approached as $v \rightarrow -\infty$.

1454 Applying this argument coordinatewise yields

$$\sigma(g_k^*(x)) = \eta_k(x), \quad \sigma(s_j^*(x)) = \alpha_j(x),$$

1455 for all $k \in [K]$ and $j \in [J]$. ■

1456 This proposition is the positive side of OvA. Unlike Additive CE or PiCCE, OvA does not distort the
1457 conditional target. At the probability level, it identifies the correct class-marginal quantities $\eta_k(x)$
1458 and the correct expert utilities $\alpha_j(x)$. The limitation of OvA is therefore not that it aims at the wrong
1459 population object. The limitation is representational: the class head becomes a coherent categorical
1460 posterior only at the optimum, not by construction at intermediate parameter values.

1461 H.3 Curvature

1462 The local geometry of OvA is as simple as its conditional risk. Because each loss term depends on
1463 only one score coordinate, the first and second derivatives separate completely.

1464 Consider first a class coordinate g_k . Its samplewise contribution is

$$L_k(g_k; y) = -b_k(y) \log \sigma(g_k) - (1 - b_k(y)) \log(1 - \sigma(g_k)).$$

1465 Differentiating with respect to g_k gives

$$\frac{\partial L_k}{\partial g_k} = \sigma(g_k) - b_k(y).$$

1466 Differentiating once more gives

$$\frac{\partial^2 L_k}{\partial g_k^2} = \sigma(g_k)(1 - \sigma(g_k)).$$

1467 Exactly the same calculation applies to an expert coordinate s_j :

$$L_j(s_j; y, m) = -d_j(y, m) \log \sigma(s_j) - (1 - d_j(y, m)) \log(1 - \sigma(s_j)),$$

1468 so

$$\frac{\partial L_j}{\partial s_j} = \sigma(s_j) - d_j(y, m), \quad \frac{\partial^2 L_j}{\partial s_j^2} = \sigma(s_j)(1 - \sigma(s_j)).$$

1469 Now observe that no class term depends on any other class coordinate or on any expert coordinate,
1470 and no expert term depends on any class coordinate or any other expert coordinate. Therefore every
1471 mixed partial derivative is zero:

$$\frac{\partial^2 \Phi^{\text{OvA}}}{\partial g_k \partial g_r} = 0 \quad (k \neq r), \quad \frac{\partial^2 \Phi^{\text{OvA}}}{\partial s_j \partial s_\ell} = 0 \quad (j \neq \ell), \quad \frac{\partial^2 \Phi^{\text{OvA}}}{\partial g_k \partial s_j} = 0.$$

1472 If we concatenate all coordinates into one vector $a = (g_1, \dots, g_K, s_1, \dots, s_J) \in \mathbb{R}^{K+J}$, the Hessian
1473 is therefore diagonal:

$$\nabla_a^2 \Phi^{\text{OvA}} = \text{Diag}(\sigma(a_i)(1 - \sigma(a_i))),$$

1474 where the diagonal entries corresponding to class coordinates are $\sigma(g_k)(1 - \sigma(g_k))$ and those
1475 corresponding to expert coordinates are $\sigma(s_j)(1 - \sigma(s_j))$.

1476 The eigenvalues of a diagonal matrix are exactly its diagonal entries. Since $0 \leq \sigma(v)(1 - \sigma(v)) \leq 1/4$
1477 for every $v \in \mathbb{R}$, we obtain

$$\lambda_{\max}(\nabla_a^2 \Phi^{\text{OvA}}) \leq \frac{1}{4}.$$

1478 This is the cleanest geometry among the baselines considered in the paper. There is no class–expert
1479 coupling, no winner-take-all effect, and no multiplicity amplification. If OvA has a weakness, it is
1480 not a weakness of local optimization geometry.

1481 H.4 Head-to-Head with the Decoupled Surrogate

1482 **Axis (i): statistical target.** OvA and the decoupled surrogate agree on the most important population
1483 statement. At the conditional optimum, both methods recover the class marginals $\eta_k(x)$ and the
1484 expert utilities $\alpha_j(x)$. In that sense, OvA is much closer to the decoupled surrogate than the earlier
1485 augmented-softmax surrogates.

1486 The difference lies in how these quantities are represented away from the optimum. The decoupled
1487 surrogate parameterizes the class side as one categorical distribution $p(x) \in \Delta^K$ for every parameter
1488 value. The vector $p(x)$ is therefore a coherent class posterior throughout training. OvA instead
1489 parameterizes the class side as K independent sigmoids. Each coordinate can be interpreted as a
1490 class marginal, but the full vector need not lie in Δ^K unless the model happens already to be at the
1491 conditional optimum.

1492 **Axis (ii): local geometry.** On the geometric axis the two methods are genuinely similar. Both are
 1493 coordinatewise, both have bounded curvature, and both avoid all of the shared-geometry pathologies
 1494 that affect the augmented-action family. The decoupled surrogate is not preferred because it is more
 1495 decoupled than OvA; there is little to improve on that axis.

1496 The decoupled surrogate is preferred because it combines this clean geometry with a categorical class
 1497 model. The routing rule in the decoupled surrogate always compares $\max_k p_k(x)$ and $\max_j u_j(x)$
 1498 on a coherent probability scale. OvA also compares quantities in $[0, 1]$, but the class side is not forced
 1499 to behave like one categorical posterior during training. Several class sigmoids can be simultaneously
 1500 large, and the largest one can overstate classifier confidence relative to the expert utilities.

1501 H.5 Worked Example: Several Class Sigmoids Above 0.5

1502 Consider $K = 3$ and suppose that during training the OvA class head outputs

$$\sigma(g_1) = 0.77, \quad \sigma(g_2) = 0.73, \quad \sigma(g_3) = 0.69.$$

1503 These values are individually admissible Bernoulli probabilities, but together they cannot form a
 1504 categorical posterior because they sum to 2.19. Nothing in the OvA objective penalizes this. Each
 1505 class coordinate is trained separately, so the model is free to regard several classes as simultaneously
 1506 likely in the one-vs-all sense.

1507 Now suppose that the best expert estimate is

$$\sigma(s_1) = 0.70.$$

1508 The OvA routing rule compares the largest class score with the best expert score. Since

$$\max_{k \in [K]} \sigma(g_k) = 0.77 > 0.70 = \max_{j \in [J]} \sigma(s_j),$$

1509 OvA predicts rather than defers.

1510 Now contrast the decoupled surrogate. A coherent class posterior for the same ambiguous input could
 1511 be

$$p = (0.40, 0.35, 0.25), \quad u_1 = 0.70.$$

1512 Then the Bayes-style comparison is immediate: $\max_k p_k = 0.40 < 0.70$, so the decoupled surrogate
 1513 defers. This does not mean that OvA is inconsistent. If the model class is rich enough and training
 1514 finds the conditional optimum, the OvA class sigmoids will match the true class-marginal probabilities
 1515 and therefore sum to one. The point is instead that the intermediate states visited during training need
 1516 not have that property. Because routing uses the current scores, this lack of simplex structure can
 1517 matter in practice.

1518 The example isolates the remaining gap between OvA and the decoupled surrogate. OvA has already
 1519 fixed the geometric side of the problem. What it has not fixed is the representation of class uncertainty.
 1520 The decoupled surrogate keeps the same clean optimization geometry while ensuring that the class
 1521 side remains a categorical posterior at every point along training.

1522 I Comparison of Excess-Risk Constants

1523 We compare the surrogate-regret transfer guarantees of the decoupled surrogate with the available
 1524 baselines, focusing on how the calibration constant scales with the size of the multi-expert action
 1525 space. Two distinct prior references need to be separated carefully. The constant $\sqrt{2(J+1)}$ comes
 1526 from the earlier comp-sum reduction analyzed by Mao et al. (2024e), which covers the Mozannar-
 1527 style augmented-action construction. It is not the constant for the later Mao25 single-stage surrogate
 1528 (Mao et al., 2025d). For the Mao25 single-stage surrogate used in this paper, the relevant excess-risk
 1529 factor scales with the augmented action dimension $K + J$.

1530 **The decoupled surrogate (Theorem 10).** The decoupled surrogate satisfies

$$\mathcal{E}_\perp - \mathcal{E}_\perp^* \leq \underbrace{\max \left\{ 2\sqrt{2}, \sqrt{\frac{2J}{\lambda}} \right\}}_{C_{\text{dec}}} \sqrt{\mathcal{E}_{\text{dec}} - \mathcal{E}_{\text{dec}}^*}.$$

1531 Whenever the per-expert weight satisfies $\lambda/J \geq 1/4$, the constant $C_{\text{dec}} = 2\sqrt{2} \approx 2.83$ is independent
 1532 of J .

1533 **Add. CE (Mozannar and Sontag, 2020).** For Add. CE, Mao et al. (2024e) derive a transfer constant
 1534 $C_{\text{CE}} = \sqrt{2(J+1)}$, which grows as $O(\sqrt{J})$. This is the $\sqrt{2(J+1)}$ factor quoted in the main paper
 1535 for the Add. CE baseline.

1536 **Mao25 (Mao et al., 2025d).** The single-stage surrogate studied by Mao et al. (2025d) should be kept
 1537 separate from the earlier comp-sum reduction above. Quantitatively, the corresponding excess-risk
 1538 constant scales with the full augmented action dimension. In the multi-expert setting considered here,
 1539 this yields

$$C_{\text{Mao25}} = K + J,$$

1540 so the dependence on the number of experts is linear: $O(J)$ for fixed K . This is a different regime
 1541 from the Add. CE bound above: Add. CE inherits the earlier $\sqrt{2(J+1)}$ comp-sum constant, whereas
 1542 Mao25 scales with the entire augmented action count.

1543 **Other surrogates.** For the remaining baselines, the cited analyses establish Bayes consistency but
 1544 do not give explicit transfer constants: (Verma et al., 2022; Cao et al., 2024; Liu et al., 2026).

1545 **Summary.**

Surrogate	Consistency type	Transfer constant	Dependence on J
Add. CE	Bayes	—	—
PiCCE	Bayes	—	—
OvA	Bayes	—	—
A-SM	Bayes	—	—
Add. CE	Quantitative	$\sqrt{2(J+1)}$	$O(\sqrt{J})$
Mao25	Quantitative	$K + J$	$O(J)$ for fixed K
Decoupled	Quantitative	$\max\{2\sqrt{2}, \sqrt{2J/\lambda}\}$	$O(1)$ for fixed λ/J

1547 Among the multi-expert surrogates with quantitative transfer constants, the decoupled surrogate is
 1548 the only one whose transfer constant does not grow with J for a fixed per-expert weight λ/J . When
 1549 $\lambda/J \geq 1/4$, the decoupled surrogate constant simplifies to $2\sqrt{2}$. Add. CE inherits a $\sqrt{2(J+1)}$
 1550 factor from the earlier deferral analysis, while the Mao25 surrogate scales with the full augmented
 1551 action dimension $K + J$. The difference reflects a structural advantage: the decoupled surrogate
 1552 estimates (η, α) through independent subproblems, so the conditional surrogate excess decomposes
 1553 into a sum of KL terms that can be converted to estimation error without cross-contamination. The
 1554 augmented-action surrogates embed classes and experts in a shared geometry whose normalization
 1555 couples the KL structure, so the quantitative transfer factor worsens as the action space grows.

1556 J Compute Resources

1557 All experiments were run on NVIDIA A100 GPUs with 40GB of memory. Each run uses a single
 1558 GPU; multi-seed results are obtained by repeating the same configuration independently across
 1559 seeds. The synthetic suites are lightweight and finish within minutes per seed, while the CIFAR-10,
 1560 CIFAR-10H, and Covertypes experiments fit comfortably within the same single-GPU setting.

1561 K Synthetic Experiments

1562 This appendix reports the retained synthetic suites used in the paper. Each suite instantiates one
 1563 mechanism from Section 4 under a distribution for which the Bayes rule is known exactly. In every
 1564 suite we specify (η, α) analytically, so the Bayes-optimal L2D action at each x is determined by the
 1565 comparison

$$\max_{k \in [K]} \eta_k(x) \quad \text{versus} \quad \max_{j \in [J]} \alpha_j(x),$$

1566 and the exact defer regret $\Delta_{\perp}(f) := \mathcal{E}_{\perp}(f) - \mathcal{E}_{\perp}^*$ can be evaluated directly.

1567 **Protocol.** All suites use linear models (no hidden layer) unless stated otherwise. Every method
 1568 receives the same feature vector X and training tuple (X, Y, M) . The ground-truth quantities $\eta(x)$
 1569 and $\alpha(x)$ are used only *after* training to evaluate the exact Bayes regret, and on the validation split to
 1570 select the checkpoint with the lowest true defer loss. This model-selection rule is applied uniformly

to every surrogate. All reported numbers are means with one standard deviation in parentheses over multiple seeds (3–5, stated per suite).

Each suite is written in the same order: *objective*, *setting*, *Bayes rule* and *link to the theory*, and *result*.

K.1 Nested Redundant Experts

Objective. Vary the overlap multiplicity while keeping the Bayes decision problem fixed, so that any performance degradation must be attributed to the surrogate itself, not to a harder task. Although the suite is designed around Proposition 4 (additive CE amplification), the nesting also stresses every other baseline for surrogate-specific reasons discussed in the analysis below.

Setting. The retained setting uses $K=16$ classes, $J=24$ experts, $n_{\text{train}}=900$, and $n_{\text{test}}=8,000$. All models are linear (no hidden layer).

Features and class posteriors. Each point draws a region indicator $R \sim \text{Bernoulli}(0.65)$. The feature vector lies in $\mathbb{R}^K$. On the defer-favorable region D ($R=1$), the label is uniform on $[K]$, $\eta_k(x)=1/K$ for all k , and $X = 3.5 \mathbf{1}_K + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 0.05^2 I_K)$; hence no class is linearly separable. On D^c ($R=0$), the class posterior is near-deterministic, $\eta_Y(x)=0.998$, and the feature for the true label is shifted to 5 (easily separable).

Expert utilities. The J experts have monotonically decreasing accuracies on D . Define a spacing variable $\rho_j := \log(1+j)/\log(J)$ for $j \in \{0, \dots, J-1\}$ (logarithmic spacing, so the first few experts are close in quality and the last ones are markedly weaker). On D ,

$$\alpha_j(x) = 0.99 - (0.99 - 0.75) \rho_j, \quad (22)$$

giving $\alpha_0=0.99$ for the strongest expert down to $\alpha_{J-1}=0.75$ for the weakest. On D^c , all experts are near-random: $\alpha_j(x) = 0.04 - (0.04 - 0.002) \rho_j$.

Nested correctness. A single latent $U \sim \text{Unif}[0, 1]$ is drawn per sample, and expert j is correct iff $U \leq \alpha_j(x)$. Because $\alpha_0 \geq \alpha_1 \geq \dots \geq \alpha_{J-1}$, this implies

$$\{M_{j+1} = Y\} \subseteq \{M_j = Y\} \quad \text{for all } j.$$

When expert j is incorrect, its prediction is drawn uniformly from $[K] \setminus \{Y\}$. The nesting guarantees that adding experts never changes the Bayes action: expert 0 is always Bayes-best on D , and classification is always best on D^c . What grows with J is only the realized overlap $|\mathcal{J}(Y, M)|$.

Bayes rule. On D : defer to expert 0. On D^c : classify. This is invariant for all J .

Why each baseline degrades. The propositions in Section 4 predict a different failure mechanism for each surrogate:

- **Add. CE** (Proposition 4). The gradient carries the factor $1+|\mathcal{J}|$, and the Hessian scales identically. With $J=24$ nested experts and high α values, a typical defer-region sample has $|\mathcal{J}| \approx 18$, so each such sample is weighted $\sim 19\times$ compared to a classify-region sample. This gives high-overlap samples disproportionate gradient and curvature relative to samples near the classify-versus-defer boundary.
- **PicCE** (Proposition 5). Only the winning correct expert receives a negative gradient; all other correct experts receive the positive push $2q_{K+j} > 0$. With 24 nested experts, as many as 23 correct experts are suppressed on samples where all experts are correct, and in general $|\mathcal{J}| - 1$ correct experts are suppressed whenever multiple experts are correct. The resulting winner-take-all lock-in concentrates all expert mass on whichever expert had the highest logit at initialization, preventing the model from learning the correct ranking $\alpha_0 > \dots > \alpha_{J-1}$.
- **A-SM** (Proposition 7). The class-logit gradient contains the expert leakage term $-\pi_r \sum_j (\psi_j - t_j)$. When many experts are correct, the residuals $\psi_j - t_j$ accumulate, injecting noise into the class update. In a linear model, class and expert logits share the same parameters through the feature matrix, so this leakage directly corrupts the classify-versus-defer boundary.
- **OvA** (Appendix H.2). The K independent sigmoid class heads are not coupled by a simplex constraint, so the class confidence $\max_k \sigma(g_k)$ can be miscalibrated. With $K=16$ uniform classes on D , the population binary targets are $1/16$ for each class, but finite-sample and optimization errors

are not constrained to preserve a simplex. The maximum over 16 independently trained sigmoids can therefore be inflated relative to the true $\max_k \eta_k = 1/16$, biasing the routing comparison toward classifying when it should defer.

• **Mao25** (Proposition 6). The gradient factor $(S_S - 1)$ is common to all acceptable actions, so the update cannot rank experts. Additionally, the MAE-like structure means $\sum_{i \in \mathcal{S}} -\partial \Phi / \partial a_i = S_S(1 - S_S) \leq 1/4$, so the total corrective signal is weak and vanishes as $S_S \rightarrow 0$, creating slow convergence toward the defer region.

Result. Table 4 reports the retained $J=24$ point. The broader suite sweeps $J \in \{16, 24, 32\}$ (Table 5).

Method	Exact defer regret	System accuracy
Decoupled	0.0002 (0.0001)	0.993 (0.001)
PiCCE	0.238 (0.116)	0.755 (0.118)
OvA	0.294 (0.020)	0.699 (0.020)
A-SM	0.304 (0.035)	0.688 (0.035)
Mao25	0.323 (0.035)	0.669 (0.036)
Add. CE	0.348 (0.038)	0.646 (0.036)

Table 4: Nested redundant suite at $J=24$. The decoupled surrogate remains essentially Bayes-optimal; all baselines degrade severely. Means with std in parentheses over 3 seeds.

	$J = 16$	$J = 24$	$J = 32$
Decoupled defer regret	0.013	0.0002	0.007
Best non-decoupled defer regret	0.226	0.238	0.293
Best non-decoupled method	PiCCE	PiCCE	OvA

Table 5: J -sweep summary. The decoupled surrogate stays near Bayes-optimal as the expert pool grows, while the best non-decoupled regret increases monotonically.

Analysis. Because the Bayes rule is unchanged across the J -sweep, the large degradation of every baseline cannot be explained by a harder routing problem; it is attributable to how each surrogate responds to redundant multiplicity. The decoupled surrogate remains near Bayes-optimal ($\Delta_{\perp} = 0.0002$ at $J=24$) because its expert loss decomposes into J independent BCE terms: each α_j is estimated coordinatewise, and the gradient $\partial \Phi^{\text{dec}} / \partial s_j = (\lambda/J)(u_j - t_j)$ carries no cross-expert coupling and no dependence on $|\mathcal{J}|$ (Proposition 9). As the J -sweep in Table 5 shows, the best non-decoupled regret *increases* with J (from 0.226 to 0.293), while the decoupled surrogate stays below 0.013 throughout, confirming that the baselines’ failures scale with the overlap multiplicity as predicted by the theory.

K.2 Rare Specialist for PiCCE

Objective. Isolate the winner-take-all starvation from Proposition 5. The construction places a specialist expert that is Bayes-preferred on a rare region R but whose correct events are mostly shared with a stronger generalist. PiCCE should suppress the specialist because only the current $\arg \max$ winner receives the reinforcing gradient $2q_{K+j^*} - 1 < 0$, while every other correct expert is pushed down by $2q_{K+j} > 0$ (eq. (7)).

Setting. We use $K=2$ classes, $J=2$ experts (generalist $j=1$, specialist $j=2$), $n_{\text{train}}=7,000$, $n_{\text{test}}=18,000$, and 5 seeds. All models are linear.

Features and class posteriors. Draw $Z \sim \text{Unif}([-1, 1]^2)$ and an independent rare indicator $R \sim \text{Bernoulli}(0.15)$. The feature vector is $X = (Z_1, Z_2, \mathbf{1}\{R\}) \in \mathbb{R}^3$. The class posterior depends on the region:

$$\max_k \eta_k(x) = \begin{cases} 0.60, & x \in R, \\ 0.90, & x \notin R, \end{cases}$$

with the maximizing class determined by $\text{sgn}(Z_1)$. Classification is therefore easy outside R ($\eta_{\text{max}}=0.90$) and uncertain on R ($\eta_{\text{max}}=0.60$).

1649 *Expert utilities.* Outside R , expert correctness is independent: $\alpha_{\text{gen}}(x)=0.45$, $\alpha_{\text{spec}}(x)=0.15$.
 1650 Neither expert is useful, and the Bayes action is to classify. On R , the marginal utilities are:

$$\alpha_{\text{spec}}(x) = 0.75 > \alpha_{\text{gen}}(x) = 0.60 = \max_k \eta_k(x) = 0.60,$$

1651 so the Bayes action is to defer to the specialist, with a tie between the generalist and the classifier
 1652 below the specialist.

1653 *Joint correctness on R .* The key design choice is that correctness is *not* independent on R . We
 1654 specify the joint law directly:

	Specialist correct	Specialist incorrect
Generalist correct	0.55	0.05
Generalist incorrect	0.20	0.20

1655 This gives $\alpha_{\text{spec}}=0.55+0.20=0.75$ and $\alpha_{\text{gen}}=0.55+0.05=0.60$ as required. Critically, when the
 1657 specialist is correct on R , the generalist is also correct with probability

$$\Pr(C_{\text{gen}}=1 \mid C_{\text{spec}}=1, x \in R) = \frac{0.55}{0.75} = 0.73.$$

1658 So 73% of the specialist’s positive training signal comes from shared-correct samples where both
 1659 experts are right.

1660 **Bayes rule.** On R (15% of data): defer to the specialist ($\alpha_{\text{spec}}=0.75 > \alpha_{\text{gen}}=0.60 = \eta_{\text{max}}=0.60$).
 1661 On R^c (85% of data): classify ($\eta_{\text{max}}=0.90 > \alpha_{\text{gen}}=0.45$).

1662 **Why PiCCE fails.** On the 73% of specialist-correct samples on R where the generalist is also correct
 1663 ($\mathcal{J}=\{1, 2\}$), PiCCE selects the winner $j^* = \arg \max_{j \in \mathcal{J}} a_{K+j}$. The gradient on the *non-winning*
 1664 correct expert is $\partial \Phi^{\text{PiCCE}} / \partial a_{K+j} = 2q_{K+j} > 0$ (Proposition 5), which pushes its logit *down*. If the
 1665 generalist starts with a marginally higher logit (by initialization noise), the specialist is suppressed on
 1666 every shared-correct sample. The gap compounds across iterations: the generalist wins more arg max
 1667 selections, receives more reinforcing gradient, and the specialist is driven to zero. On the 27% of
 1668 specialist-correct events that are specialist-only ($C_{\text{gen}}=0$), the specialist does receive a negative
 1669 gradient, but $15\% \times 27\%=4\%$ of total training data is not enough to overcome the opposite signal
 1670 from the dominant shared-correct mass.

1671 **Why the decoupled surrogate succeeds.** The decoupled surrogate trains each expert head with
 1672 an independent BCE loss: $\partial \Phi^{\text{dec}} / \partial s_j = (\lambda/J)(u_j - t_j)$, where $u_j = \sigma(s_j)$ and $t_j = \mathbf{1}\{M_j=Y\}$. On
 1673 a shared-correct sample, *both* experts receive the reinforcing gradient $u_j - 1 < 0$ independently.
 1674 No arg max selection occurs, so the generalist’s logit does not compete with the specialist’s. Both
 1675 utilities are estimated faithfully, and the routing comparison $\sigma(s_{\text{spec}}) \gtrless \sigma(s_{\text{gen}})$ learns $0.75 > 0.60$
 1676 from the data.

1677 **Metrics.** We report two specialist-specific diagnostics alongside the standard defer regret and system
 1678 accuracy:

- 1679 • *Specialist selection on R :* $\Pr(r(X)=\text{spec} \mid X \in R)$, computed over all test samples in R
 1680 (including those that classify; classifying samples count as not selecting the specialist).
- 1681 • *Shared-correct routing:* $\Pr(r(X)=\text{spec} \mid X \in R, C_{\text{gen}}=1, C_{\text{spec}}=1)$, i.e. specialist selection
 1682 restricted to the shared-correct subset. This is the hardest case: both experts are right, and only the
 1683 ranking $\alpha_{\text{spec}} > \alpha_{\text{gen}}$ distinguishes them.

1684 **Result.** Table 6 reports the retained result (means with std in parentheses over 5 seeds).

1685 **Analysis.** The table reveals three distinct failure modes.

1686 *PiCCE (winner-take-all starvation).* PiCCE has reasonable coverage (0.821, close to the Bayes-
 1687 optimal 0.85), so it does learn the classify-versus-defer boundary. However, its specialist selection is
 1688 exactly 0 on R : when it defers, it always picks the generalist. The shared-correct routing rate is also 0,
 1689 confirming that on the decisive subset where both experts are correct, PiCCE never learns to prefer
 1690 the specialist. This is the direct empirical realization of Proposition 5: the generalist captures the
 1691 arg max early and the positive-gradient push $2q_{K+\text{spec}} > 0$ suppresses the specialist permanently.

Method	Defer regret	System acc.	Coverage	Spec. sel. on R	Shared-correct routing
Decoupled	0.003 (0.002)	0.875 (0.002)	0.849	0.993 (0.007)	0.994 (0.007)
OvA	0.003 (0.001)	0.875 (0.003)	0.850	0.988 (0.023)	0.987 (0.026)
A-SM	0.003 (0.002)	0.875 (0.003)	0.850	0.988 (0.017)	0.988 (0.017)
Mao25	0.026 (0.002)	0.852 (0.004)	1.000	0.000 (0.000)	0.000 (0.000)
PiCCE	0.035 (0.001)	0.843 (0.002)	0.821	0.000 (0.000)	0.000 (0.000)
Add. CE	0.324 (0.067)	0.554 (0.069)	0.128	1.000 (0.000)	1.000 (0.000)

Table 6: Rare-specialist suite. *Coverage* is $\Pr(r(X)=\text{classify})$; Bayes-optimal coverage is 0.85. PiCCE defers but never to the specialist; Mao25 never defers at all; The decoupled surrogate, OvA, and A-SM recover the specialist almost perfectly.

1692 *Mao25 (no deferral learned).* Mao25 has coverage 1.0: it *never* defers and always classifies.
1693 Consequently, its specialist selection is trivially 0 (there are no deferral events). Its system accuracy
1694 of 0.852 is approximately the accuracy of a Bayes-optimal classifier that ignores experts: $0.85 \times$
1695 $0.90 + 0.15 \times 0.60 = 0.855$. This is consistent with the MAE-like optimization pathology from
1696 Proposition 6: the corrective gradient on acceptable actions has magnitude $S_S(1-S_S) \leq 1/4$, which
1697 vanishes when $S_S \approx 0$ early in training. Mao25 never escapes the classify-everything initialization.

1698 *Add. CE (global collapse).* Add. CE has coverage 0.128 (it defers 87% of the time) and defer
1699 regret 0.324. Even with only $J=2$ experts, the gradient amplification of Proposition 4 is sufficient to
1700 destabilize training in this low- K regime: the factor $1+|\mathcal{J}|$ reaches 3 on shared-correct samples, and
1701 the optimizer overshoots the defer boundary.

1702 *The decoupled surrogate, OvA, and A-SM (correct routing).* All three methods achieve near-Bayes-
1703 optimal defer regret (0.003), coverage close to 0.85, and specialist selection ≥ 0.988 on R . The
1704 task has only $J=2$ experts, so the A-SM coupling (Proposition 7) and the OvA miscalibration are
1705 negligible here. This confirms that the PiCCE failure is specific to the winner-take-all mechanism
1706 and not a general difficulty of the task.

1707 K.3 Shared Acceptability and Expert Ranking for Mao25

1708 **Objective.** Isolate the ranking limitation of Proposition 6: the Mao25 gradient factor (S_S-1) is the
1709 same for all acceptable actions, so the samplewise update cannot distinguish the best expert from a
1710 merely acceptable one. We construct a task where the main difficulty is expert ranking *within* a defer
1711 region under heavy shared acceptability.

1712 **Setting.** We use $K=10$ classes, $J=4$ experts, $n_{\text{train}}=1,000$, $n_{\text{test}}=12,000$, and 3 seeds. All models
1713 are linear.

1714 *Features and class posteriors.* Each point draws a defer-region indicator $D \sim \text{Bernoulli}(0.60)$.
1715 The feature vector lies in $\mathbb{R}^{K+J} = \mathbb{R}^{14}$. On D , a sector $Q \sim \text{Unif}([J])$ is drawn, the label is uniform
1716 on $[K]$ ($\eta_k=1/10$ for all k), and

$$X_{1:K} = 3.1 \mathbf{1}_K + \varepsilon_{\text{cls}}, \quad X_{K+Q} = 2.2 + \varepsilon_Q, \quad \varepsilon \sim \mathcal{N}(0, 0.05^2).$$

1717 The first K features are uninformative for the class (all near 3.1); the sector feature X_{K+Q} identifies
1718 which expert is locally best. On D^c , the class posterior is near-deterministic ($\eta_Y=0.998$, feature of
1719 the true label shifted to 5).

1720 *Joint expert correctness on D .* Inside D , the identity of the best expert changes with the sector Q .
1721 The joint correctness of the $J=4$ experts is drawn from the following event model:

Event	Probability	Who is correct
All correct	0.10	all 4 experts
Q + one random other	0.78	Q and one uniformly chosen $j \neq Q$
Q only	0.05	only Q
One random non- Q only	0.03	one uniformly chosen $j \neq Q$
None correct	0.04	no expert

1723 The marginal utilities on D are therefore

$$\alpha_Q(x) = 0.10 + 0.78 + 0.05 = 0.93, \quad \alpha_{j \neq Q}(x) = 0.10 + \frac{0.78}{3} + \frac{0.03}{3} = 0.37.$$

On D^c , all experts have utility $\alpha_j=0.05$.

Two properties of this design are essential. First, *shared acceptability is high*: on 88% of defer-region samples ($0.10+0.78$) at least two experts are correct, so the acceptable set $\mathcal{S}(y, m)$ typically contains the true class plus multiple experts. Second, *expert ranking matters*: the Bayes-optimal expert changes with the sector Q , and the gap between the best and non-best utilities is large (0.93 versus 0.37).

Bayes rule. On D (60% of data): defer to the sector-best expert Q ($\alpha_Q=0.93 \gg \eta_{\max}=0.10$). On D^c (40% of data): classify ($\eta_Y=0.998 \gg \alpha_j=0.05$).

Bayes-optimal system accuracy: $0.60 \times 0.93 + 0.40 \times 0.998 = 0.957$; Bayes-optimal coverage: 0.40.

Why Mao25 fails at ranking. Proposition 6 shows that the Mao25 gradient on an acceptable action $i \in \mathcal{S}$ is $q_i(S_S-1)$, where $S_S = \sum_{c \in \mathcal{S}} q_c$ is the total softmax mass on the acceptable set. The factor (S_S-1) is the same for every acceptable action: the best expert Q and a non-best expert $j \neq Q$ receive the same gradient magnitude (up to the softmax weight q_i , which is controlled by the model, not the data). This means the samplewise loss provides no direct signal that Q should be preferred over other acceptable experts.

Across samples, the model can still learn the ranking indirectly because expert Q appears in the acceptable set much more often than the other experts on sector Q . However, this signal comes from marginal frequency across samples rather than from an explicit within-sample preference among acceptable experts. The result is degraded but non-trivial ranking ability, as measured by the best-expert selection rate.

Metrics. We report *best-expert selection on D*: among test samples in D where the model defers, the fraction that selects the sector-best expert Q . With $J=4$ symmetric sectors, chance level (uniform random among experts) is 0.25. A model that picks uniformly among *correct* experts would achieve ≈ 0.48 ($0.10 \times \frac{1}{4} + 0.78 \times \frac{1}{2} + 0.05 \times 1 + 0.03 \times 0 + 0.04 \times \frac{1}{4}$). Perfect ranking gives 1.0.

Result. Table 7 reports the retained result (means with std in parentheses over 3 seeds).

Method	Defer regret	System acc.	Coverage	Best expert sel. on D
Decoupled	0.0008 (0.001)	0.957 (0.003)	0.397	1.000
A-SM	0.064 (0.020)	0.894 (0.018)	0.330	1.000
OvA	0.082 (0.076)	0.876 (0.078)	0.349	1.000
PiCCE	0.129 (0.045)	0.829 (0.043)	0.383	0.661
Add. CE	0.151 (0.033)	0.808 (0.031)	0.241	1.000
Mao25	0.154 (0.043)	0.804 (0.045)	0.324	0.750

Table 7: Pair-shared ranking suite. Chance-level best-expert selection is 0.25; uniform-among-correct is 0.48. Mao25 reaches 0.750, well above chance but far below the 1.000 achieved by the decoupled surrogate and three other baselines, confirming that the loss provides partial but degraded ranking signal.

Analysis. The decoupled surrogate reaches the Bayes-optimal system accuracy (0.957) with coverage $0.397 \approx 0.40$ and perfect best-expert selection. Its independent BCE heads estimate each α_j faithfully, so the routing comparison $\max_j \sigma(s_j) \geq \max_k p_k$ is well-calibrated on both the classify/defer boundary and the expert ranking.

Mao25. Mao25 achieves best-expert selection of 0.750. This is well above chance (0.25) and above the uniform-among-correct baseline (0.48), indicating that the linear model does extract partial ranking from the sector feature. However, it is significantly below the 1.000 achieved by the decoupled surrogate, A-SM, OvA, and Add. CE, all of which have loss terms that produce per-expert ranking signal. The gap from 0.750 to 1.000 is the empirical cost of the (S_S-1) factor being common to all acceptable actions (Proposition 6). Combined with imperfect coverage (0.324 vs optimal 0.40), this yields a defer regret of 0.154 — nearly $200\times$ the decoupled surrogate regret.

PiCCE. PiCCE has even worse ranking (0.661) with high variance across seeds (one seed drops to 0.50), consistent with the winner-take-all lock-in from Proposition 5: which expert “wins” depends on initialization noise, producing unstable ranking.

1764 *A-SM and OvA.* Both achieve perfect ranking (1.000) and reasonable coverage, but their defer
 1765 regret (0.064 and 0.082) is still 80–100 $\times$ the decoupled surrogate value. The ranking is correct, but
 1766 the classify-versus-defer boundary is less well-calibrated: A-SM coverage is 0.330 (optimal: 0.40),
 1767 indicating over-deferral. For A-SM this is consistent with the class–expert coupling (Proposition 7)
 1768 perturbing the class logits.

1769 *Add. CE.* Add. CE also achieves perfect ranking but has the lowest coverage (0.241), reflecting
 1770 severe over-deferral from the gradient amplification of Proposition 4: with $J=4$ experts and shared
 1771 acceptability on 88% of defer samples, $|\mathcal{J}|$ is typically 2–4, inflating the gradient by factors of 3–5.

1772 K.4 Logit-Space Geometry Diagnostics

1773 The task-level suites above measure end-to-end performance. This section complements them with
 1774 *direct measurements of the differential quantities* predicted by Propositions 4, 5, and 7.

1775 **Protocol.** For each diagnostic we train a linear model (no hidden layer) to convergence on a synthetic
 1776 L2D task, then compute per-sample gradients and Hessians of the surrogate loss on held-out test
 1777 logits. Using trained (rather than random) logits ensures the measurements reflect the geometry at
 1778 the operating point the optimizer actually reaches. All quantities are computed on the *augmented*
 1779 action vector $a = (a_1, \dots, a_K, a_{K+1}, \dots, a_{K+J})$, i.e. on the logit space of the surrogate itself.
 1780 Each diagnostic is repeated across multiple seeds (stated per table); we report means with standard
 1781 deviations in parentheses across all per-sample measurements pooled over seeds.

1782 **Add. CE curvature inflation (Proposition 4).** Proposition 4 predicts that the Hessian of the additive
 1783 CE surrogate Φ^{CE} has largest eigenvalue $\lambda_{\max} \leq (1 + |\mathcal{J}|)/2$, growing linearly in the number of
 1784 correct experts $|\mathcal{J}|$. The decoupled surrogate, by contrast, has $\lambda_{\max} \leq 1/2$ for the class block and
 1785 $\lambda/(4J)$ per expert entry, independently of $|\mathcal{J}|$ ((12)).

1786 *Setup.* We use the nested redundant-expert construction (Section K.1) with $J \in \{1, 4, 8, 16, 32\}$
 1787 to control the overlap multiplicity $|\mathcal{J}|$. For each value of J , we train an Add. CE and a decoupled
 1788 surrogate model ($\lambda = 1.0$) on $n_{\text{train}}=3,500$ samples and evaluate on $n_{\text{test}}=6,000$. We restrict to test
 1789 samples where at least one expert is correct ($|\mathcal{J}| > 0$). For each such sample, we compute (i) the
 1790 gradient norm $\|\nabla_a \Phi\|_2$ and (ii) the top Hessian eigenvalue $\lambda_{\max}(\nabla_a^2 \Phi)$ via the full $(K+J) \times (K+J)$.
 1791 Results are pooled over ~ 900 qualifying samples per (J, method) combination across 5 independent
 1792 seeds.

Method	$ \mathcal{J} $	Gradient norm $\ \nabla_a \Phi\ _2$	Top eigenvalue $\lambda_{\max}(\nabla_a^2 \Phi)$
Add. CE	1	0.586 (0.305)	0.821 (0.084)
Add. CE	4	0.609 (0.266)	1.093 (0.067)
Add. CE	8	0.616 (0.292)	1.101 (0.116)
Add. CE	16	0.633 (0.287)	1.125 (0.186)
Add. CE	32	0.613 (0.282)	1.117 (0.181)
Decoupled	1	0.704 (0.222)	0.368 (0.111)
Decoupled	4	0.573 (0.250)	0.358 (0.137)
Decoupled	8	0.551 (0.280)	0.361 (0.125)
Decoupled	16	0.534 (0.290)	0.360 (0.134)
Decoupled	32	0.510 (0.297)	0.359 (0.132)

Table 8: Per-sample curvature as overlap multiplicity $|\mathcal{J}|$ varies (5 seeds, ~ 900 samples each). The Add. CE top eigenvalue grows from 0.82 at $|\mathcal{J}|=1$ to 1.12 at $|\mathcal{J}|=32$, consistent with the multiplicative $(1+|\mathcal{J}|)$ factor in Proposition 4, while also reflecting the dependence of the softmax covariance on q . The decoupled surrogate remains flat at ≈ 0.36 for all $|\mathcal{J}|$, consistent with the constant $\leq 1/2$ bound from (12).

1793 *Reading the table.* The CE eigenvalue jumps from 0.82 to 1.09 as $|\mathcal{J}|$ goes from 1 to 4, then
 1794 saturates near 1.12 for larger $|\mathcal{J}|$. The saturation is expected: at $|\mathcal{J}|=32$ the theoretical bound is
 1795 $(1 + 32)/2 = 16.5$, but the *realised* eigenvalue is much smaller because the softmax probabilities
 1796 q_i become very small when spread over $K+32$ actions. The key observation is the sharp increase
 1797 and high-overlap saturation of $\lambda_{\max}$ for Add. CE, versus the *complete independence* from $|\mathcal{J}|$ for the
 1798 decoupled surrogate.

PiCCE starvation gradient (Proposition 5). Proposition 5 predicts that under PiCCE, every correct expert other than the winner j^* receives a strictly positive gradient $\partial\Phi^{\text{PiCCE}}/\partial a_{K+k} = 2q_{K+k} > 0$, pushing its logit *downward* despite being correct. Under the decoupled surrogate, each expert gradient is $(\lambda/J)(u_j - t_j)$, which is *negative* (pulling toward deferral) whenever the expert is correct ($t_j = 1$) and its sigmoid output $u_j < 1$.

Setup. We use the PiCCE specialist task (Section K.2) with $K=2$ classes and $J=2$ experts, where samples routinely have both experts correct. For each method (PiCCE and the decoupled surrogate, $\lambda=1.0$) we train on $n_{\text{train}}=4,500$ and evaluate on $n_{\text{test}}=7,000$. We select test samples where $|\mathcal{J}| > 1$ (multiple correct experts). For PiCCE, the “suppressed expert” is any $k \in \mathcal{J}$ with $k \neq j^*$, where $j^* = \arg \max_{j \in \mathcal{J}} a_{K+j}$ is the winning expert (determined by the trained logit magnitudes). For the decoupled surrogate, we measure the gradient on *all* correct expert logits (since the decoupled surrogate has no winner selection). Results pool 784 decoupled-surrogate gradient samples and 392 PiCCE gradient samples across 5 seeds.

Method	Mean gradient $\partial\Phi/\partial a_{K+k}$	Positive-gradient rate
Decoupled	−0.245 (0.103)	0.000
PiCCE	+0.240 (0.076)	1.000

Table 9: Gradient on suppressed correct expert logits (5 seeds; 784 decoupled-surrogate and 392 PiCCE measurements). PiCCE produces a *positive* gradient on 100% of suppressed experts — pushing them away from deferral, exactly as Proposition 5 predicts ($2q_{K+k} > 0$). The decoupled surrogate always produces a *negative* gradient, pulling every correct expert toward deferral.

Reading the table. The positive-gradient rate is 1.000 for PiCCE and 0.000 for the decoupled surrogate, with zero variance: the predicted sign is not a statistical tendency but an exact per-sample property. The mean PiCCE gradient on the suppressed expert (+0.240) is comparable in magnitude to the decoupled surrogate gradient (−0.245), so the starvation is not a negligible effect — it is an active repulsive force of the same scale as the decoupled surrogate’s attractive signal.

A-SM class–expert coupling (Proposition 7). Proposition 7 shows that the A-SM gradient on each class logit a_r ($r \neq k^*$) includes a leakage term $-\pi_r \sum_j (\psi_j - t_j)$ from the expert estimates. At second order, this coupling produces a non-zero mixed class–expert Hessian block $H_{\text{mix}} = \partial^2 \Phi^{\text{ASM}} / (\partial a_{1:K} \partial a_{K+1:K+J})$. The decoupled surrogate’s Hessian is block-diagonal ((12)), so its mixed block is *identically* zero.

Setup. We vary $J \in \{1, 3, 5, 9, 17\}$ by adding distractor experts (always incorrect) alongside one correct expert. For each J , we train an A-SM and a decoupled surrogate model ($\lambda=1.0$) on $n_{\text{train}}=4,500$ and evaluate on $n_{\text{test}}=7,000$. For each test sample we assemble the full $(K+J) \times (K+J)$ Hessian via second-order autograd, extract the $K \times J$ upper-right block H_{mix} , and compute its operator norm $\|H_{\text{mix}}\|_{\text{op}}$. We report the mean operator norm over 128 test samples, averaged across 5 seeds. For the decoupled surrogate, the mixed block is analytically zero by construction (class logits w and expert logits s_j enter disjoint terms in $\Phi_{\lambda}^{\text{dec}}$), so the code assigns 0.0 directly.

Method	Experts J	Mixed-block norm $\ H_{\text{mix}}\ _{\text{op}}$
A-SM	1	0.122 (0.004)
A-SM	3	0.207 (0.011)
A-SM	5	0.263 (0.009)
A-SM	9	0.351 (0.009)
A-SM	17	0.475 (0.012)
Decoupled	1	0.000 (0.000)
Decoupled	3	0.000 (0.000)
Decoupled	5	0.000 (0.000)
Decoupled	9	0.000 (0.000)
Decoupled	17	0.000 (0.000)

Table 10: Mixed class–expert Hessian block operator norm (5 seeds, 128 samples each). A-SM coupling grows monotonically from 0.12 at $J=1$ to 0.48 at $J=17$, consistent with the $O(\sqrt{J})$ scaling from Proposition 7. The decoupled surrogate is identically zero for all J , confirming the block-diagonal structure of (12).

1829 *Reading the table.* The A-SM coupling norm grows roughly as $\sqrt{J}$: from 0.12 at $J=1$ to 0.48
1830 at $J=17 \approx 0.12\sqrt{17} \approx 0.49$. This scaling matches the theoretical $O(\sqrt{J})$ bound derived in
1831 Appendix G.4. The growth means that adding experts — even *distractor* experts that are never correct
1832 — injects progressively more expert-side noise into the class gradient. The decoupled surrogate
1833 is immune: its zero mixed block guarantees that expert estimation errors cannot perturb the class
1834 probability estimates, regardless of J .

1835 L CIFAR-10 with Synthetic Experts

1836 The synthetic experiments of Section K use linear models on low-dimensional Gaussian data, which
1837 isolates surrogate pathologies in a setting where the Bayes solution is known. We now test whether
1838 the same effects persist on a real vision task.

1839 **Setup.** We train a ResNet-12 backbone (end-to-end, no pre-training) on CIFAR-10 ($K=10$ classes)
1840 with standard data augmentation (random crop, horizontal flip). All methods share the same backbone
1841 architecture and optimiser (AdamW, lr 10^{-3} , 80 epochs). A held-out validation split (5 000 samples)
1842 is used to select the best checkpoint (lowest validation deferral loss) and, for the decoupled surrogate,
1843 to choose the per-expert weight $\lambda/J = 1/2$. For the decoupled surrogate, the linear head is a *split*
1844 *head*: two independent projections $f_{\text{cls}}: \mathbb{R}^{256} \rightarrow \mathbb{R}^K$ and $f_{\text{exp}}: \mathbb{R}^{256} \rightarrow \mathbb{R}^J$, matching the decoupled
1845 loss structure. All other methods use a single $\mathbb{R}^{256} \rightarrow \mathbb{R}^{K+J}$ head. The total number of parameters
1846 is the same across all methods.

1847 **Redundant expert suite.** We construct J synthetic experts with *nested* correctness (Section K.1): a
1848 single latent $U \sim \text{Uniform}(0, 1)$ per sample determines all expert outcomes, with expert j correct iff
1849 $U < \alpha_j$ where $\alpha_j = 0.70 - 0.01j$. This ensures that the first 8 experts in a $J=32$ run are *identical*
1850 to those in a $J=8$ run, so performance changes are attributable purely to the surrogate’s response to
1851 added experts.

Table 11: CIFAR-10 redundant expert suite (mean $\pm$ std over 4 seeds). Best values per J -block in **bold**; second-best underlined. “Cls. Acc.” = classifier accuracy (ignoring the routing decision). “Cov.” = coverage (fraction classified by the model, not deferred).

Method	J	Validation Metrics			
		Sys. Acc. $\uparrow$	Defer Loss $\downarrow$	Cls. Acc.	Cov.
Decoupled	8	0.918 $\pm$.001	0.082 $\pm$.001	<u>0.909</u> $\pm$.001	0.961 $\pm$.004
Classifier only	8	<u>0.910</u> $\pm$.002	<u>0.090</u> $\pm$.002	0.910 $\pm$.002	1.000 $\pm$.000
OvA	8	0.901 $\pm$.001	0.099 $\pm$.001	0.875 $\pm$.001	0.857 $\pm$.015
PiCCE	8	0.858 $\pm$.007	0.142 $\pm$.007	0.888 $\pm$.008	0.616 $\pm$.032
A-SM	8	0.860 $\pm$.002	0.140 $\pm$.002	0.831 $\pm$.002	0.774 $\pm$.014
Add. CE	8	0.815 $\pm$.002	0.185 $\pm$.002	0.902 $\pm$.004	0.529 $\pm$.006
Mao25	8	0.703 $\pm$.000	0.297 $\pm$.000	0.105 $\pm$.024	0.000 $\pm$.000
Decoupled	16	0.919 $\pm$.003	0.081 $\pm$.003	<u>0.909</u> $\pm$.002	0.961 $\pm$.004
Classifier only	16	<u>0.911</u> $\pm$.001	<u>0.089</u> $\pm$.001	0.911 $\pm$.001	1.000 $\pm$.000
OvA	16	0.890 $\pm$.002	0.110 $\pm$.002	0.859 $\pm$.004	0.829 $\pm$.014
PiCCE	16	0.860 $\pm$.011	0.140 $\pm$.011	0.895 $\pm$.006	0.633 $\pm$.064
A-SM	16	0.827 $\pm$.003	0.174 $\pm$.003	0.772 $\pm$.003	0.662 $\pm$.035
Add. CE	16	0.790 $\pm$.013	0.210 $\pm$.013	0.895 $\pm$.003	0.502 $\pm$.035
Mao25	16	0.703 $\pm$.000	0.297 $\pm$.000	0.111 $\pm$.010	0.000 $\pm$.000
Decoupled	32	0.919 $\pm$.002	0.081 $\pm$.002	<u>0.908</u> $\pm$.003	0.957 $\pm$.005
Classifier only	32	<u>0.908</u> $\pm$.002	<u>0.092</u> $\pm$.002	0.908 $\pm$.002	1.000 $\pm$.000
OvA	32	0.880 $\pm$.003	0.120 $\pm$.003	0.844 $\pm$.003	0.798 $\pm$.015
PiCCE	32	0.859 $\pm$.008	0.141 $\pm$.008	0.880 $\pm$.014	0.636 $\pm$.027
A-SM	32	0.783 $\pm$.006	0.218 $\pm$.006	0.680 $\pm$.009	0.484 $\pm$.039
Add. CE	32	0.712 $\pm$.015	0.288 $\pm$.015	0.872 $\pm$.009	0.305 $\pm$.022
Mao25	32	0.703 $\pm$.000	0.297 $\pm$.000	0.105 $\pm$.025	0.000 $\pm$.000

1852 **Analysis.** The decoupled surrogate is the only method in our comparison that improves on a
1853 *standalone classifier*. At every J , the decoupled surrogate achieves higher system accuracy than
1854 the classifier-only baseline: +0.8 pp at $J=8$, +0.8 pp at $J=16$, and +1.1 pp at $J=32$. Every other

1855 L2D surrogate *degrades* below the classifier-only baseline, confirming that the surrogate pathologies
1856 identified in the synthetic experiments (Sections K.1–K.3) are not artefacts of the linear setting.

1857 *The decoupled surrogate preserves classifier quality; other surrogates corrupt it.* The decoupled sur-
1858 surrogate maintains classifier accuracy at 90.8–90.9% across all J , matching the classifier-only baseline.
1859 By contrast, A-SM drops from 83.1% ($J=8$) to 68.0% ($J=32$)—a 15 pp collapse consistent with
1860 expert gradients contaminating the class head through the shared softmax normalizer (Proposition 7).
1861 OvA also degrades (87.5% $\rightarrow$ 84.4%). This supports the interpretation that the decoupled surrogate’s
1862 decoupled loss protects the classifier: the BCE expert loss trains only the expert head, leaving the
1863 class head free to learn optimal class probabilities.

1864 *For most baselines, deferral actively hurts.* When system accuracy falls *below* classifier accuracy,
1865 routing decisions are destructive. At $J=8$, PiCCE achieves 88.8% classifier accuracy but only 85.8%
1866 system accuracy (-3.0 pp from deferral); Add. CE similarly loses 8.7 pp. Both route a substantial
1867 share of samples (PiCCE 38.4%, Add. CE 47.1%) to experts with $\approx 70\%$ accuracy—far worse than
1868 the classifier.

1869 *Add. CE and A-SM degrade monotonically with J .* Add. CE system accuracy drops from 81.5%
1870 ($J=8$) to 71.2% ($J=32$), matching the gradient amplification of Proposition 4: the $(1+|J|)$ factor
1871 grows with the redundant pool, pushing coverage down to 30.5%. A-SM shows a similar pattern
1872 (86.0% $\rightarrow$ 78.3%), driven by the class–expert coupling of Proposition 7.

1873 *Mao25 fails to learn.* Across all J , Mao25 converges to 100% deferral within the first few epochs,
1874 achieving exactly the expert oracle accuracy ($\approx 70.3\%$). The $O(1/(K+J))$ gradient magnitude
1875 through the softmax (Section F.3) prevents meaningful parameter updates on CIFAR-10’s $K=10$
1876 classes.

1877 *The decoupled surrogate is J -robust.* The decoupled surrogate system accuracy stays at 91.8–91.9%
1878 from $J=8$ to $J=32$ (a change of only 0.1 pp), compared to 2.1 pp for OvA and 10.3 pp for Add. CE.
1879 This is consistent with the theoretical prediction: the decoupled surrogate’s J -independent gradient
1880 structure (Proposition 9) translates directly to scalability on real image features.

1881 M CIFAR-10H with Real Human Annotators

1882 The synthetic-expert experiments above let us control the expert accuracy profile exactly, but they
1883 leave open the question of whether the same surrogate pathologies appear when the annotators are real
1884 humans with correlated, noisy labels. We therefore evaluate all methods on CIFAR-10H (Peterson
1885 et al., 2019), a dataset that pairs every CIFAR-10 test image with soft human annotations collected
1886 from $\approx 2,700$ Amazon Mechanical Turk workers.

1887 **Setup.** We first pretrain a ResNet-12 backbone on the full CIFAR-10 training set (50 000 images,
1888 80 epochs, SGD, lr 0.05, momentum 0.9, weight decay 5×10^{-4}) and then *freeze* it. Each L2D method
1889 trains only a linear head on top of the frozen 256-dimensional features. We split the CIFAR-10 test
1890 set (10 000 images annotated by CIFAR-10H) into 6 000 train / 2 000 val / 2 000 test. All methods
1891 share the same pretrained backbone and data split.

1892 **Annotator selection.** From the CIFAR-10H pool we retain only annotators with at least 200 labelled
1893 images. For each run we subsample $J \in \{5, 10, 20\}$ annotators uniformly at random; for images not
1894 labelled by a selected annotator, we impute a smoothed class-prior prediction (Laplace smoothing,
1895 $\alpha=1$). We repeat each configuration over 5 independent annotator draws (seeds 0–4) and report
1896 mean with std in parentheses across draws. All methods use AdamW (lr 10^{-3} , 30 epochs). For the
1897 decoupled surrogate, $\lambda=1/2$ and weight decay 10^{-4} ; all other methods use weight decay 10^{-4} .

1898 **Analysis.** *The decoupled surrogate leads at every pool size.* At $J=5$ the margin is narrow (0.960
1899 vs. 0.959 for OvA), but the decoupled surrogate’s advantage widens as the pool grows: at $J=20$ the
1900 decoupled surrogate reaches 0.961 while OvA drops to 0.956 (+0.5 pp). This mirrors the pattern
1901 from synthetic experts (Table 11): the decoupled surrogate scales better because its per-expert BCE
1902 terms are unaffected by pool size.

1903 *The decoupled surrogate preserves classifier quality; baselines collapse.* The decoupled surrogate’s
1904 classifier accuracy stays at 0.889–0.892 across all J , matching the standalone classifier baseline
1905 (0.889). A-SM maintains 0.890 at $J=5$ but collapses to 0.470 at $J=20$ (std 0.435, indicating
1906 complete instability); Add. CE similarly drops from 0.889 to 0.742; OvA degrades to 0.799 (std

Table 12: CIFAR-10H with real human annotators (test metrics, mean with std over 5 annotator draws). Best values per J -block in **bold**; second-best underlined. “Cls. Acc.” = classifier accuracy (ignoring the routing decision). “Cov.” = coverage (fraction classified by the model, not deferred).

Method	J	Test Metrics			
		Sys. Acc. $\uparrow$	Defer Loss $\downarrow$	Cls. Acc.	Cov.
Decoupled	5	0.960 (0.005)	0.039 (0.005)	0.892 (0.001)	0.517 (0.036)
OvA	5	<u>0.959</u> (0.005)	<u>0.039</u> (0.004)	0.889 (0.001)	0.375 (0.016)
A-SM	5	0.959 (0.004)	0.040 (0.004)	<u>0.890</u> (0.002)	0.341 (0.035)
PiCCE	5	0.954 (0.003)	0.045 (0.003)	0.889 (0.001)	0.390 (0.020)
Add. CE	5	0.953 (0.004)	0.045 (0.004)	0.889 (0.002)	0.248 (0.037)
Mao25	5	0.950 (0.005)	0.047 (0.005)	0.116 (0.133)	0.000 (0.000)
Classifier only	5	0.889 (0.000)	0.109 (0.001)	0.889 (0.000)	1.000 (0.000)
Decoupled	10	0.958 (0.003)	0.042 (0.004)	0.892 (0.003)	0.517 (0.038)
PiCCE	10	<u>0.957</u> (0.004)	<u>0.043</u> (0.004)	0.890 (0.002)	0.400 (0.006)
OvA	10	0.954 (0.008)	0.046 (0.008)	0.799 (0.203)	0.285 (0.164)
A-SM	10	0.952 (0.007)	0.047 (0.007)	0.747 (0.313)	0.170 (0.134)
Add. CE	10	0.952 (0.004)	0.047 (0.004)	<u>0.888</u> (0.001)	0.248 (0.004)
Mao25	10	0.948 (0.003)	0.052 (0.003)	0.077 (0.024)	0.000 (0.000)
Classifier only	10	0.889 (0.000)	0.111 (0.000)	0.889 (0.000)	1.000 (0.000)
Decoupled	20	0.961 (0.004)	0.039 (0.004)	<u>0.889</u> (0.001)	0.530 (0.038)
OvA	20	<u>0.956</u> (0.006)	<u>0.043</u> (0.006)	0.890 (0.002)	0.335 (0.047)
PiCCE	20	<u>0.953</u> (0.002)	0.047 (0.002)	0.890 (0.001)	0.399 (0.022)
Add. CE	20	0.951 (0.003)	0.049 (0.003)	0.742 (0.331)	0.183 (0.103)
A-SM	20	0.949 (0.005)	0.051 (0.005)	0.470 (0.435)	0.002 (0.004)
Mao25	20	0.948 (0.004)	0.051 (0.004)	0.045 (0.014)	0.000 (0.000)
Classifier only	20	0.888 (0.000)	0.112 (0.000)	0.888 (0.000)	1.000 (0.000)

0.203) at $J=10$. All three methods compensate by over-deferring—A-SM reaches $\geq 99\%$ deferral at $J=20$ —so their system accuracy is sustained by expert answers, not by the classifier. This confirms on real human annotators the same softmax-coupling (Proposition 7) and gradient-amplification (Proposition 4) pathologies seen in synthetic settings.

The decoupled surrogate maintains healthy coverage. The decoupled surrogate’s coverage is stable at ≈ 0.52 , meaning it classifies about half the samples itself and defers the rest—a balanced routing that produces system accuracy well above either component alone.

Mao25 defers everything. As in the synthetic setting, Mao25 converges to 100% deferral and never learns a classifier (classifier accuracy ≤ 0.116). Its system accuracy (≈ 0.950) is entirely the expert oracle accuracy, confirming the vanishing-gradient pathology of Proposition 6 on real human-annotator data.

N Covertypes with Model Experts

The preceding experiments use either synthetic accuracy profiles (Section L) or real human annotators (Section M). This final benchmark replaces both with *real pretrained models*: the experts are machine-learning classifiers with different architectures, feature views, and training biases. This is the most common deployment scenario for learning to defer—a system that routes each input to the most reliable model in a pool.

Setup. We use Forest CoverType (Blackard and Dean, 1999), a 7-class tabular benchmark with 54 features (continuous topographic variables, binary wilderness indicators, and binary soil-type indicators). A single stratified 60/20/20 train/validation/test split is made once; only the continuous features are standardized using train-set statistics. The L2D backbone is a shared two-layer MLP ($54 \rightarrow 256 \rightarrow 128$, ReLU, batch normalization, no dropout).

Expert pool. The pool contains $J=5$ pretrained models, trained once on the training split and then frozen:

- (i) a global histogram-gradient-boosted tree on all features;
- (ii) a low-elevation specialist (reweighted toward low elevations and wilderness areas 1–2);

- 1933 (iii) a high-elevation specialist (reweighted toward high elevations and wilderness areas 3–4);
 1934 (iv) a topography-view MLP trained only on the continuous variables;
 1935 (v) a context-view random forest trained only on wilderness and soil indicators.
- 1936 The expert oracle accuracy on the validation split is 0.946, a +9.8 pp gain over the best single expert
 1937 (0.848), confirming that routing is nontrivial. The data split and expert-pool training seed are held
 1938 fixed; only the downstream L2D training seed changes across runs.
- 1939 **Protocol.** All methods use AdamW with cosine learning-rate decay. All the approaches use a linear
 1940 head, lr 10^{-3} , weight decay 10^{-4} , and 40 epochs. We report results for per-expert weight $\lambda/J = 1/2$.
 1941 Checkpoints are selected by the lowest validation true deferral loss.

Table 13: Covertypes with model experts ($J=5$, mean $\pm$ std over 4 seeds). Best values in **bold**; second-best underlined. “Cls. Acc.” = classifier accuracy (ignoring routing). “Cov.” = coverage (fraction classified, not deferred).

Method	Sys. Acc. $\uparrow$	Defer Loss $\downarrow$	Cls. Acc.	Cov.
Decoupled	0.934 $\pm$.000	0.066 $\pm$.000	0.941 $\pm$.001	0.712 $\pm$.040
Classifier only	<u>0.929</u> $\pm$.001	<u>0.071</u> $\pm$.001	<u>0.929</u> $\pm$.001	1.000 $\pm$.000
PiCCE	0.908 $\pm$.003	0.092 $\pm$.003	0.911 $\pm$.002	0.449 $\pm$.007
OvA	0.907 $\pm$.001	0.093 $\pm$.001	0.906 $\pm$.000	0.437 $\pm$.011
A-SM	0.899 $\pm$.001	0.101 $\pm$.001	0.892 $\pm$.001	0.439 $\pm$.007
Add. CE	0.899 $\pm$.001	0.101 $\pm$.001	0.907 $\pm$.001	0.098 $\pm$.002
Mao25	0.877 $\pm$.001	0.123 $\pm$.001	0.351 $\pm$.023	0.000 $\pm$.000

1942 **Analysis.** *The decoupled surrogate is the only L2D method in our comparison that improves over*
 1943 *the standalone classifier.* The decoupled surrogate reaches 0.934 system accuracy, +0.5 pp above the
 1944 classifier-only baseline (0.929) and +2.6 pp above the best baseline (PiCCE at 0.908). Every other
 1945 L2D surrogate degrades below the standalone classifier, confirming that their routing decisions are
 1946 harmful.

1947 *The decoupled surrogate preserves and even improves classifier quality.* The decoupled surrogate
 1948 achieves the highest classifier accuracy (0.941), surpassing even the standalone classifier (0.929),
 1949 showing that the expert BCE loss does not interfere with class learning. By contrast, Mao25’s
 1950 classifier collapses to 0.351 and A-SM degrades to 0.892.

1951 *The decoupled surrogate maintains the highest coverage among L2D methods.* The decoupled
 1952 surrogate classifies 71.2% of samples directly, far more than PiCCE (44.9%), A-SM (43.9%), or Add.
 1953 CE (9.8%). The baselines over-defer because their class heads are too degraded to be confidently
 1954 preferred over an expert.

1955 *Mao25 collapses to full deferral.* As in CIFAR-10 and CIFAR-10H, Mao25 converges to 100%
 1956 deferral (zero coverage) and never learns a useful classifier, matching the vanishing-gradient prediction
 1957 of Proposition 6.

1958 *All surrogate pathologies transfer to model experts.* The method ranking observed in synthetic and
 1959 human-annotator experiments holds on tabular data with real model experts: The decoupled surrogate
 1960 $> \text{PiCCE} \approx \text{OvA} > \text{A-SM} \approx \text{Add. CE} > \text{Mao25}$. This demonstrates that the decoupled surrogate’s
 1961 advantages are not specific to vision backbones or to any particular type of expert.

1962 NeurIPS Paper Checklist

1963 1. Claims

1964 Question: Do the main claims made in the abstract and introduction accurately reflect the
 1965 paper’s contributions and scope?

1966 Answer: [\[Yes\]](#)

1967 Justification: The abstract and introduction state the two-axis (statistical target, optimization
 1968 geometry) analysis of five augmented-action surrogates, the decoupled surrogate, and
 1969 the consistency bound with J -independent constant for fixed λ/J ; all are established in
 1970 Sections 4–6 and the appendix.

1971 Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: A dedicated Limitations paragraph follows the Conclusion (Section 8). It states that we see no limitations specific to the decoupled surrogate beyond those inherent to multi-expert L2D itself—reliance on observed expert predictions over labeled training data and the assumption that each expert remains available at inference—and identifies relaxing those assumptions (missing or non-stationary experts, partial annotation, availability constraints, calibration of human trust) as future work that can build on top of the decoupled formulation.

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- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

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Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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Justification: All propositions and the main consistency theorem are stated with their distributional assumptions in the main text; complete proofs, together with worked examples, are given in Appendices B–C.5.

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- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
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